Microsoft

Code, Collaborate, Accelerate: The New Era of Power BI Development

Rui Romano

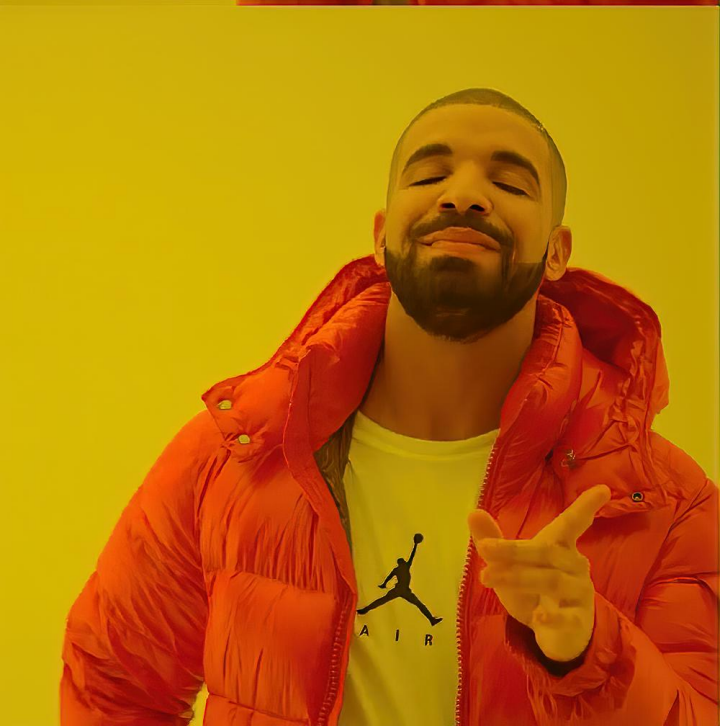
PM @ Power BI

There might be a better way…

Classified as Microsoft Confidential

Power BI in 2023

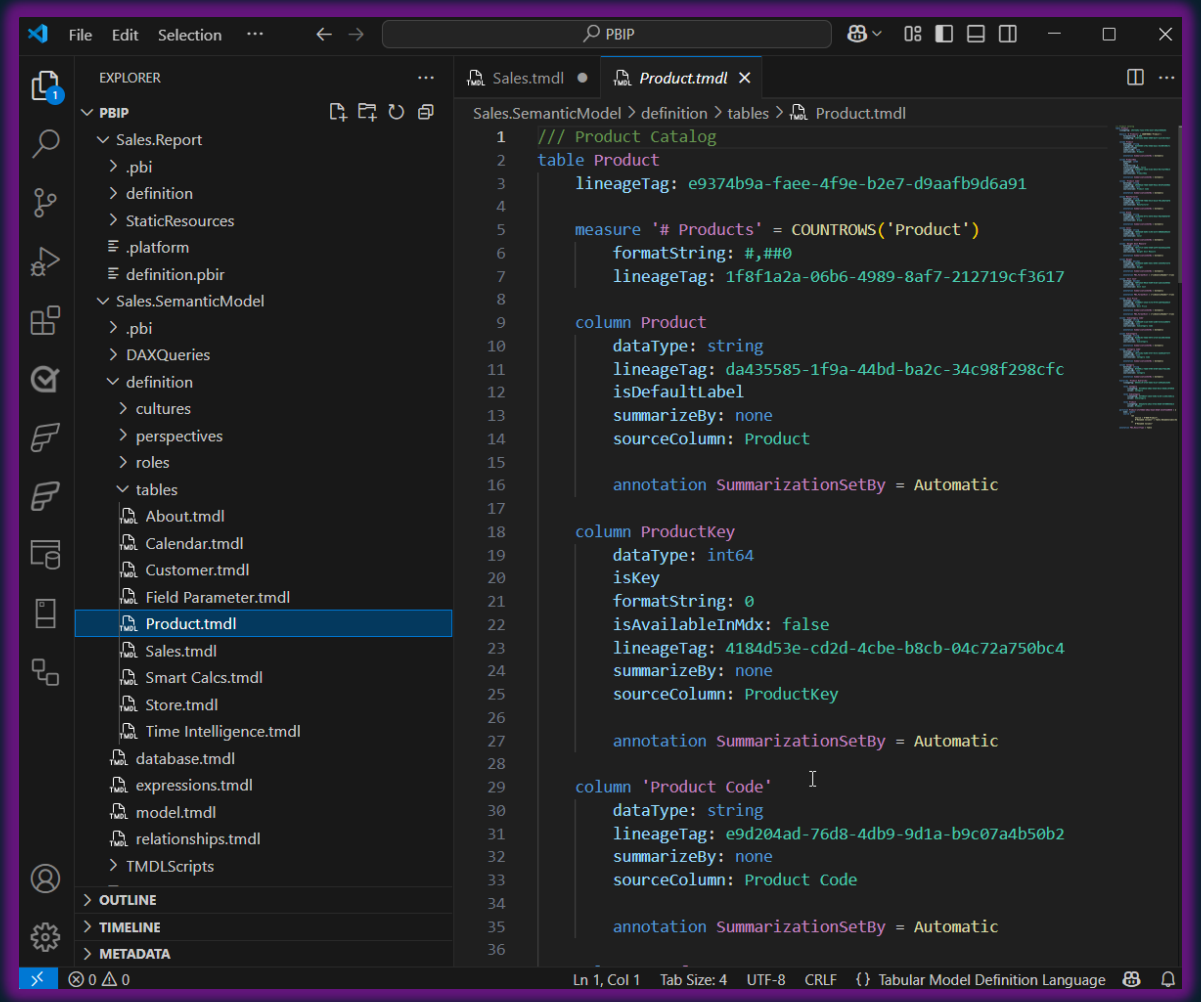
- OneDrive for co-development
  - Workaround with external tools
- No code experiences
- Low-code CI/CD



Power BI in 2025

- Co-development
- Code experiences + Gen AI
- CRUD APIs
- Pro code CI/CD

New Era for Power BI Developers



{ } Code

Open file formats.

⚡ Accelerate

Efficiency boost with AI.

🤝 Collaborate

Work together to achieve more.

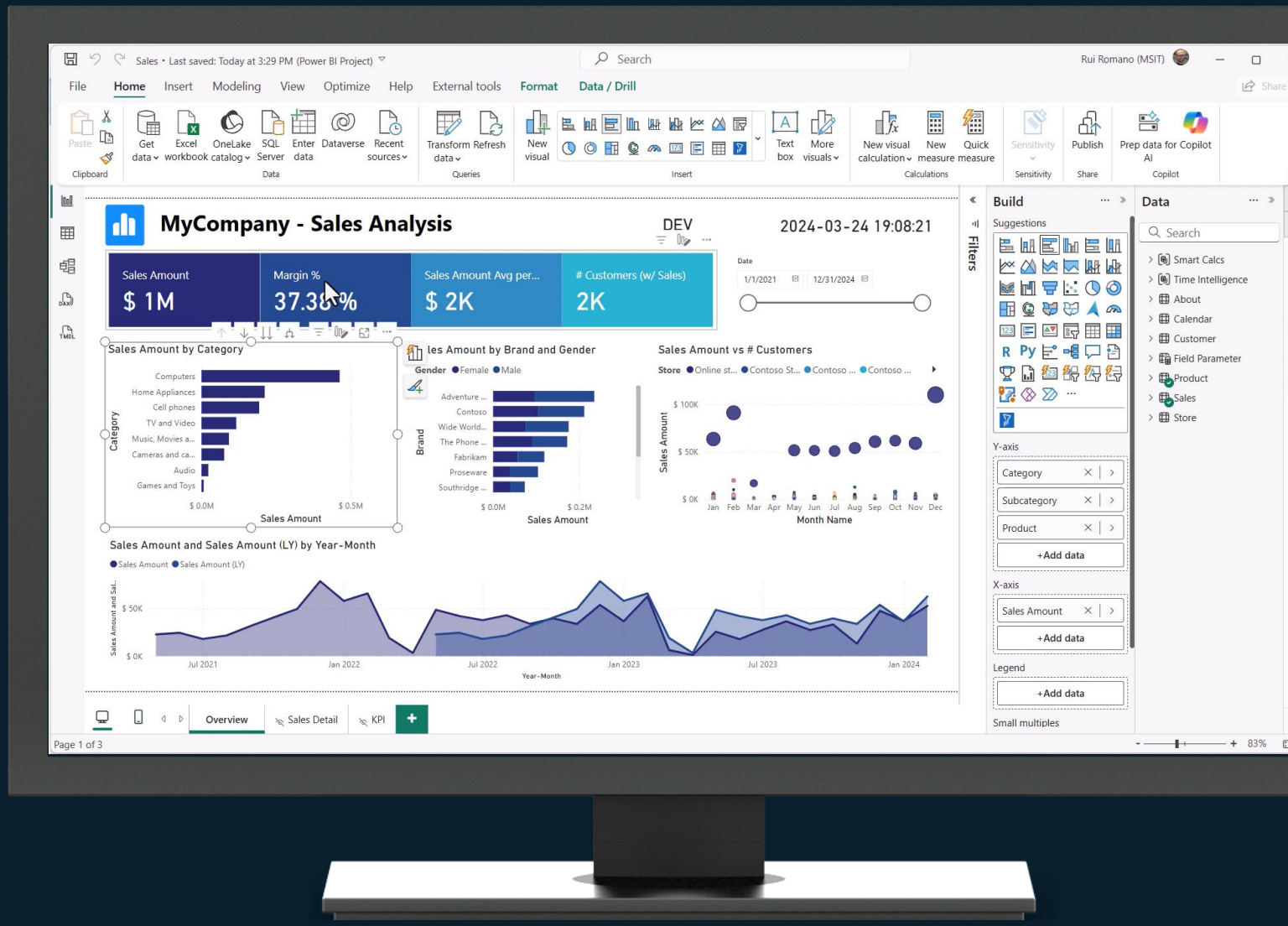
Developer mode / Power BI Project files (PBIP)

GA 2025

The code-behind Power BI

- Open code experience with TMDL & PBIR
- Unblock co-development
- Efficiency Boost
- CICD: test & deploy

It's a foundation for modular Power BI development with full control!



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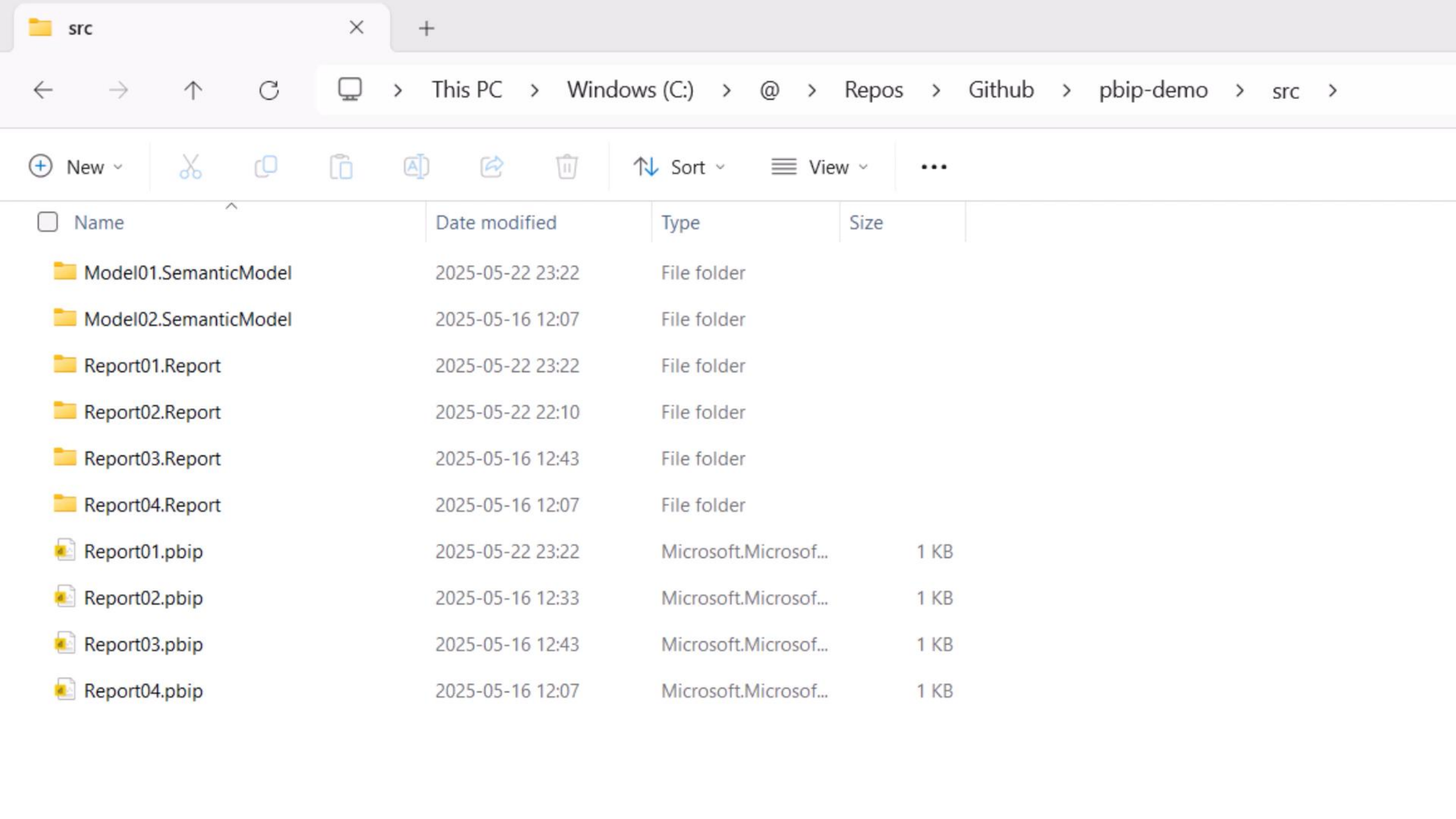
Demo

PBIP folder

Copy files around

Metadata publish with Fabric Studio


src
This PC
Windows (C:)
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Repos
Github
pbip-demo
src
New
Sort
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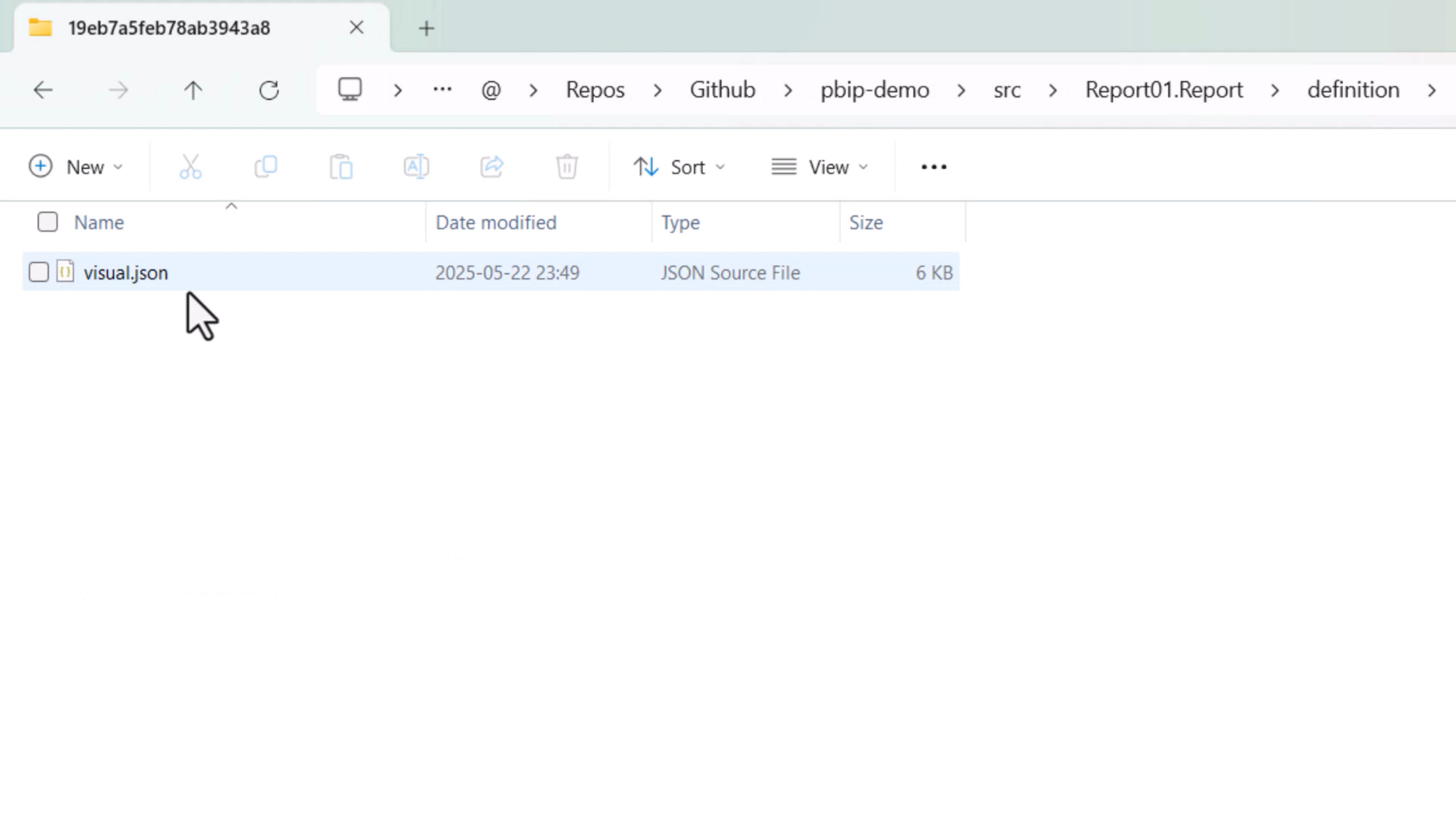
Model01.SemanticModel 2025-05-22 23:22 File folder
Model02.SemanticModel 2025-05-16 12:07 File folder
Report01.Report 2025-05-22 23:22 File folder
Report02.Report 2025-05-22 22:10 File folder
Report03.Report 2025-05-16 12:43 File folder
Report04.Report 2025-05-16 12:07 File folder
Report01.pbip 2025-05-22 23:22 Microsoft.Microsof... 1 KB
Report02.pbip 2025-05-16 12:33 Microsoft.Microsof... 1 KB
Report03.pbip 2025-05-16 12:43 Microsoft.Microsof... 1 KB
Report04.pbip 2025-05-16 12:07 Microsoft.Microsof... 1 KB


19eb7a5feb78ab3943a8
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JSON Source File
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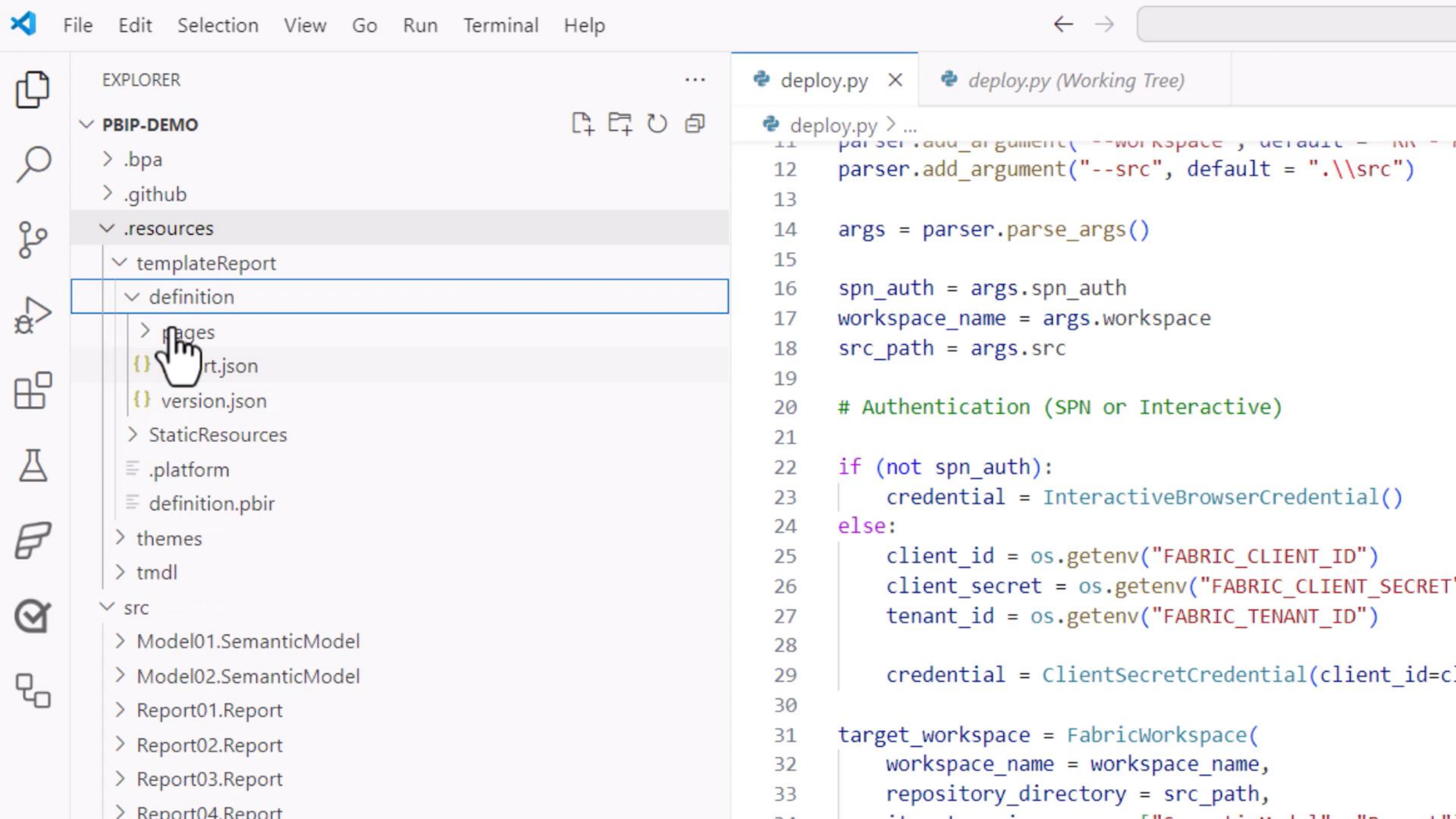
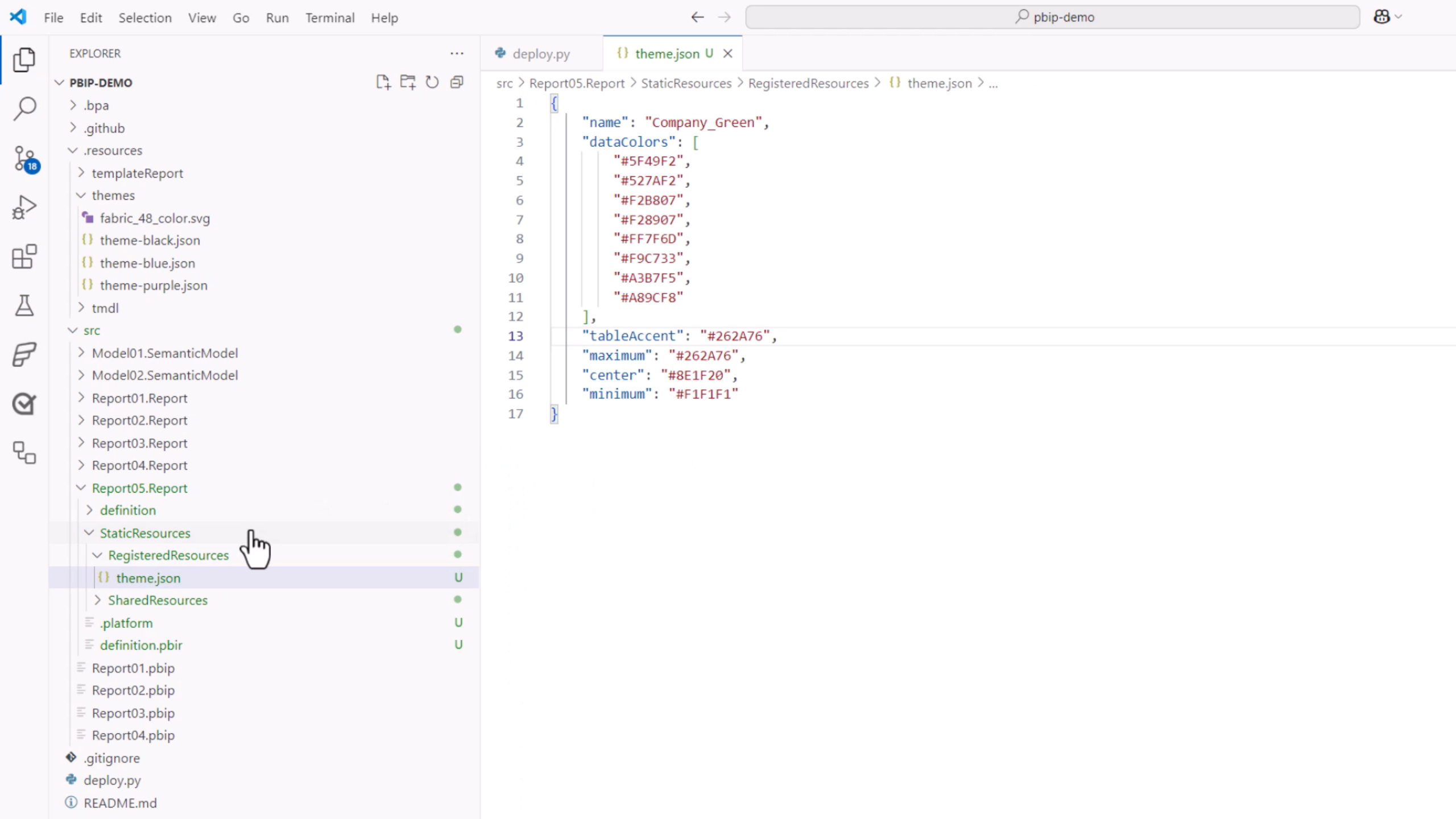

File Edit Selection View Go Run Terminal Help
EXPLORER
PBIP-DEMO
.bpa
.github
.resources
templateReport
definition
pages
rt.json
version.json
StaticResources
.platform
definition.pbir
themes
tmdl
src
Model01.SemanticModel
Model02.SemanticModel
Report01.Report
Report02.Report
Report03.Report
Report04.Report
deploy.py
deploy.py (Working Tree)
deploy.py > ...
parser.add_argument("--src", default = ".\\src")
args = parser.parse_args()
spn_auth = args.spn_auth
workspace_name = args.workspace
src_path = args.src
# Authentication (SPN or Interactive)
if (not spn_auth):
credential = InteractiveBrowserCredential()
else:
client_id = os.getenv("FABRIC_CLIENT_ID")
client_secret = os.getenv("FABRIC_CLIENT_SECRET
tenant_id = os.getenv("FABRIC_TENANT_ID")
credential = ClientSecretCredential(client_id=c
target_workspace = FabricWorkspace(
workspace_name = workspace_name,
repository_directory = src_path,


File Edit Selection View Go Run Terminal Help
pbip-demo
EXPLORER
PBIP-DEMO
.bpa
.github
.resources
templateReport
themes
fabric_48_color.svg
theme-black.json
theme-blue.json
theme-purple.json
tmdl
src
Model01.SemanticModel
Model02.SemanticModel
Report01.Report
Report02.Report
Report03.Report
Report04.Report
Report05.Report
definition
StaticResources
RegisteredResources
theme.json U
SharedResources
.platform U
definition.pbir U
Report01.pbip
Report02.pbip
Report03.pbip
Report04.pbip
.gitignore
deploy.py
README.md
deploy.py
theme.json U
src > Report05.Report > StaticResources > RegisteredResources > theme.json > ...
{
"name": "Company_Green",
"dataColors": [
"#5F49F2",
"#527AF2",
"#F2B807",
"#F28907",
"#FF7F6D",
"#F9C733",
"#A3B7F5",
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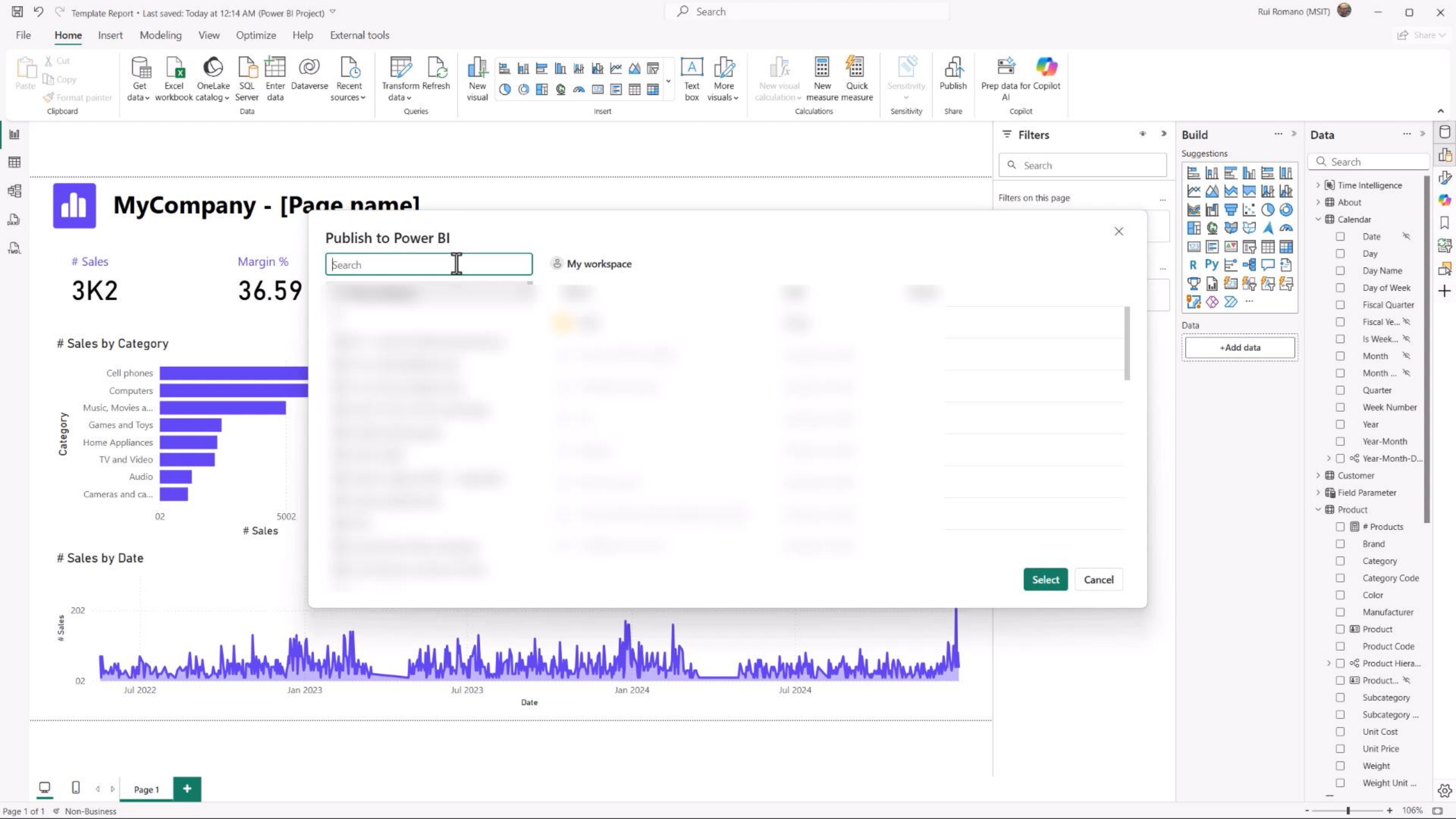
"minimum": "#F1F1F1"
}


Template Report • Last saved: Today at 12:14 AM (Power BI Project)
Search
Rui Romano (MSIT)
File
Home
Insert
Modeling
View
Optimize
Help
External tools
Share
Cut
Copy
Paste
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Sensitivity
Publish
Share
Prep data for Copilot AI
Copilot
MyCompany - [Page name]
Sales
3K2
Margin %
36.59
Sales by Category
Cell phones
Computers
Music, Movies a...
Games and Toys
Home Appliances
TV and Video
Audio
Cameras and ca...
Category
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5002
Sales
Sales by Date
202
02
Sales
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Date
Publish to Power BI
Search
My workspace
Select
Cancel
Filters
Search
Filters on this page
Build
Suggestions
Data
+Add data
Data
Search
Time Intelligence
About
Calendar
Date
Day
Day Name
Day of Week
Fiscal Quarter
Fiscal Ye...
Is Week...
Month
Month ...
Quarter
Week Number
Year
Year-Month
Year-Month-D...
Customer
Field Parameter
Product
Products
Brand
Category
Category Code
Color
Manufacturer
Product
Product Code
Product Hiera...
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Subcategory
Subcategory ...
Unit Cost
Unit Price
Weight
Weight Unit ...
Page 1
Page 1 of 1
Non-Business
106%

File Edit Selection View Go Run Terminal Help

Search

EXTENSIONS

Search Extensions in Marketplace

INSTALLED 25

Fabric Studio
Fabric Studio extension for VSCode
Gerhard Brueckl

Git Graph 87ms
View a Git Graph of your repository, and perform Git acti...
mhutchie

GitHub Actions
GitHub Actions workflows and runs for github.com hoste...
GitHub

GitHub Copilot 609ms
Your AI pair programmer
GitHub

GitHub Copilot Chat 540ms
AI chat features powered by Copilot
GitHub

Jupyter
Jupyter notebook support, interactive programming and ...
Microsoft

Jupyter Cell Tags
Jupyter Cell Tags support for VS Code
Microsoft

Jupyter Keymap
Jupyter keymaps for notebooks

RECOMMENDED 4

markdownlint 8.9M 4.5
Markdown linting and style checking for Visual Studio Co...
David Anson Install

Docker 45.3M 4.5
Makes it easy to create, manage, and debug containerize...
Microsoft Install

Git History 14.2M 4.5
View git log, file history, compare branches or commits
Don Jayamanne Install

Extension: Fabric Studio

Fabric Studio

Gerhard Brueckl gbrueckl.at | 5,078 | (1) | Sponsor

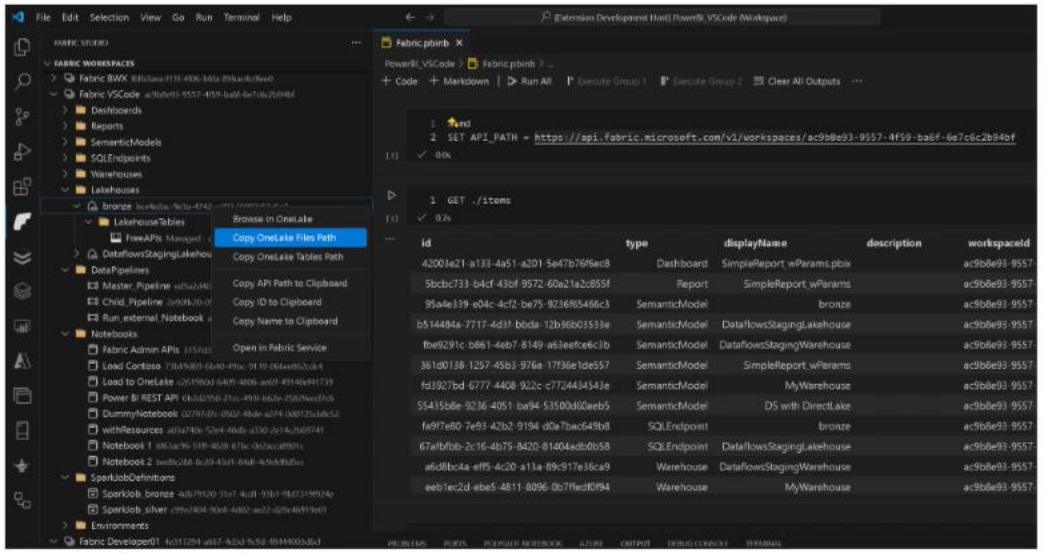
Fabric Studio extension for VSCode

Disable Uninstall Auto Update

DETAILS FEATURES CHANGELOG

Fabric Studio

version v1.14.0 installs 5.1k vs marketplace rate limited by upstream service rating 5.0/5 (1)



A VSCode extension for managing your Fabric tenant using the Fabric REST API and modify Fabric items directly from within VSCode.

Installation

The extensions can be installed directly from within VSCode by searching for this extension

Installation

Identifier	gerhardbrueckl.fabricstu
Version	1.14.0
Last Updated	2025-05-22, 10:55:27
Source	VSIX
Size	1.57MB

Marketplace

Published	2024-05-21, 10:02:25
Last Released	2025-05-24, 20:56:00

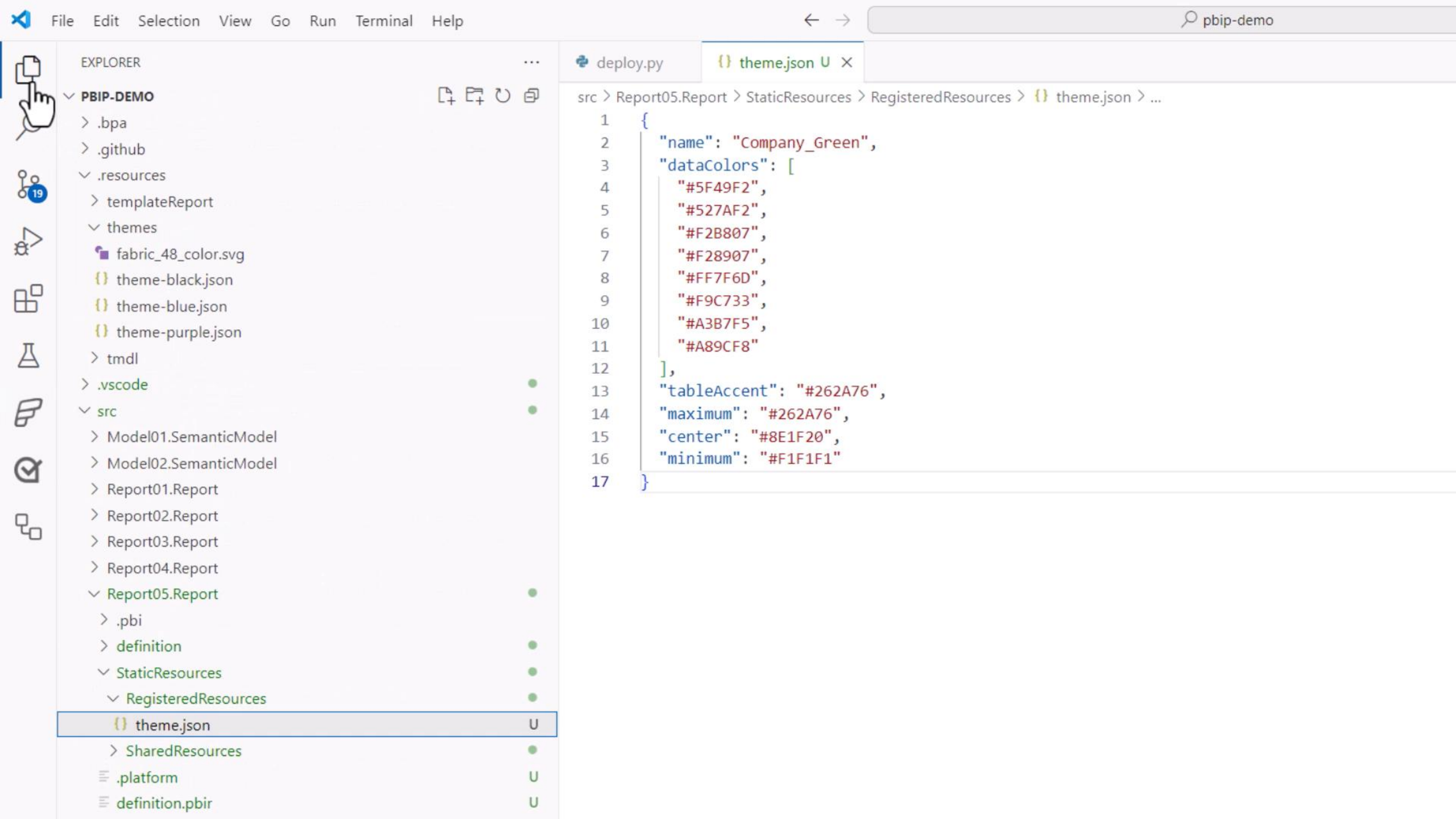
Categories

Azure Extension Packs Other

Resources

Marketplace
Issues
Repository
License
Gerhard Brueckl

⊗ 0 ⚠ 0


File Edit Selection View Go Run Terminal Help
pbip-demo
EXPLORER
PBIP-DEMO
.bpa
.github
.resources
templateReport
themes
fabric_48_color.svg
theme-black.json
theme-blue.json
theme-purple.json
tmdl
.vscode
src
Model01.SemanticModel
Model02.SemanticModel
Report01.Report
Report02.Report
Report03.Report
Report04.Report
Report05.Report
.pbi
definition
StaticResources
RegisteredResources
theme.json U
SharedResources
.platform U
definition.pbir U
deploy.py
theme.json U
src > Report05.Report > StaticResources > RegisteredResources > theme.json > ...
{
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"#527AF2",
"#F2B807",
"#F28907",
"#FF7F6D",
"#F9C733",
"#A3B7F5",
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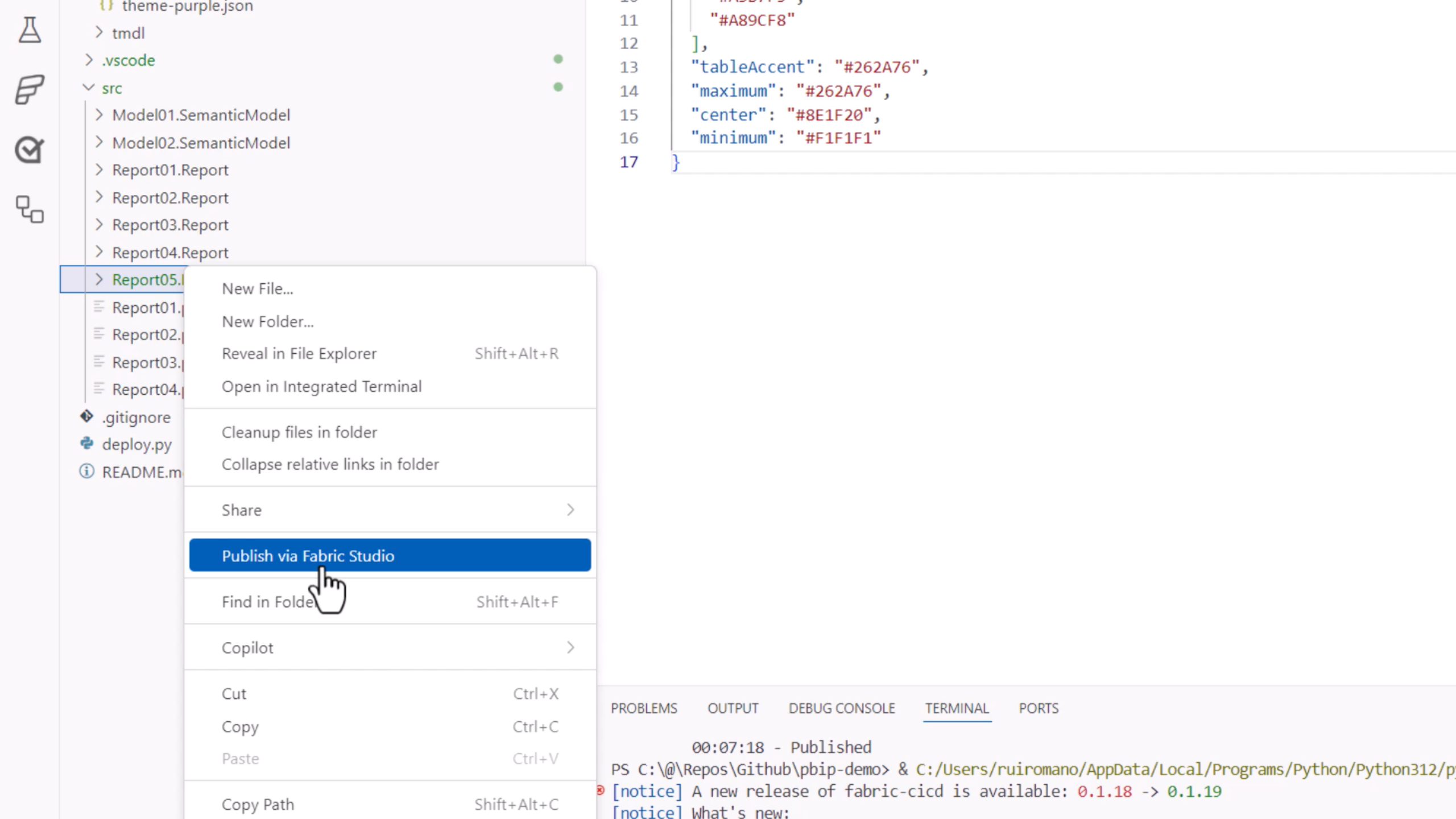
theme-purple.json
tmdl
.vscode
src
Model01.SemanticModel
Model02.SemanticModel
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Report03.Report
Report04.Report
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Report02.
Report03.
Report04.
.gitignore
deploy.py
README.m
"#A89CF8"
],
"tableAccent": "#262A76",
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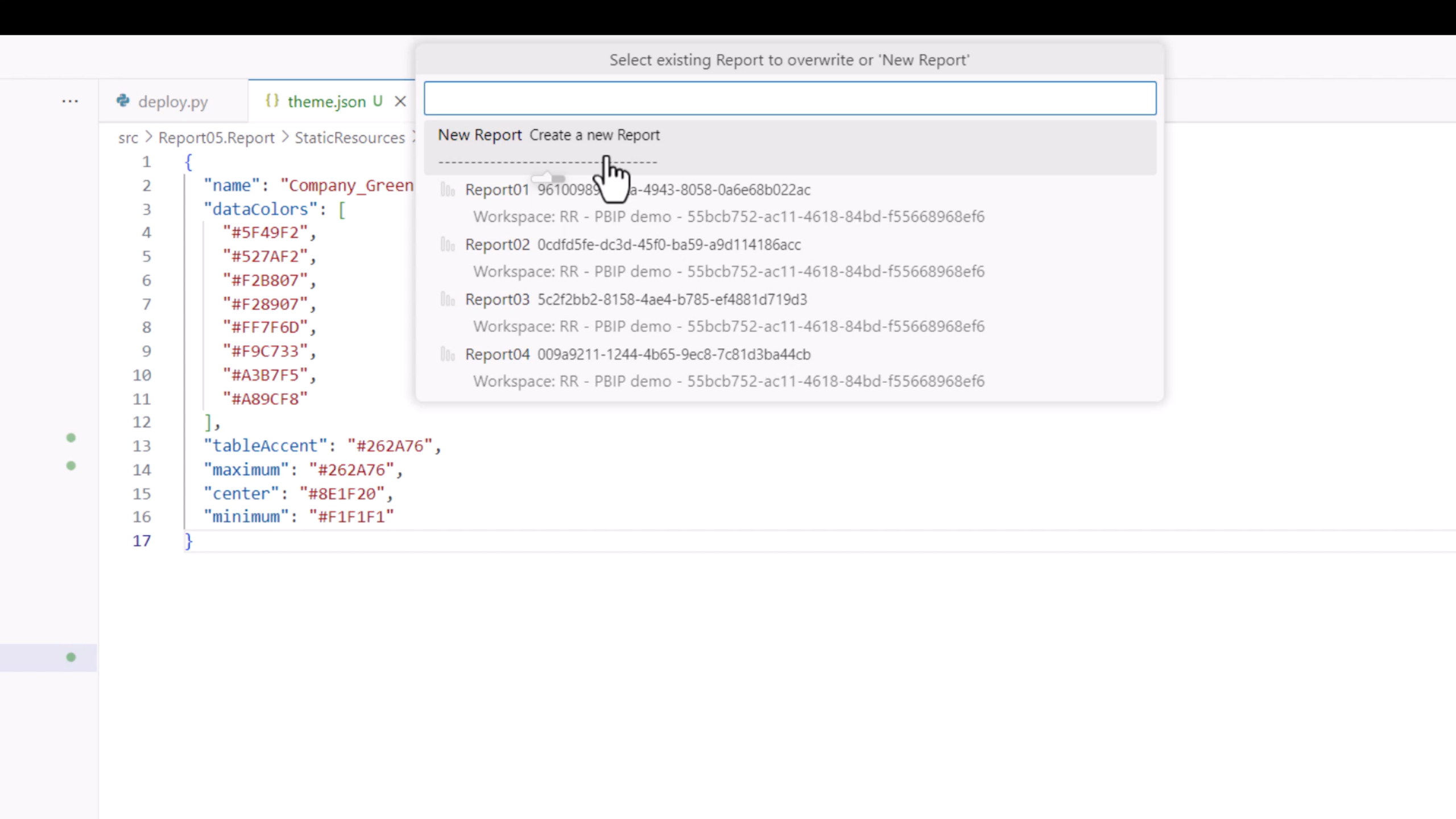
"center": "#8E1F20",
"minimum": "#F1F1F1"
}
New File...
New Folder...
Reveal in File Explorer Shift+Alt+R
Open in Integrated Terminal
Cleanup files in folder
Collapse relative links in folder
Share
Publish via Fabric Studio
Find in Folde Shift+Alt+F
Copilot
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Copy Path Shift+Alt+C
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
00:07:18 - Published
PS C:\@\Repos\Github\pbip-demo> & C:/Users/ruiromano/AppData/Local/Programs/Python/Python312/p
[notice] A new release of fabric-cicd is available: 0.1.18 -> 0.1.19
[notice] What's new:


Select existing Report to overwrite or 'New Report'
New Report Create a new Report
Report01 96100989 a-4943-8058-0a6e68b022ac
Workspace: RR - PBIP demo - 55bcb752-ac11-4618-84bd-f55668968ef6
Report02 0cdfd5fe-dc3d-45f0-ba59-a9d114186acc
Workspace: RR - PBIP demo - 55bcb752-ac11-4618-84bd-f55668968ef6
Report03 5c2f2bb2-8158-4ae4-b785-ef4881d719d3
Workspace: RR - PBIP demo - 55bcb752-ac11-4618-84bd-f55668968ef6
Report04 009a9211-1244-4b65-9ec8-7c81d3ba44cb
Workspace: RR - PBIP demo - 55bcb752-ac11-4618-84bd-f55668968ef6
deploy.py
theme.json U
src > Report05.Report > StaticResources
{
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"dataColors": [
"#5F49F2",
"#527AF2",
"#F2B807",
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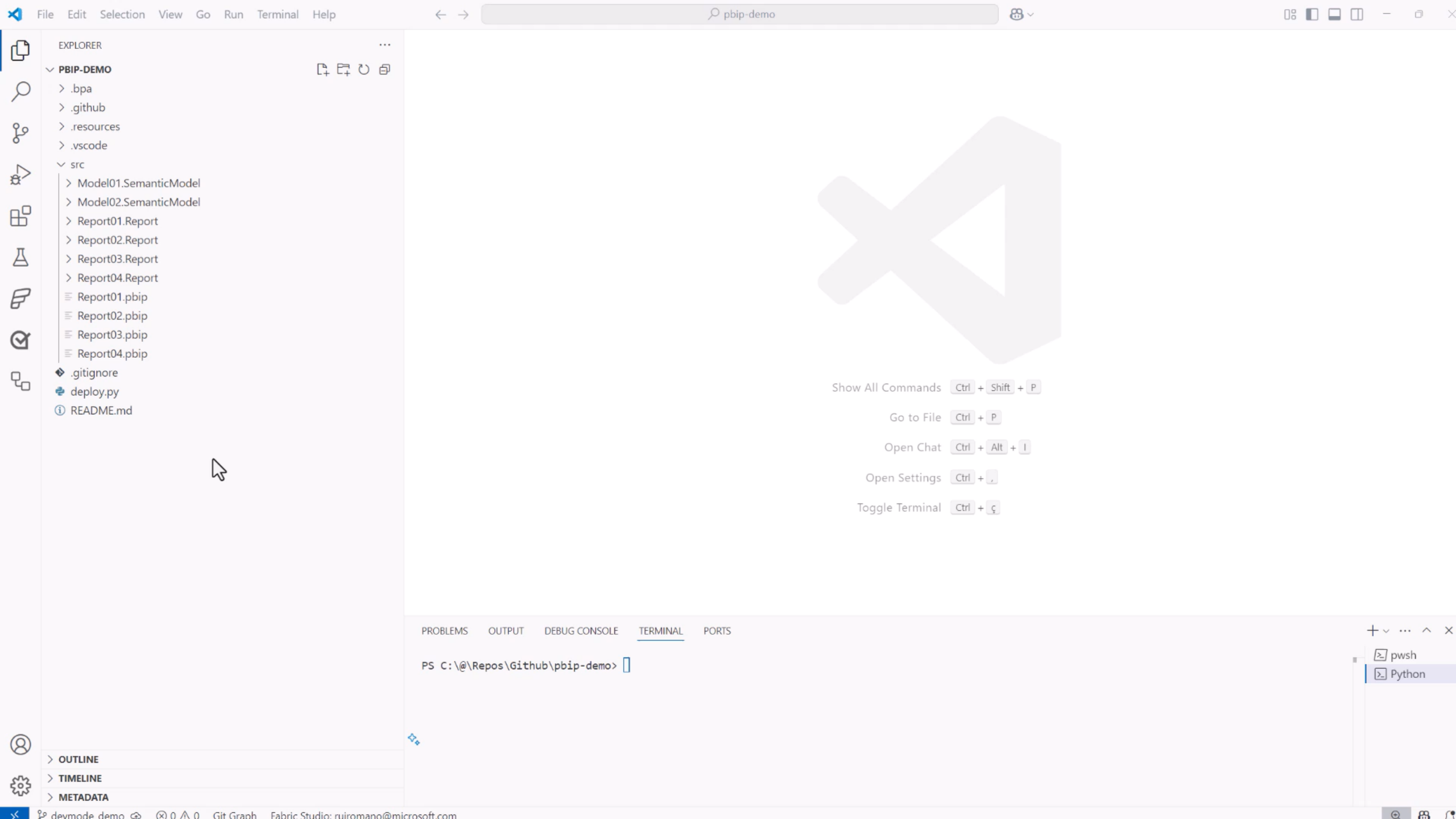
"#A3B7F5",
"#A89CF8"
],
"tableAccent": "#262A76",
"maximum": "#262A76",
"center": "#8E1F20",
"minimum": "#F1F1F1"
}

Demo

GitHub Copilot for project docs


File Edit Selection View Go Run Terminal Help
pbip-demo
EXPLORER
PBIP-DEMO
.bpa
.github
.resources
.vscode
src
Model01.SemanticModel
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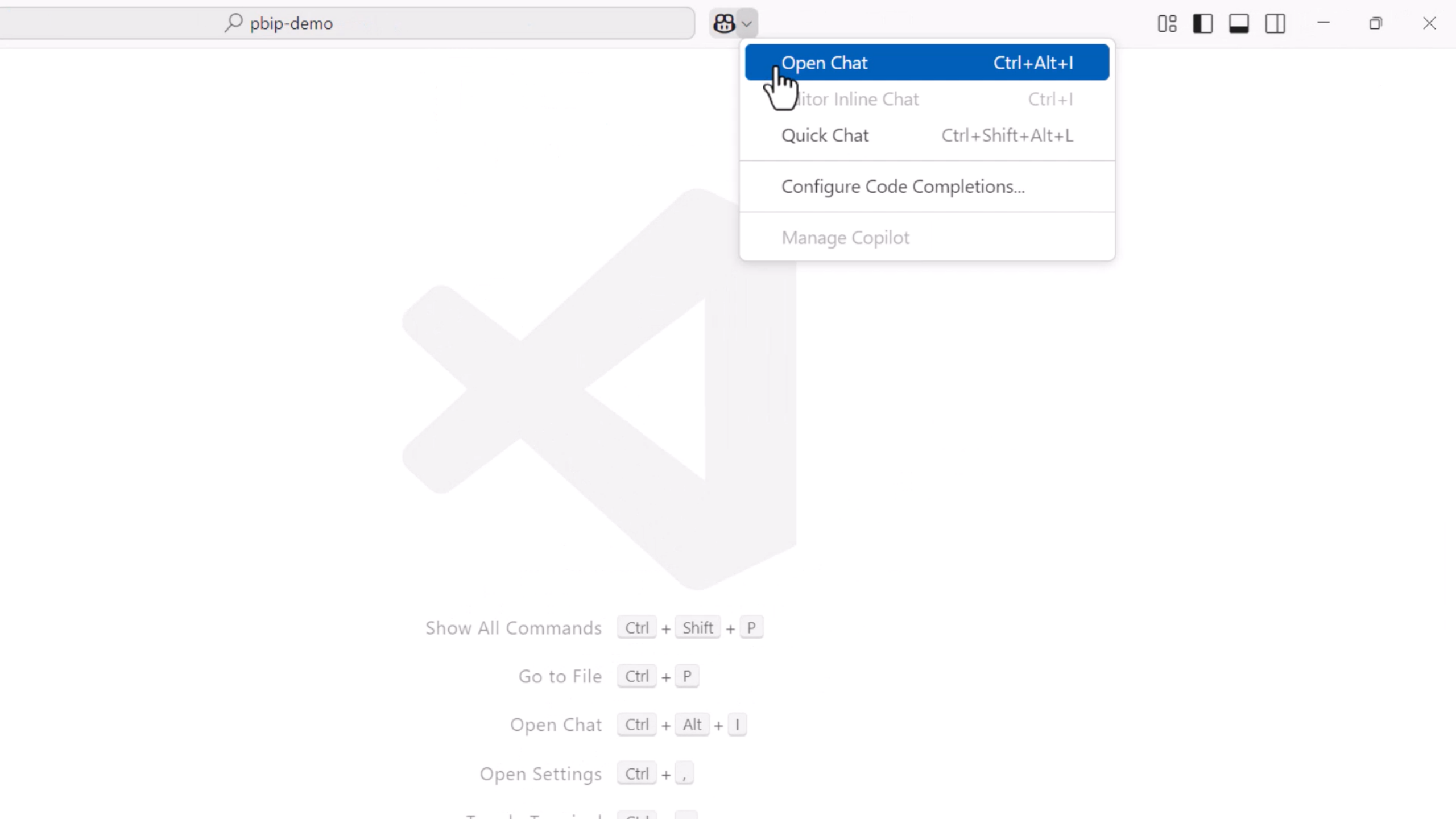
Report02.Report
Report03.Report
Report04.Report
Report01.pbip
Report02.pbip
Report03.pbip
Report04.pbip
.gitignore
deploy.py
README.md
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Open Chat Ctrl + Alt + I
Open Settings Ctrl + ,
Toggle Terminal Ctrl + ç
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\@\Repos\Github\pbip-demo>
pwsh
Python
OUTLINE
TIMELINE
METADATA
devmode_demo
Git Graph
Fabric Studio: ruiromano@microsoft.com


pbip-demo
Open Chat Ctrl+Alt+I
itor Inline Chat Ctrl+I
Quick Chat Ctrl+Shift+Alt+L
Configure Code Completions...
Manage Copilot
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Open Chat Ctrl + Alt + I
Open Settings Ctrl + ,

or type # to attach context

pwsh

Python

Ask

Edit

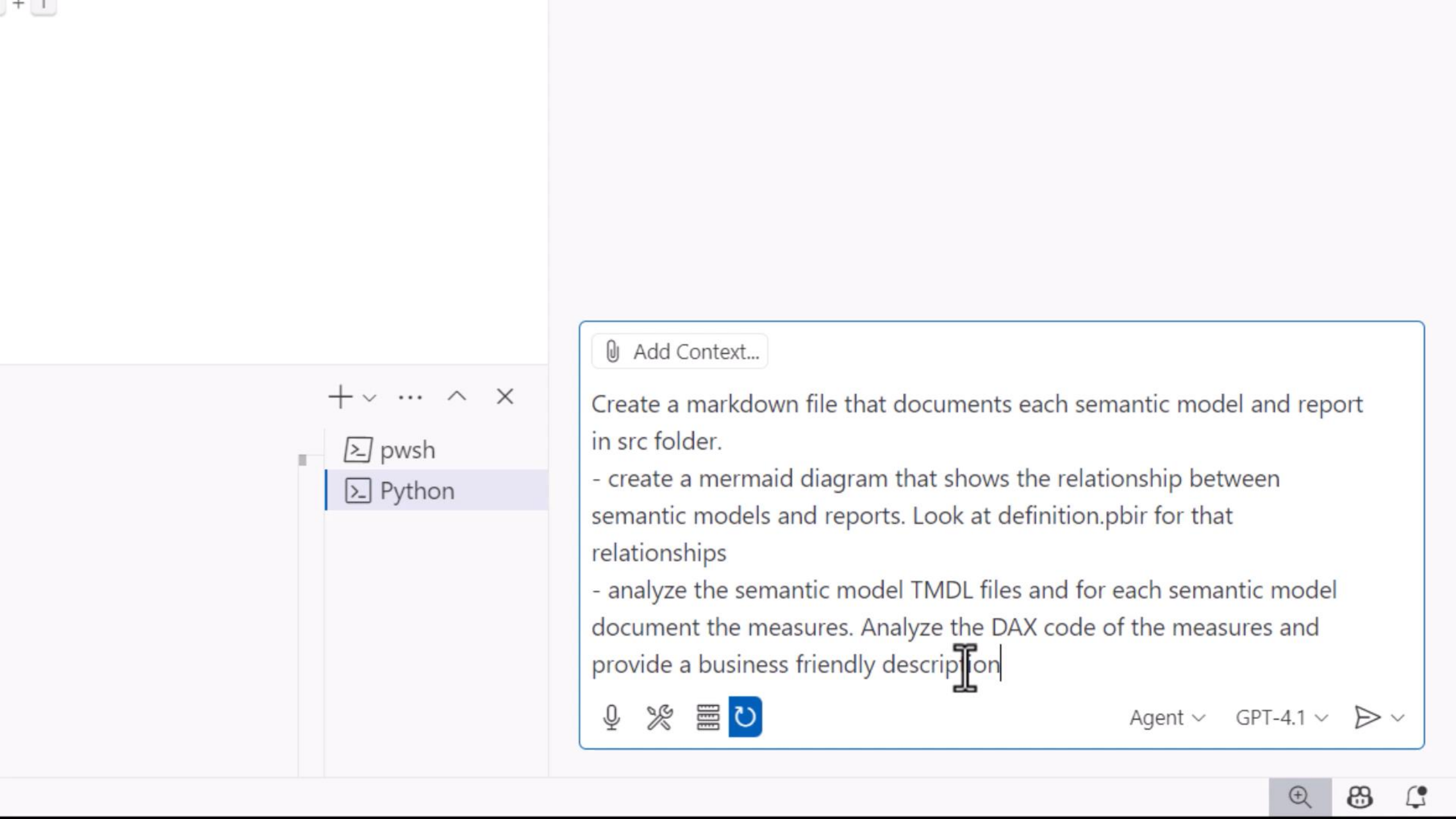
Agent

Add Context...

Edit files in your workspace in agent mode

Agent

T-4.1


Add Context...
Create a markdown file that documents each semantic model and report in src folder.
- create a mermaid diagram that shows the relationship between semantic models and reports. Look at definition.pbir for that relationships
- analyze the semantic model TMDL files and for each semantic model document the measures. Analyze the DAX code of the measures and provide a business friendly description
Agent
GPT-4.1
pwsh
Python

Demo

Batch PBIR edit with GitHub Copilot

Report01 • Last saved: Today at 10:11 PM (Power BI Project)

Search

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File Home Insert Modeling View Optimize Help External tools

Share

Paste Cut Copy Format painter — Clipboard

Get data Excel workbook OneLake catalog SQL Server Enter data Dataverse Recent sources — Data

Transform data Refresh — Queries

New visual Text box More visuals — Insert

New visual calculation New measure Quick measure — Calculations

Sensitivity — Sensitivity

Publish — Share

Prep data for AI — Copilot

MyCompany - Report 01

DEV

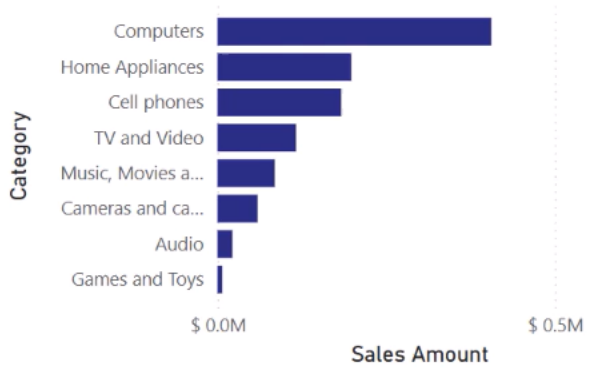
2024-03-24 19:08:21

Sales Amount	Margin %	Sales Amount Avg per...	# Customers (w/ Sales)
$ 1M	36.59 %	$ 2K	2K

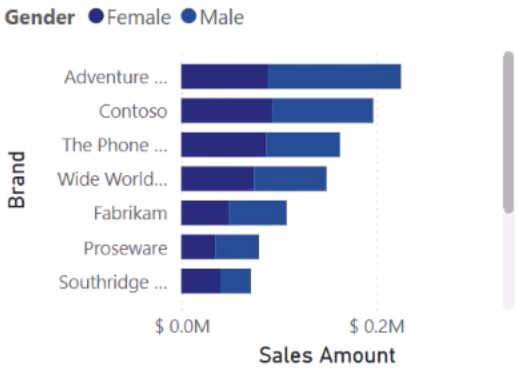
Date

1/1/2021 — 12/31/2025

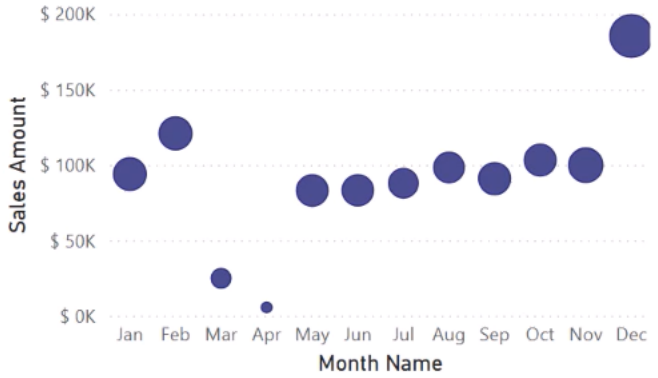
Sales Amount by Category



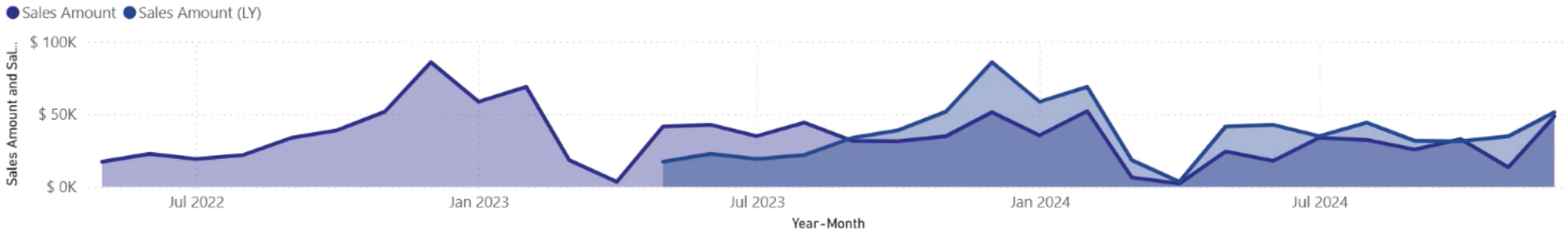
Sales Amount by Brand and Gender



Sales Amount vs # Customers



Sales Amount and Sales Amount (LY) by Year-Month



Filters

Build

Suggestions

Data

+Add data

Data

Search

Time Intelligence
About
Calendar
Customer
Field Parameter
Product
Sales
Store

Page 1

Page 1 of 1

123%

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Themes | Page view | Scale to fit | Mobile layout | Mobile | Gridlines | Snap to grid | Lock objects | Filters | Pane manager | Data | Bookmarks | Sync slicers | Build | Selection | Page visuals | Format | Performance analyzer | Show panes

MyCompany - Report 01

DEV

2024-03-24 19:08:21

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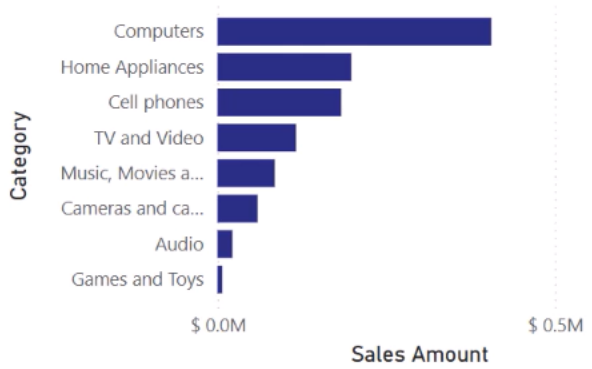
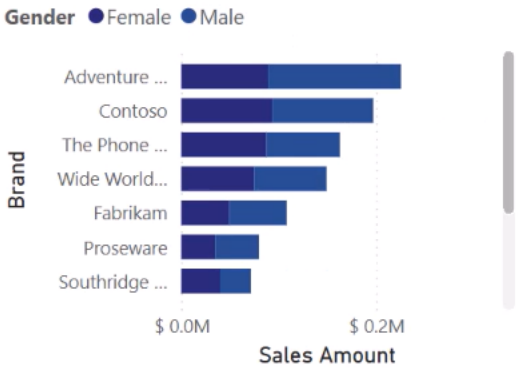
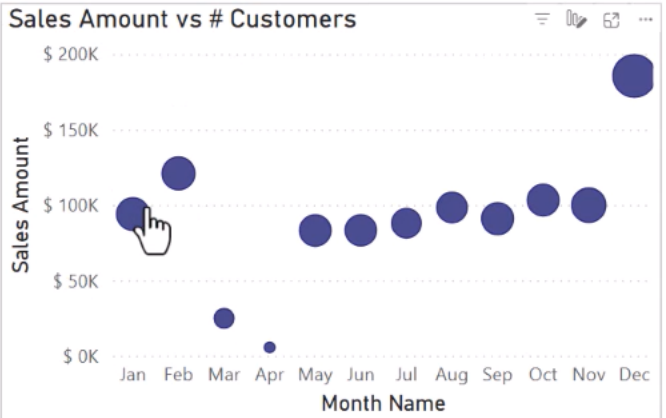
Sales Amount by Category



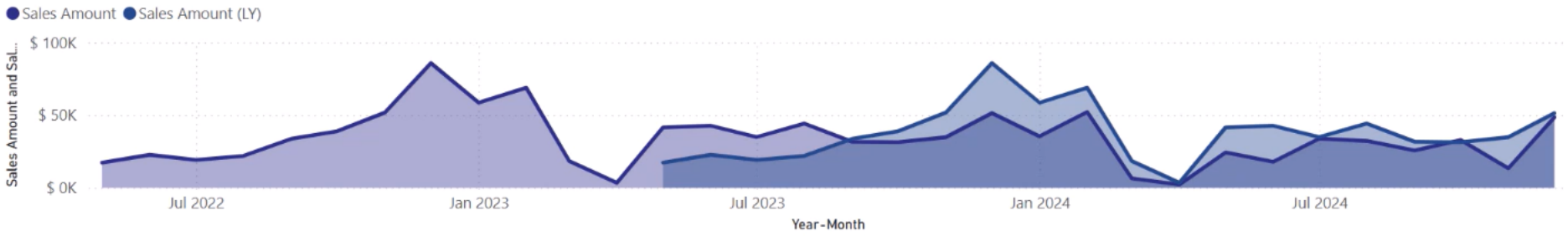
Sales Amount by Brand and Gender



Sales Amount vs # Customers



Sales Amount and Sales Amount (LY) by Year-Month



Page 1

Build

Suggestions

Data

+Add data

Data

Search

Time Intelligence
About
Calendar
Customer
Field Parameter
Product
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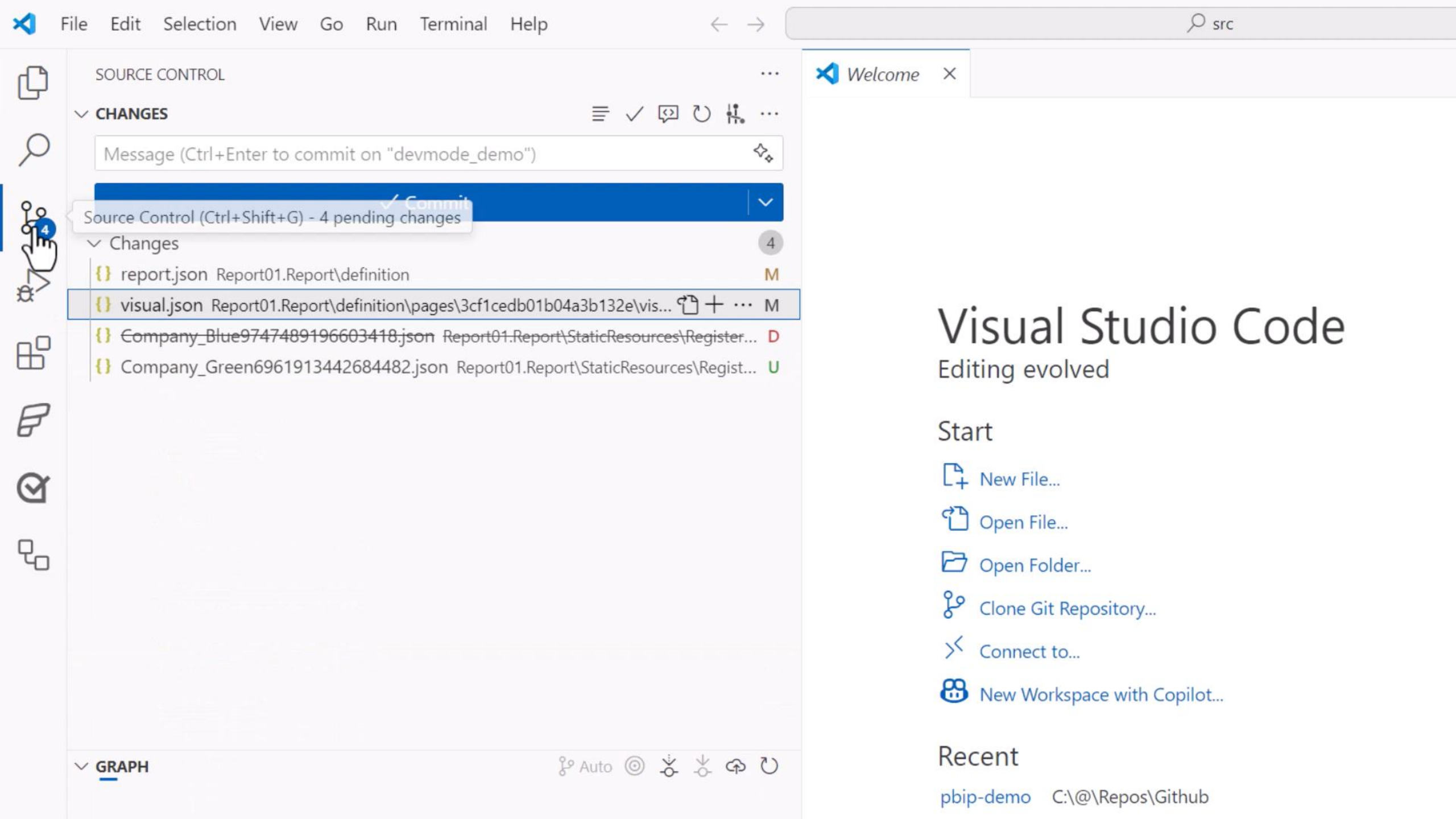
Store

Filters

Page 1 of 1

123%

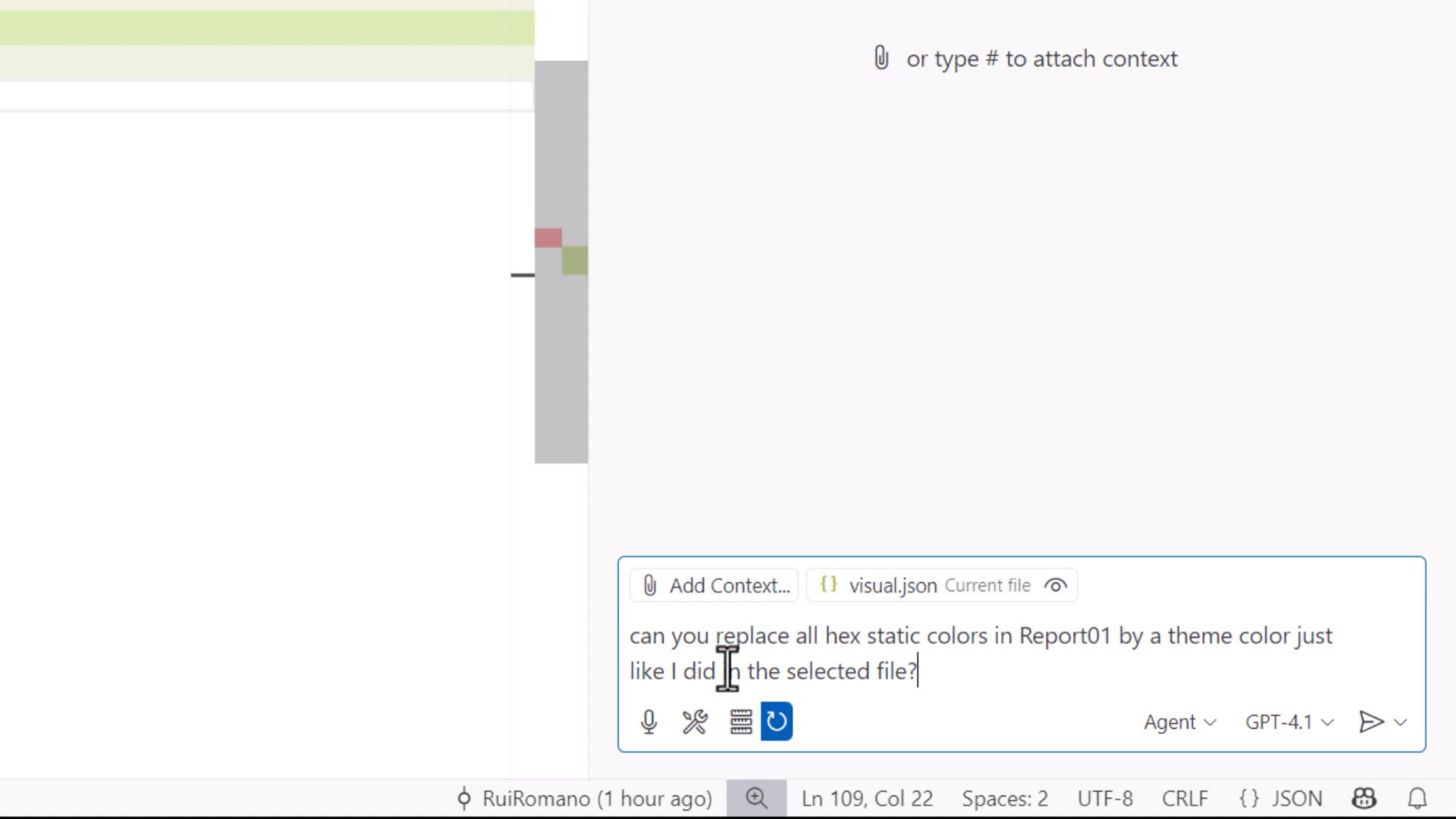

File
Edit
Selection
View
Go
Run
Terminal
Help
src
SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "devmode_demo")
Commit
Source Control (Ctrl+Shift+G) - 4 pending changes
Changes
4
report.json Report01.Report\definition M
visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b132e\vis... M
Company_Blue9747489196603418.json Report01.Report\StaticResources\Register... D
Company_Green6961913442684482.json Report01.Report\StaticResources\Regist... U
GRAPH
Auto
Welcome
Visual Studio Code
Editing evolved
Start
New File...
Open File...
Open Folder...
Clone Git Repository...
Connect to...
New Workspace with Copilot...
Recent
pbip-demo C:\@\Repos\Github

src

visual.json (Working Tree) M

1.Report > definition > pages > 3cf1cedb01b04a3b132e > visuals > 5868707ac858bcbe007a > visual.json > { } visual > { } objects > [ ] dataPoint > { } 0 > { } properties > { } fill > { } solid > { } color > { } expr >

```
87  12    "visual": {
88  14      "query": {
89  80        "sortDefinition": {
90  81          "sort": [
91  82            {
92  92            },
93  93            "direction": "Descending"
94  94          }
95  95        ]
96  96      }
97  97    },
98  98    "objects": {
99  99      "dataPoint": [
100 100       {
101 101         "properties": {
102 102           "fill": {
103 103             "solid": {
104 104               "color": {
105 105                 "expr": {
106 -                   "Literal": {
107 -                     "Value": "'#282D83'"
    106+                "ThemeDataColor": {
    107+                  "ColorId": 2,
    108+                  "Percent": 0
108 109                 }
109 110               }
110 111             }
111 112           }
112 113         }
113 114       }
114 115     }
115 116   ]
```


or type # to attach context
Add Context...
visual.json Current file
can you replace all hex static colors in Report01 by a theme color just like I did in the selected file?
Agent
GPT-4.1
RuiRomano (1 hour ago)
Ln 109, Col 22
Spaces: 2
UTF-8
CRLF
{} JSON

File Edit Selection View Go Run Terminal Help

src

SOURCE CONTROL

CHANGES

Message (Ctrl+Enter to commit on "devmode_demo")

Commit

Changes 8

- report.json Report01.Report\definition M
- visual.json Report01.Report\definition\pages\3cf1ced... M
- visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b... M
- visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b... M
- visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b... M
- visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b... M
- ~~Company_Blue9747489196603418.json~~ ~~Report01.Report\Stati...~~ D
- Company_Green6961913442684482.json Report01.Report\Sta... U

GRAPH

Auto

- init RuiRomano — devmode_demo
- fff RuiRomano
- logo resource RuiRomano — origin/main
- readme RuiRomano
- folder refactoring RuiRomano
- docs RuiRomano
- bpa docs RuiRomano
- readme RuiRomano
- src path RuiRomano
- readme RuiRomano
- resources RuiRomano
- dd RuiRomano
- fix bpa paths RuiRomano
- invali paths RuiRomano
- badge RuiRomano
- bpa, reports RuiRomano
- readme RuiRomano

4d6ecd065 M | visual.json ...\19eb7a5feb78ab3943a8 M | visual.json ...\611011a7ca2045d99da7 M

3943a8 > {} visual.json > {} visual > {} objects > [ ] dataPoint > {} 0 > {} properties > {} fill > {} solid > {} color > {} expr

```
12    "visual": {
101     "objects": {
102       "dataPoint": [
103         {
104           "properties": {
111             "Percent": 0
112           }
113         }
114       }
115     }
116   },
117   "selector": {
118     "data": [
119       {
120         "scopeId": {
121           "Comparison": {
122             "ComparisonKind": 0,
123             "Left": {
124               "Column": {
125                 "Expression": {
126                   "SourceRef": {
127                     "Entity": "Customer"
128                   }
129                 },
130                 "Property": "Gender"
131               }
132             },
133             "Right": {
134               "Literal": {
135                 "Value": "'Male'"
136               }
137             }
138           }
139         }
140       }
141     ]
142   }
143 },
144 {
145   "properties": {
146     "fill": {
147       "solid": {
148         "color": {
149
```

CHAT

- Read {} visual.json, lines 1 to 101
- Read {} visual.json, lines 1 to 161
- Read {} visual.json, lines 1 to 121
- Read {} visual.json, lines 1 to 181
- Read {} visual.json, lines 1 to 41
- Read {} visual.json, lines 1 to 41
- Read {} visual.json, lines 101 to 108
- Read {} visual.json, lines 161 to 208
- Read {} visual.json, lines 121 to 127

{} visual.json

{} visual.json

{} visual.json

{} visual.json +6 -4

Checked {} visual.json, {} visual.json, {} visual.json, {} visual.json, no problems found

All static hex colors in the visuals of Report01 have been replaced with theme colors using the ThemeDataColor format, just like in your selected file. All changes are valid and error-free.

If you need this applied to more files or want to review the specific changes, let me know!

4 files changed

- {} visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b132e\visual...
- {} visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b132e\visual...
- {} visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b132e\visual...
- {} visual.json Report01.Report\definition\pages\3cf1cedb01b04a3b132e\visual...

Add Context... {} visual.json Current file

Edit files in your workspace in agent mode

Agent GPT-4.1

devmode_demo* 0 0 Git Graph

Ln 109, Col 1 Spaces: 2 UTF-8 CRLF {} JSON

MyCompany - Report 01

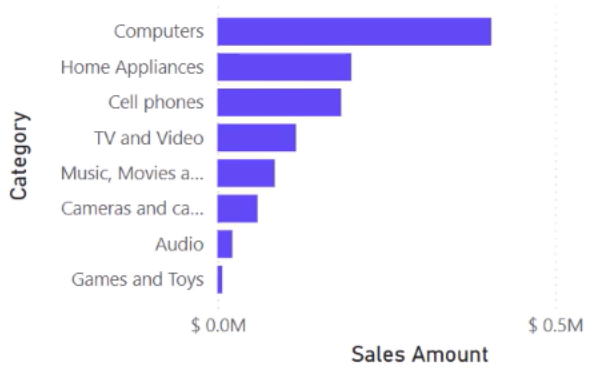
DEV

2024-03-24 19:08:21

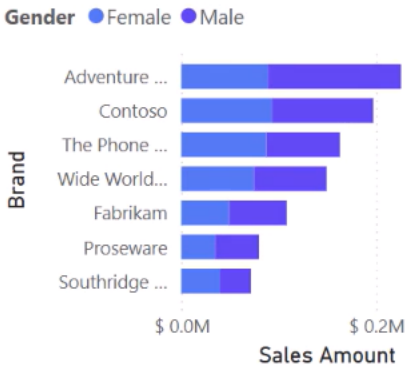
Sales Amount	Margin %	Sales Amount Avg per...	# Customers (w/ Sales)
$ 1M	36.59 %	$ 2K	2K

Date
1/1/2021
12/31/2025

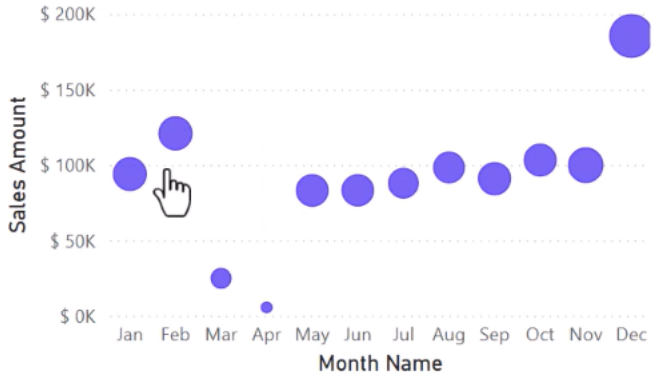
Sales Amount by Category



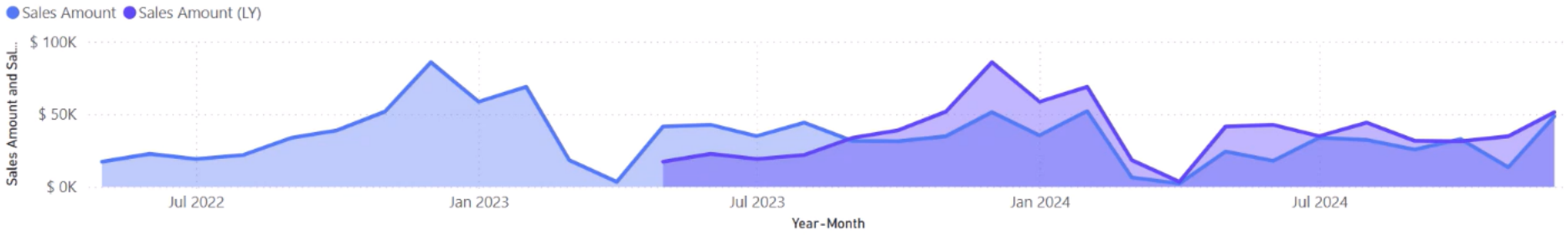
Sales Amount by Brand and Gender



Sales Amount vs # Customers



Sales Amount and Sales Amount (LY) by Year-Month



Page 1

Filters

Build
Suggestions
Data
+Add data

Data
Search
Time Intelligence
About
Calendar
Customer
Field Parameter
Product
Sales
Store

🧠 Recap

- PBIP as code-behind of your development
  - Not only for source control
- Templatize developments
- Leverage APIs and/or Fabric Studio for metadata publishing
- Leverage Gen AI to accelerate developments
- Explore Fabric Studio by Gerhard Brueckl
  - https://github.com/gbrueckl/FabricStudio
- Learn with a sample: https://aka.ms/pbip-sample

Classified as Microsoft Confidential

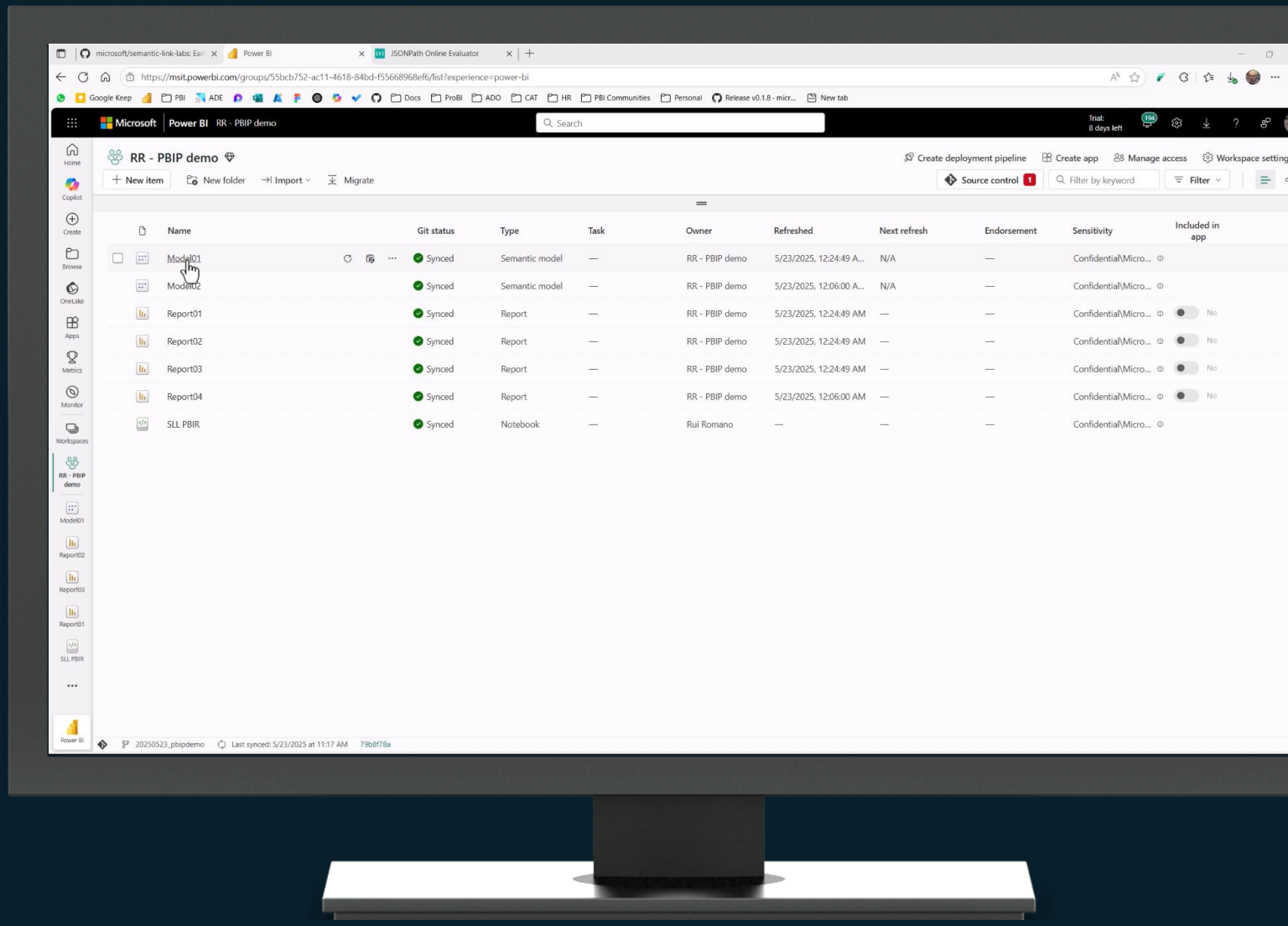
Automation in Service with Semantic Link Labs

Available Now!

Make challenging developer/admin tasks incredibly simple.

Many more scenarios:

- Bulk update published items
- Analyze your model memory
- Best practice analysis
- Troubleshooting
- Run queries and commands
- …



https://github.com/microsoft/semantic-link-labs

Classified as Microsoft Confidential

Demo

Report authoring automation in Service

Semantic Link Labs

Read the documentation on ReadTheDocs!

Read the Wiki!

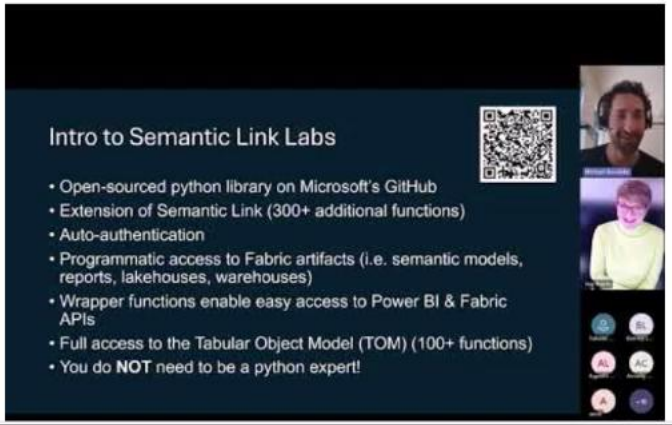
See code examples!

Semantic Link Labs is a Python library designed for use in Microsoft Fabric notebooks. This library extends the capabilities of Semantic Link offering additional functionalities to seamlessly integrate and work alongside it. The goal of Semantic Link Labs is to simplify technical processes, empowering people to focus on higher level activities and allowing tasks that are better suited for machines to be efficiently handled without human intervention.

If you encounter any issues, please raise a bug.

If you have ideas for new features/functions, please request a feature.

Check out the video below for an introduction to Semantic Link, Semantic Link Labs and demos of key features!



No packages published

Contributors 17

+ 3 contributors

Deployments 91

github-pages 6 months ago

+ 90 deployments

Languages

```
%pip install semantic-link-labs
```

PySpark (Python)

```
from sempy_labs.report import connect_report

# Replace non theme colors with built in function

with connect_report(report='Report01', readonly=True) as rpt:

    rpt.update_to_theme_colors(mapping = {
        '#000CFA': 2,
        '#071A40': 3,
        '#247452': 2
        })

    # Many others:

    # rpt.list_bookmarks
    # rpt.list_pages
    # rpt.add_new_visual
    # rpt.add_image

```

- Command executed in 4 sec 612 ms by Rui Romano on 10:26:46 AM, 5/23/25

PySpark (Python)

```
from sempy_labs.report import connect_report

# Easily query, set and remove JSON from PBIR using https://jsonpath.com/

with connect_report(report='Report01', readonly = True) as rpt:

    output = rpt.get("**/visual.json", json_path = "$..color.expr.Literal.Value")

    # set

    #  rpt.set_json(file_path = "**/visual.json"
    #             , json_path = "$..color.expr"
    #             , json_value = {
    #                 "color": {
    #                     "expr": {
    #                         "ThemeDataColor": {
    #                         "ColorId": 0,
```

```
%pip install semantic-link-labs
```

PySpark (Python)

```
from sempy_labs.report import connect_report

# Replace non theme colors with built in function

with connect_report(report='Report01', readonly=True) as rpt:

    rpt.update_to_theme_colors(mapping = {
        '#000CFA': 2,
        '#071A40': 3,
        '#247452': 2
        })

    # Many others:

    # rpt.list_bookmarks
    # rpt.list_pages
    # rpt.add_new_visual
    # rpt.add_image

```

[21] ✓ 4 sec - Command executed in 4 sec 666 ms by Rui Romano on 10:36:38 AM, 5/23/25

PySpark (Python)

Log

```
~ definition/pages/3cf1cedb01b04a3b132e/visuals/19eb7a5feb78ab3943a8/visual.json
  ~ visual.objects
  ~ dataPoint:
    @@ -1,4 +1,4 @@
    -  {"properties": {"fill": {"solid": {"color": {"expr": {"Literal": {"Value": "'#000CFA'"}}}}}}, "selector": {"data": [{"scopeId": {"Comparison": {"ComparisonKind": 0, "Left": {"Column": {"Expression": {"SourceRef": {"Entity": "Customer"}}, "Property": "Gender"}}, "Right": {"Literal": {"Value": "'Female'"}}}}}]}}
    +  {"properties": {"fill": {"solid": {"color": {"expr": {"ThemeDataColor": {"ColorId": 2, "Percent": 0}}}}}}, "selector": {"data": [{"scopeId": {"Comparison": {"ComparisonKind": 0, "Left": {"Column": {"Expression": {"SourceRef": {"Entity": "Customer"}}, "Property": "Gender"}}, "Right": {"Literal": {"Value": "'Female'"}}}}}]}}
~ definition/pages/3cf1cedb01b04a3b132e/visuals/5868707ac858bcbe007a/visual.json
  ~ visual.objects
  ~ dataPoint:
    @@ -1,3 +1,3 @@
    -  {"properties": {"fill": {"solid": {"color": {"expr": {"Literal": {"Value": "'#000CFA'"}}}}}}}
    +  {"properties": {"fill": {"solid": {"color": {"expr": {"ThemeDataColor": {"ColorId": 2, "Percent": 0}}}}}}}
```

```python
from sempy_labs.report import connect_report

# Easily query, set and remove JSON from PBIR using https://jsonpath.com/

with connect_report(report='Report01', readonly = True) as rpt:

    output = rpt.get("**/visual.json", json_path = "$..color.expr.Literal.Value")

    # set

    #  rpt.set_json(file_path = "**/visual.json"
    #             , json_path = "$..color.expr"
    #             , json_value = {
    #                 "color": {
    #                     "expr": {
    #                         "ThemeDataColor": {
    #                         "ColorId": 0,
    #                         "Percent": 0
    #                         }
    #                     }
    #                 }
    #             })

    # remove

    # rpt.remove(file_path="**/page.json", json_path="$.objects.outspace")

output
```

- Command executed in 4 sec 675 ms by Rui Romano on 10:19:41 AM, 5/23/25

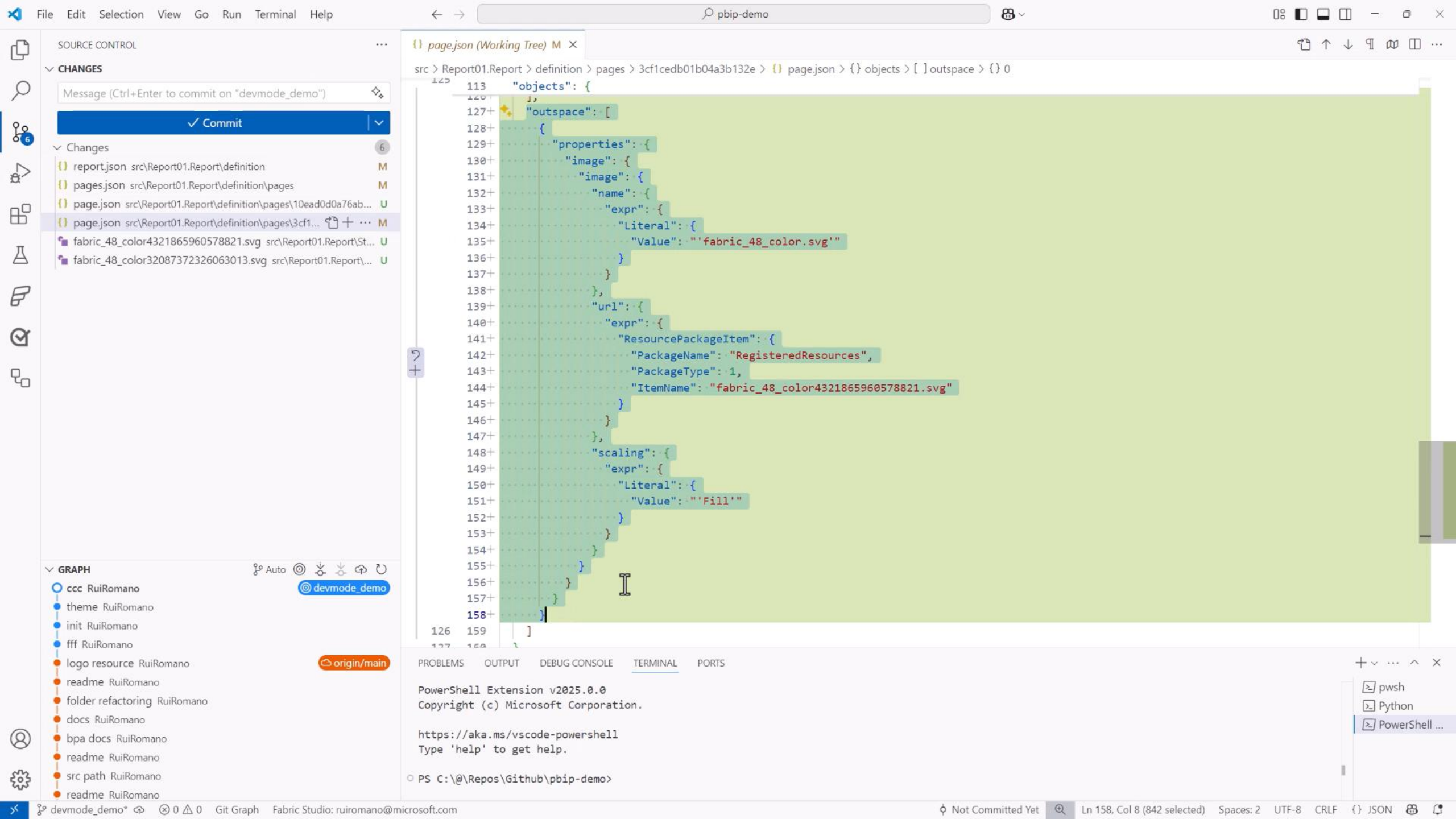
PySpark (Python)

```python
import sempy.fabric as fabric

workspace_id = fabric.get_workspace_id()

items = fabric.list_items()

# Filter only reports

items = items[items['Type'] == "Report"]

for _, report_item in items.iterrows():

    report_id = report_item['Id']
    report_name = report_item['Display Name']

```

```python
from sempy_labs.report import connect_report

# Easily query, set and remove JSON from PBIR using https://jsonpath.com/

with connect_report(report='Report01', readonly = True) as rpt:

    output = rpt.get("**/visual.json", json_path = "$..color.expr.Literal.Value")

    # set

    #  rpt.set_json(file_path = "**/visual.json"
    #             , json_path = "$..color.expr"
    #             , json_value = {
    #                 "color": {
    #                     "expr": {
    #                         "ThemeDataColor": {
    #                         "ColorId": 0,
    #                         "Percent": 0
    #                         }
    #                     }
    #                 }
    #             })

    # remove

    # rpt.remove(file_path="**/page.json", json_path="$.objects.outspace")

output
```

[22] ✓ 7 sec - Command executed in 7 sec 846 ms by Rui Romano on 10:42:16 AM, 5/23/25

Log


File Edit Selection View Go Run Terminal Help
pbip-demo
SOURCE CONTROL
CHANGES
Message (Ctrl+Enter to commit on "devmode_demo")
Commit
Changes 6
report.json src\Report01.Report\definition M
pages.json src\Report01.Report\definition\pages M
page.json src\Report01.Report\definition\pages\10ead0d0a76ab... U
page.json src\Report01.Report\definition\pages\3cf1... M
fabric_48_color4321865960578821.svg src\Report01.Report\St... U
fabric_48_color32087372326063013.svg src\Report01.Report\... U
page.json (Working Tree) M
src > Report01.Report > definition > pages > 3cf1cedb01b04a3b132e > page.json > {} objects > [ ] outspace > {} 0
"objects": {
"outspace": [
{
"properties": {
"image": {
"image": {
"name": {
"expr": {
"Literal": {
"Value": "'fabric_48_color.svg'"
}
}
},
"url": {
"expr": {
"ResourcePackageItem": {
"PackageName": "RegisteredResources",
"PackageType": 1,
"ItemName": "fabric_48_color4321865960578821.svg"
}
}
},
"scaling": {
"expr": {
"Literal": {
"Value": "'Fill'"
}
}
}
}
}
}
}
]
GRAPH
Auto
ccc RuiRomano
devmode_demo
theme RuiRomano
init RuiRomano
fff RuiRomano
logo resource RuiRomano
origin/main
readme RuiRomano
folder refactoring RuiRomano
docs RuiRomano
bpa docs RuiRomano
readme RuiRomano
src path RuiRomano
readme RuiRomano
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PowerShell Extension v2025.0.0
Copyright (c) Microsoft Corporation.
https://aka.ms/vscode-powershell
Type 'help' to get help.
PS C:\@\Repos\Github\pbip-demo>
pwsh
Python
PowerShell ...
devmode_demo*
Git Graph
Fabric Studio: ruiromano@microsoft.com
Not Committed Yet
Ln 158, Col 8 (842 selected)
Spaces: 2
UTF-8
CRLF
JSON

```
        for _, page in report_pages.iterrows():

            page_name = page['Page Name']

            rpt.set_json(file_path = f"definition/pages/{page_name}/page.json"
                , json_path = "$.objects.outspace"
                , json_value =
                    [
                    {
                        "properties": {
                        "image": {
                            "image": {
                            "name": {
                                "expr": {
                                "Literal": {
                                    "Value": f"'{image_path}'"
                                }
                                }
                            },
                            "url": {
                                "expr": {
                                "ResourcePackageItem": {
                                    "PackageName": "RegisteredResources",
                                    "PackageType": 1,
                                    "ItemName": image_path
                                }
                                }
                            },
                            "scaling": {
                                "expr": {
                                "Literal": {
                                    "Value": "'Fill'"
                                }
                                }
                            }
                            }
                        },
                        "transparency": {
                            "expr": {
                            "Literal": {
                                "Value": "0D"
                            }
                            }
                        }
                        }
                    }
                    ]
```

Home | Copilot | Create | Browse | OneLake | Apps | Metrics | Monitor | Learn | Workspaces | RR - PBIP demo | Report02 | Report03 | Report01 | SLL PBIR | Power BI

Explorer

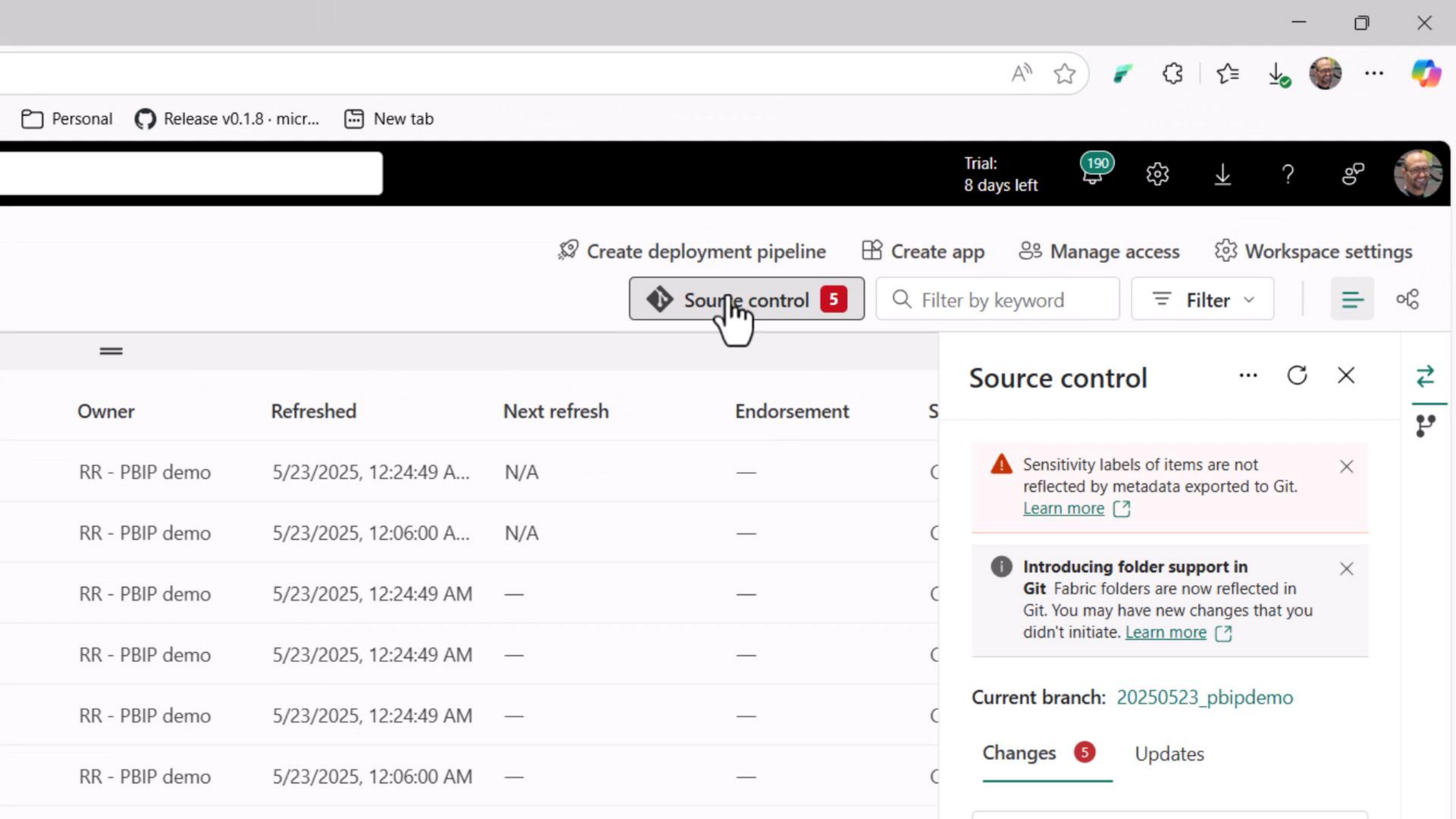
Data items | Resources

No data sources added

Add data items

```
~ resourcePackages
@@ -1,4 +1,4 @@
- {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Company_Green7643675511799916.json", "path": "Company_Green7643675511799916.json", "type": "CustomTheme"}]}
+ {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Company_Green7643675511799916.json", "path": "Company_Green7643675511799916.json", "type": "CustomTheme"},
{"name": "wallpapper.svg", "path": "wallpapper.svg", "type": "Image"}]}
The report definition has been updated successfully.
Processing report 'Report03'
+ StaticResources/RegisteredResources/wallpapper.svg
~ definition/pages/3cf1cedb01b04a3b132e/page.json
~ objects
+ outspace: [{"properties": {"image": {"image": {"name": {"expr": {"Literal": {"Value": "'wallpapper.svg'"}}}, "url": {"expr": {"ResourcePackageItem": {"PackageName": "RegisteredResources",
"PackageType": 1, "ItemName": "wallpapper.svg"}}}, "scaling": {"expr": {"Literal": {"Value": "'Fill'"}}}}}, "transparency": {"expr": {"Literal": {"Value": "0D"}}}}}]
~ definition/report.json
~ resourcePackages
@@ -1,4 +1,4 @@
- {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Company_Blue620697091731989.json", "path": "Company_Blue620697091731989.json", "type": "CustomTheme"}]}
+ {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Company_Blue620697091731989.json", "path": "Company_Blue620697091731989.json", "type": "CustomTheme"}, {"name":
"wallpapper.svg", "path": "wallpapper.svg", "type": "Image"}]}
The report definition has been updated successfully.
Processing report 'Report04'
+ StaticResources/RegisteredResources/wallpapper.svg
~ definition/pages/3dc12e426243950a56ac/page.json
~ objects
+ outspace: [{"properties": {"image": {"image": {"name": {"expr": {"Literal": {"Value": "'wallpapper.svg'"}}}, "url": {"expr": {"ResourcePackageItem": {"PackageName": "RegisteredResources",
"PackageType": 1, "ItemName": "wallpapper.svg"}}}, "scaling": {"expr": {"Literal": {"Value": "'Fill'"}}}}}, "transparency": {"expr": {"Literal": {"Value": "0D"}}}}}]
~ definition/pages/708d7c04691a88810062/page.json
~ objects
+ outspace: [{"properties": {"image": {"image": {"name": {"expr": {"Literal": {"Value": "'wallpapper.svg'"}}}, "url": {"expr": {"ResourcePackageItem": {"PackageName": "RegisteredResources",
"PackageType": 1, "ItemName": "wallpapper.svg"}}}, "scaling": {"expr": {"Literal": {"Value": "'Fill'"}}}}}, "transparency": {"expr": {"Literal": {"Value": "0D"}}}}}]
~ definition/report.json
~ resourcePackages
@@ -1,4 +1,4 @@
- {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Artboard_10897277198378641.png", "path": "Artboard_10897277198378641.png", "type": "Image"}]}
+ {"name": "RegisteredResources", "type": "RegisteredResources", "items": [{"name": "Artboard_10897277198378641.png", "path": "Artboard_10897277198378641.png", "type": "Image"}, {"name":
"wallpapper.svg", "path": "wallpapper.svg", "type": "Image"}]}
The report definition has been updated successfully.
```

```
1  %pip install "./builtin/semantic_link_labs-0.9.12-py3-none-any.whl"
2
3  from sempy_labs.report import connect_report
```


Personal
Release v0.1.8 · micr...
New tab
Trial:
8 days left
190
Create deployment pipeline
Create app
Manage access
Workspace settings
Source control 5
Filter by keyword
Filter
Owner
Refreshed
Next refresh
Endorsement
RR - PBIP demo
5/23/2025, 12:24:49 A...
N/A
RR - PBIP demo
5/23/2025, 12:06:00 A...
N/A
RR - PBIP demo
5/23/2025, 12:24:49 AM
RR - PBIP demo
5/23/2025, 12:24:49 AM
RR - PBIP demo
5/23/2025, 12:24:49 AM
RR - PBIP demo
5/23/2025, 12:06:00 AM
Source control
Sensitivity labels of items are not reflected by metadata exported to Git.
Learn more
Introducing folder support in Git Fabric folders are now reflected in Git. You may have new changes that you didn't initiate. Learn more
Current branch: 20250523_pbipdemo
Changes 5
Updates

🧠 Recap

- Consider Semantic Link Labs for bulk changes in Service
- It’s an advanced tool, things can go wrong – use Fabric Git as backup
- Lots of other applications!
  - Learn more here: https://github.com/microsoft/semantic-link-labs

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Tabular Model Definition Language (TMDL)

📄 Declarative & Readable

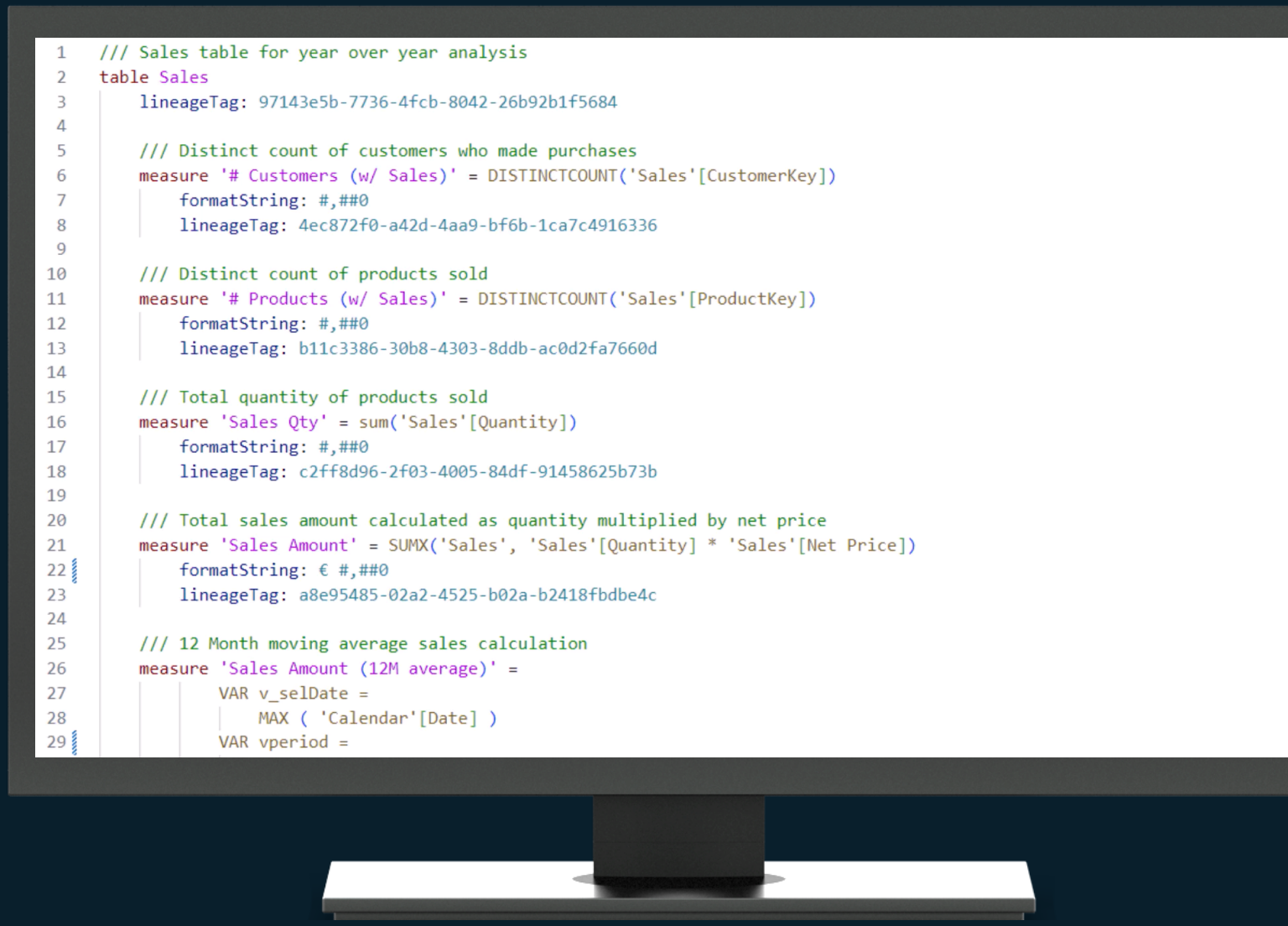
Define semantic models using clean, human-friendly text files

🤝 Built for Collaboration

Track changes, review diffs, and manage versions effortlessly with Git

♻️ Modular & Reusable

Reuse model components across projects to drive consistency and efficiency



```
/// Sales table for year over year analysis
table Sales
    lineageTag: 97143e5b-7736-4fcb-8042-26b92b1f5684

    /// Distinct count of customers who made purchases
    measure '# Customers (w/ Sales)' = DISTINCTCOUNT('Sales'[CustomerKey])
        formatString: #,##0
        lineageTag: 4ec872f0-a42d-4aa9-bf6b-1ca7c4916336

    /// Distinct count of products sold
    measure '# Products (w/ Sales)' = DISTINCTCOUNT('Sales'[ProductKey])
        formatString: #,##0
        lineageTag: b11c3386-30b8-4303-8ddb-ac0d2fa7660d

    /// Total quantity of products sold
    measure 'Sales Qty' = sum('Sales'[Quantity])
        formatString: #,##0
        lineageTag: c2ff8d96-2f03-4005-84df-91458625b73b

    /// Total sales amount calculated as quantity multiplied by net price
    measure 'Sales Amount' = SUMX('Sales', 'Sales'[Quantity] * 'Sales'[Net Price])
        formatString: € #,##0
        lineageTag: a8e95485-02a2-4525-b02a-b2418fbdbe4c

    /// 12 Month moving average sales calculation
    measure 'Sales Amount (12M average)' =
            VAR v_selDate =
                MAX ( 'Calendar'[Date] )
            VAR vperiod =
```

https://aka.ms/pbidevmode-tmdl

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Semantic Modeling as Code experience

Available Now!

Experience a modern developer-friendly code experience for semantic models

🖍️ Code Highlighting

Clear visuals for TMDL and Power Query (PQ)

💡 IntelliSense

Smarter suggestions for faster modeling

🧭 Breadcrumb Navigation

Stay oriented in large models

⚠️ Error Diagnostics

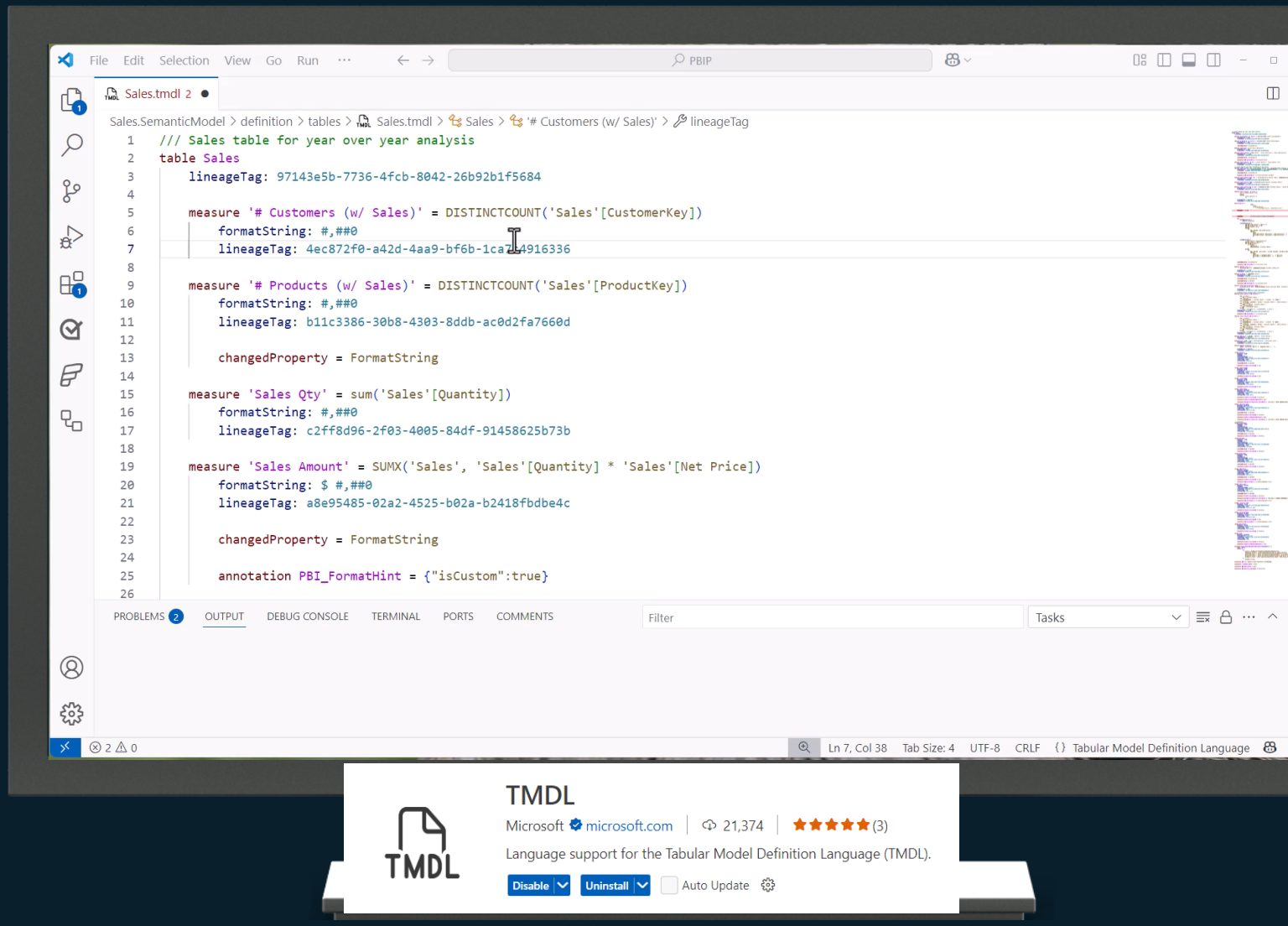
Catch issues early with real-time feedback

🛠️ Code Actions

Quick fixes and refactoring at your fingertips

🎨 Code Formatting

Keep your code clean and consistent



https://aka.ms/vscode-tmdl

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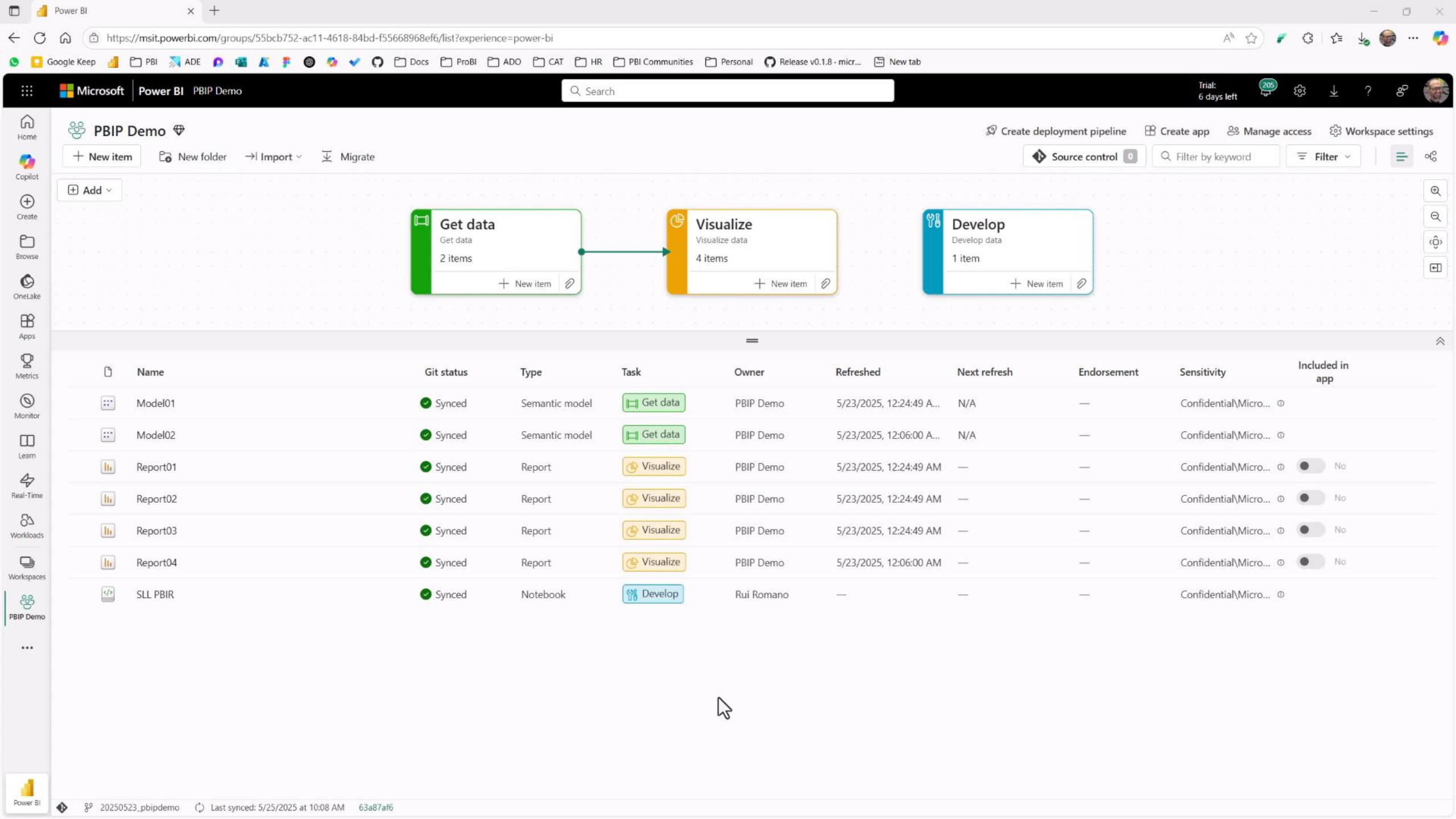
Demo

TMDL using VS Code and GitHub Copilot


Power BI
https://msit.powerbi.com/groups/55bcb752-ac11-4618-84bd-f55668968ef6/list?experience=power-bi
Microsoft | Power BI PBIP Demo
Search
Trial: 6 days left
PBIP Demo
Create deployment pipeline
Create app
Manage access
Workspace settings
New item
New folder
Import
Migrate
Source control 0
Filter by keyword
Filter
Add
Get data
Get data
2 items
New item
Visualize
Visualize data
4 items
New item
Develop
Develop data
1 item
New item
Name
Git status
Type
Task
Owner
Refreshed
Next refresh
Endorsement
Sensitivity
Included in app
Model01 Synced Semantic model Get data PBIP Demo 5/23/2025, 12:24:49 A... N/A — Confidential\Micro...
Model02 Synced Semantic model Get data PBIP Demo 5/23/2025, 12:06:00 A... N/A — Confidential\Micro...
Report01 Synced Report Visualize PBIP Demo 5/23/2025, 12:24:49 AM — — Confidential\Micro... No
Report02 Synced Report Visualize PBIP Demo 5/23/2025, 12:24:49 AM — — Confidential\Micro... No
Report03 Synced Report Visualize PBIP Demo 5/23/2025, 12:24:49 AM — — Confidential\Micro... No
Report04 Synced Report Visualize PBIP Demo 5/23/2025, 12:06:00 AM — — Confidential\Micro... No
SLL PBIR Synced Notebook Develop Rui Romano — — — Confidential\Micro...
20250523_pbipdemo
Last synced: 5/25/2025 at 10:08 AM
63a87af6

File Edit Selection View Go Run Terminal Help

Search

FABRIC STUDIO

FABRIC WORKSPACES

RR - Semantic Model dependencies (Auto binding) [tst] 55...

Extension: Fabric Studio

Fabric Studio

Gerhard Brueckl gbrueckl.at | 5,078 | (1)

Fabric Studio extension for VSCode

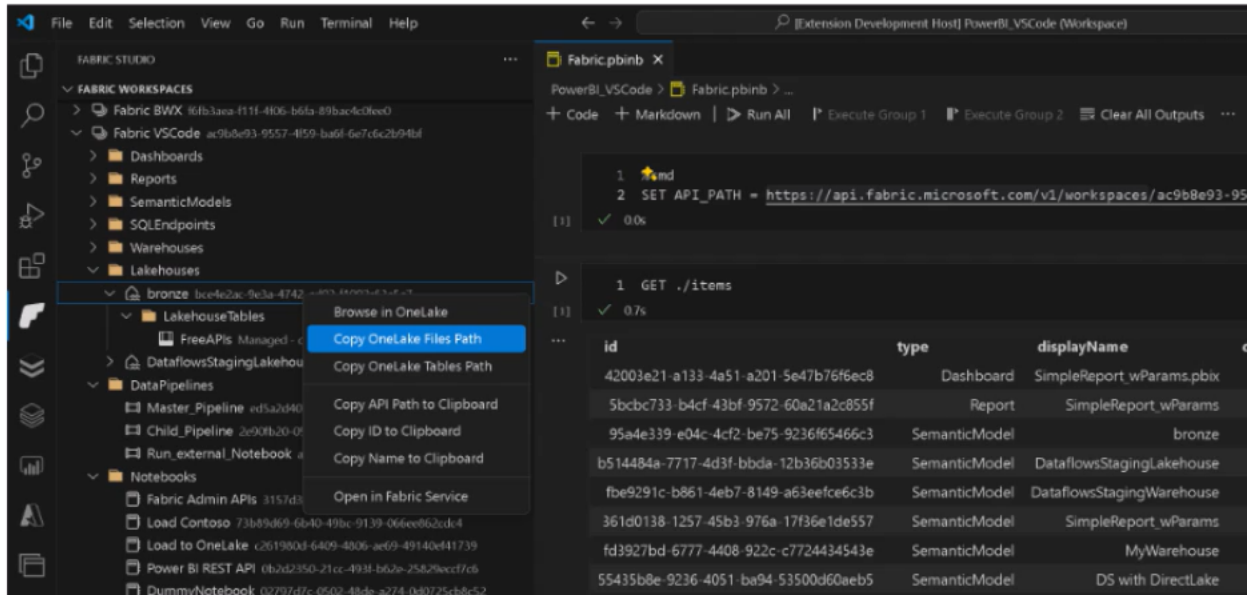
Disable | Uninstall | Auto Update

Fabric Studio

DETAILS FEATURES CHANGELOG

Fabric Studio

version v1.14.0 | installs 5.1k | vs marketplace rate limited by upstream service | rating 5.0/5 (1



File Edit Selection View Go Run Terminal Help

Search

FABRIC STUDIO

FABRIC WORKSPACES

PBIP Demo 55bcb752-ac11-4618-84bd-f55668968ef6

Notebooks

Reports

SemanticModels

Model01 9ce828f2-648b-4a13-9f47-90d2138d811d

Definitio

Connecti

Model02

Role Assign

Managed Pr

Spark Settin

PBIP Demo -

PBIP Demo -

Open in Fabric API Notebook

Download as PowerBI Project (PBIP)

Edit using TMDL

Delete

Copy API Path to Clipboard

Copy ID to Clipboard

Copy Name to Clipboard

Copy Properties to Clipboard

Open in Fabric Service

DEPLOYMENT PIPELINES

CONNECTIONS

CAPACITIES

ADMIN

Tenant Settings

Extension: Fabric Studio

Fabric Studio

Fabric Studio

Gerhard Brueckl gbrueckl.at | 5,078 | (1)

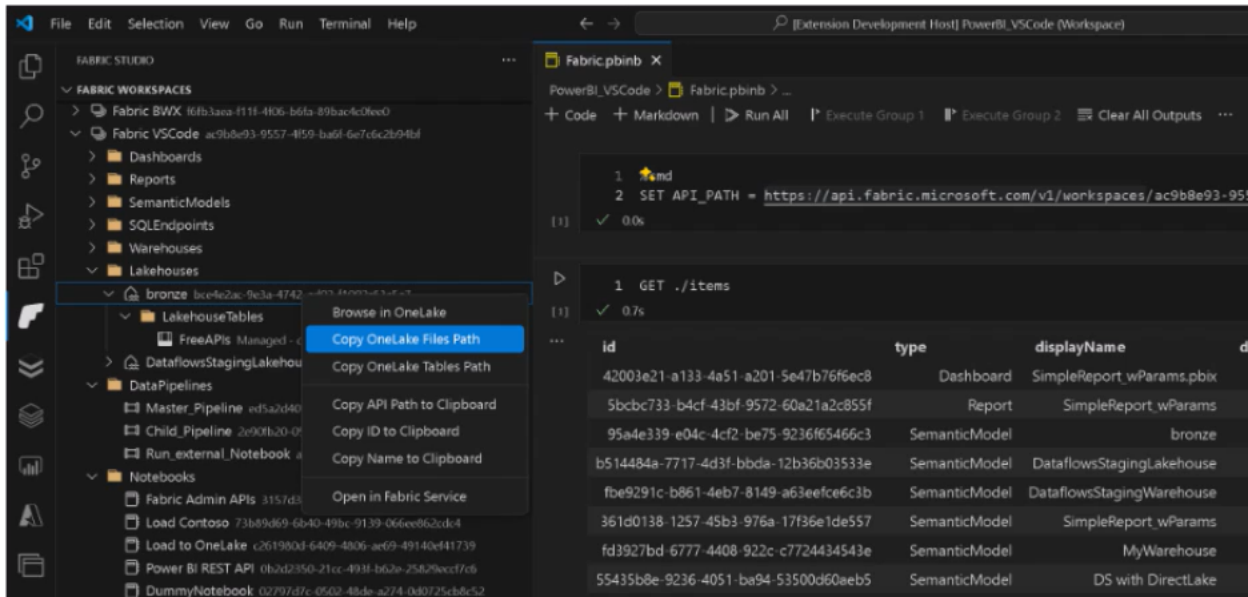
Fabric Studio extension for VSCode

Disable Uninstall Auto Update

DETAILS FEATURES CHANGELOG

Fabric Studio

version v1.14.0 installs 5.1k vs marketplace rate limited by upstream service rating 5.0/5 (1)



File Edit Selection View Go Run Terminal Help

Untitled (Workspace)

EXPLORER

UNTITLED (WORKSPACE)

Fabric - PBIP Demo - Model01 M

- definition
  - cultures
  - perspectives
  - roles
  - tables
    - About.tmdl
    - Calendar.tmdl
    - Customer.tmdl
    - Field Parameter.tmdl
    - Product.tmdl
    - Sales.tmdl
    - Store.tmdl
    - Time Intelligence.tmdl
  - database.tmdl
  - expressions.tmdl
  - model.tmdl
  - relationships.tmdl
  - .platform
  - definition.pbism

Extension: Fabric Studio

Fabric Studio

Gerhard Brueckl gbrueckl.at | 5,078 | (1)

Fabric Studio extension for VSCode

Disable Uninstall Auto Update

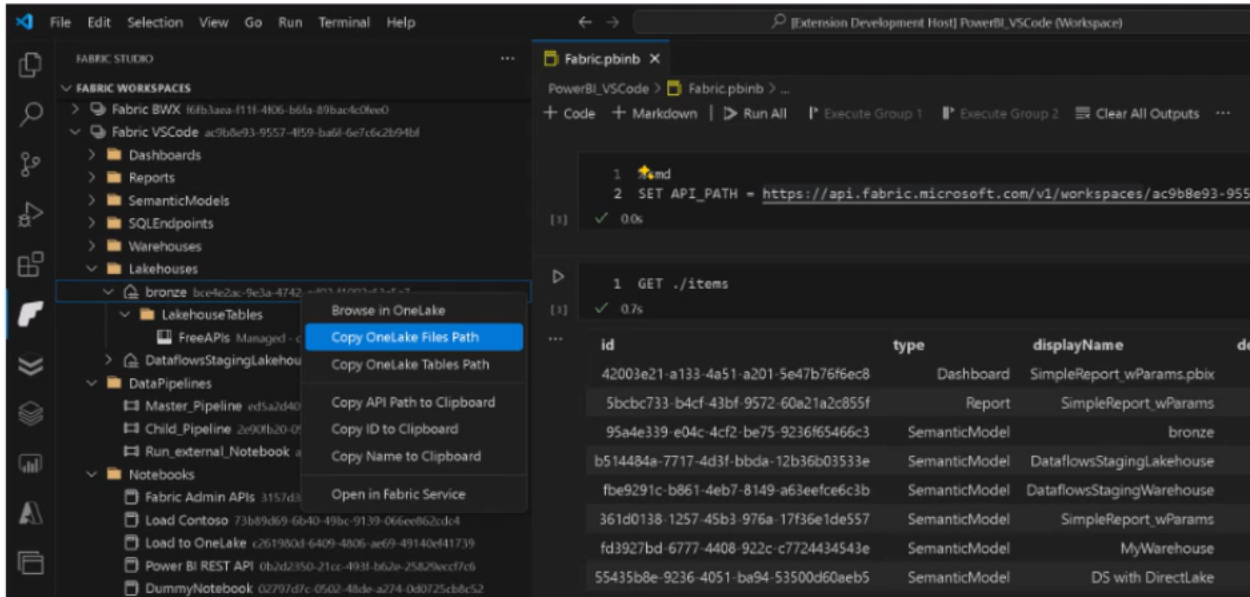
Fabric Studio

DETAILS FEATURES CHANGELOG

Fabric Studio

\workspaces\55bcb752-ac11-4618-84bd-f55668968ef6\SemanticModels\Model01\definition\tables\Product.tmdl

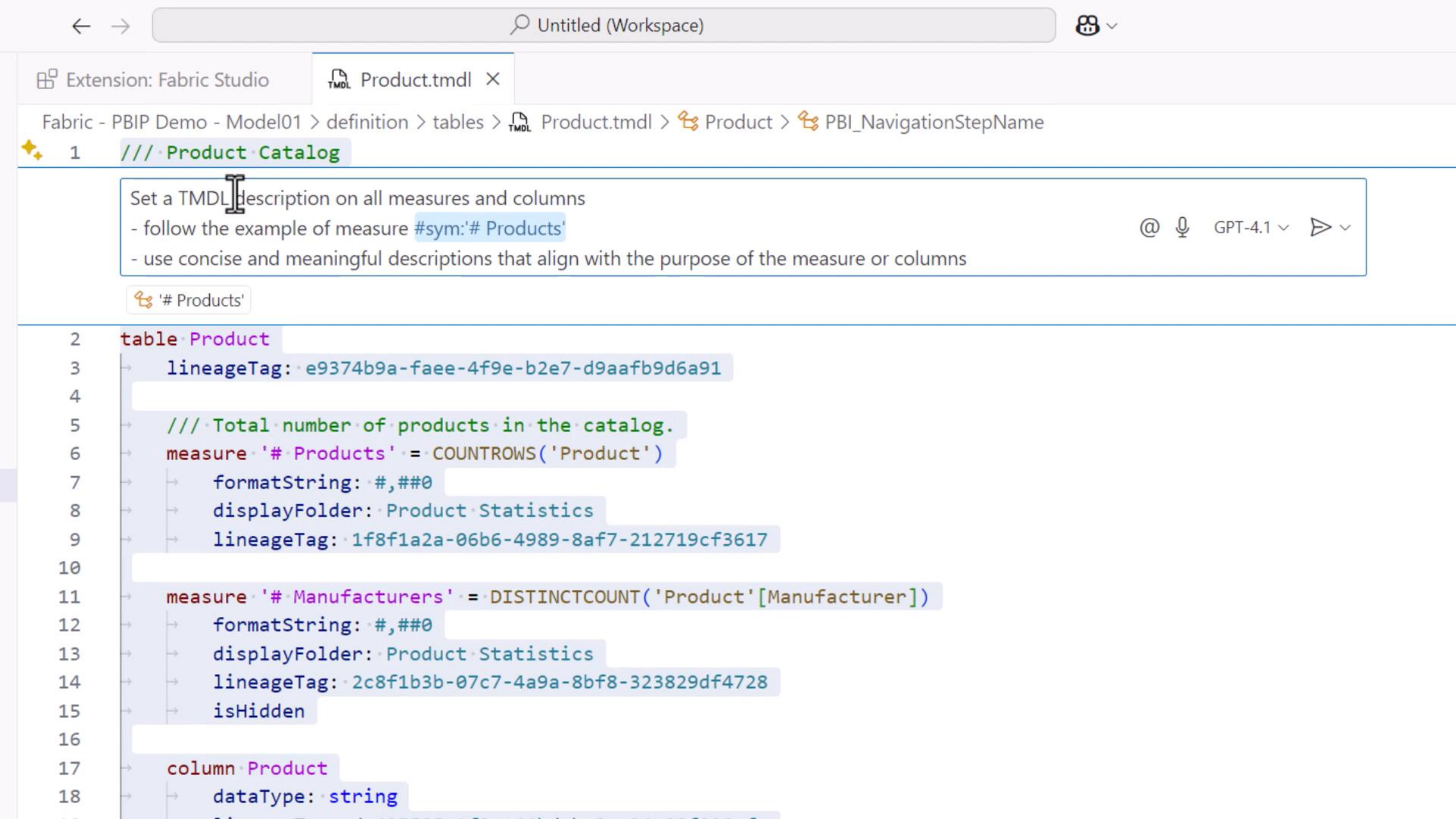
ating 5.0/5 (1)



```
1   /// Product Catalog
2   table Product
3       lineageTag: e9374b9a-faee-4f9e-b2e7-d9aafb9d6a91
4
5       meas
6
7           displayFolder: Product Statistics
8           lineageTag: 1f8f1a2a-06b6-4989-8af7-212719cf3617
9       measure '# Manufacturers' = DISTINCTCOUNT('Product'[Manufacturer])
            formatString: #,##0
            displayFolder: Product Statistics
            lineageTag: 2c8f1b3b-07c7-4a9a-8bf8-323829df4728
10
11      column Product
12          dataType: string
13          lineageTag: da435585-1f9a-44bd-ba2c-34c98f298cfc
14          isDefaultLabel
15          summarizeBy: distinctCount
16          sourceColumn: Product
17
18          annotation SummarizationSetBy = Automatic
19
```

displayFolder

Defines the display folder for the Measure, for use by clients.


Untitled (Workspace)
Extension: Fabric Studio
Product.tmdl
Fabric - PBIP Demo - Model01 > definition > tables > Product.tmdl > Product > PBI_NavigationStepName
/// Product Catalog
Set a TMDL description on all measures and columns
- follow the example of measure #sym:'# Products'
- use concise and meaningful descriptions that align with the purpose of the measure or columns
GPT-4.1
'# Products'
table Product
lineageTag: e9374b9a-faee-4f9e-b2e7-d9aafb9d6a91
/// Total number of products in the catalog.
measure '# Products' = COUNTROWS('Product')
formatString: #,##0
displayFolder: Product Statistics
lineageTag: 1f8f1a2a-06b6-4989-8af7-212719cf3617
measure '# Manufacturers' = DISTINCTCOUNT('Product'[Manufacturer])
formatString: #,##0
displayFolder: Product Statistics
lineageTag: 2c8f1b3b-07c7-4a9a-8bf8-323829df4728
isHidden
column Product
dataType: string

Untitled (Workspace)

Extension: Fabric Studio | Product.tmdl | Store - United States.tmdl 9+

Fabric - PBIP Demo - Model01 > definition > roles > Store - United States.tmdl > 'Store - United States' > 'Store - Sweden' > PBI_Id

```
1   role 'Store - United States'
```

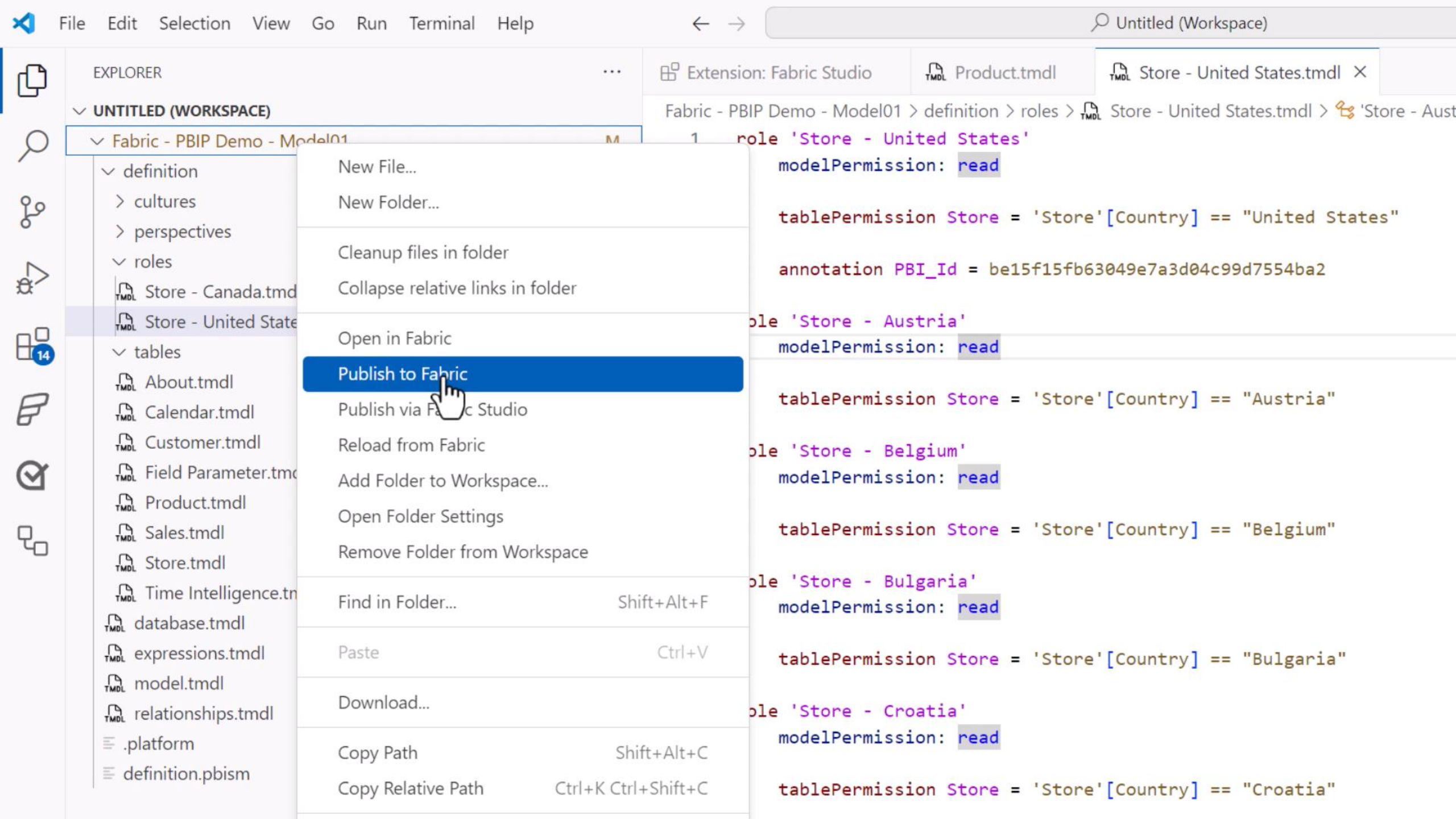
create new roles for each country in the EU. Follow the same pattern of the US role

remove the PBI_Id annotation

Ask Copilot @ GPT-4.1

Accept Close

```
8   role 'Store - Austria'
9       modelPermission: read
10
11      tablePermission Store = 'Store'[Country] == "Austria"
12
13  role 'Store - Belgium'
14      modelPermission: read
15
16      tablePermission Store = 'Store'[Country] == "Belgium"
17
18  role 'Store - Bulgaria'
19      modelPermission: read
20
21      tablePermission Store = 'Store'[Country] == "Bulgaria"
22
23  role 'Store - Croatia'
24      modelPermission: read
25
26      tablePermission Store = 'Store'[Country] == "Croatia"
27
28  role 'Store - Cyprus'
29      modelPermission: read
30
31      tablePermission Store = 'Store'[Country] == "Cyprus"
```


File Edit Selection View Go Run Terminal Help
Untitled (Workspace)
EXPLORER
UNTITLED (WORKSPACE)
Fabric - PBIP Demo - Model01
definition
cultures
perspectives
roles
Store - Canada.tmdl
Store - United State
tables
About.tmdl
Calendar.tmdl
Customer.tmdl
Field Parameter.tmdl
Product.tmdl
Sales.tmdl
Store.tmdl
Time Intelligence.tmdl
database.tmdl
expressions.tmdl
model.tmdl
relationships.tmdl
.platform
definition.pbism
New File...
New Folder...
Cleanup files in folder
Collapse relative links in folder
Open in Fabric
Publish to Fabric
Publish via Fabric Studio
Reload from Fabric
Add Folder to Workspace...
Open Folder Settings
Remove Folder from Workspace
Find in Folder... Shift+Alt+F
Paste Ctrl+V
Download...
Copy Path Shift+Alt+C
Copy Relative Path Ctrl+K Ctrl+Shift+C
Extension: Fabric Studio
Product.tmdl
Store - United States.tmdl
Fabric - PBIP Demo - Model01 > definition > roles > Store - United States.tmdl > 'Store - Aust
role 'Store - United States'
modelPermission: read
tablePermission Store = 'Store'[Country] == "United States"
annotation PBI_Id = be15f15fb63049e7a3d04c99d7554ba2
ble 'Store - Austria'
modelPermission: read
tablePermission Store = 'Store'[Country] == "Austria"
ble 'Store - Belgium'
modelPermission: read
tablePermission Store = 'Store'[Country] == "Belgium"
ble 'Store - Bulgaria'
modelPermission: read
tablePermission Store = 'Store'[Country] == "Bulgaria"
ble 'Store - Croatia'
modelPermission: read
tablePermission Store = 'Store'[Country] == "Croatia"

TMDL view – Modeling scripting experience

Available Now!

👀 See Everything

Dive deep into your entire semantic model metadata

⚡ Work Smarter, Faster

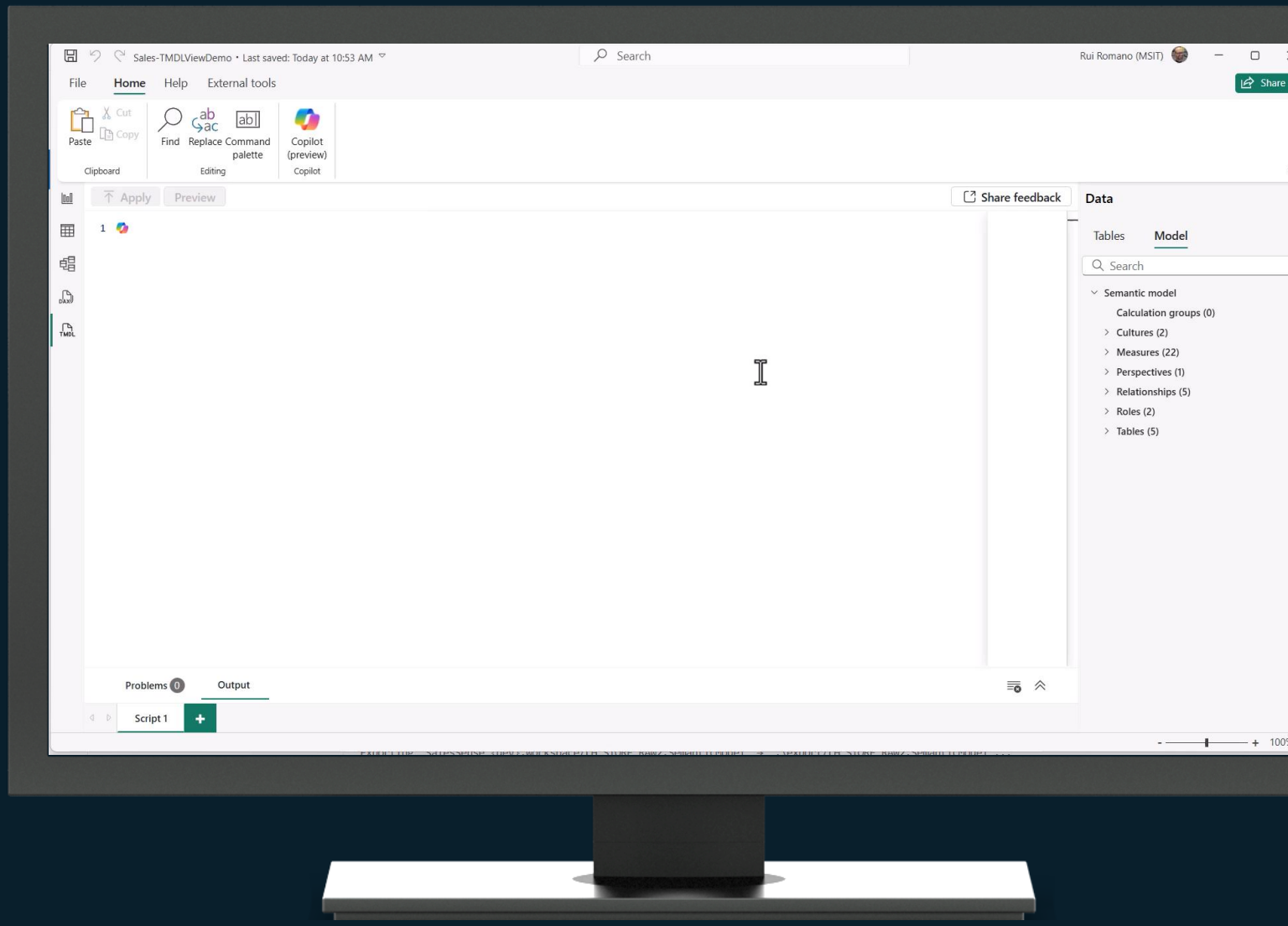
Make bulk edits with a powerful code editor - manually or with AI 🤖

🛠️ Take Full Control

Edit advanced objects and properties not available in the UI

♻️ Reuse Everywhere

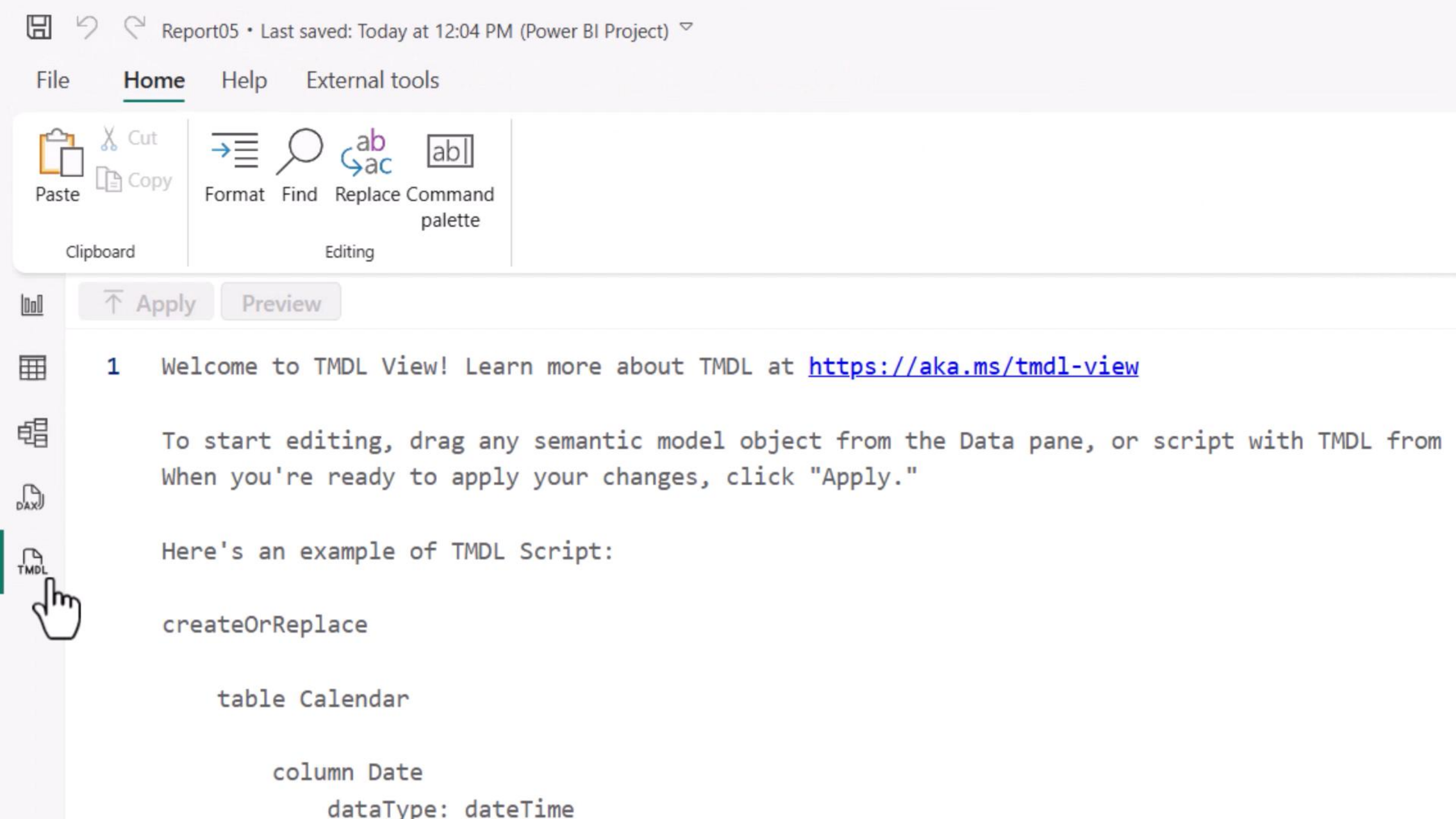
Leverage TMDL scripts to streamline modeling across multiple projects



Classified as Microsoft Confidential

Demo

TMDL view with GitHub Copilot


Report05 • Last saved: Today at 12:04 PM (Power BI Project)
File
Home
Help
External tools
Cut
Copy
Paste
Clipboard
Format
Find
Replace
Command palette
Editing
Apply
Preview
DAX
TMDL
1 Welcome to TMDL View! Learn more about TMDL at https://aka.ms/tmdl-view
To start editing, drag any semantic model object from the Data pane, or script with TMDL from
When you're ready to apply your changes, click "Apply."
Here's an example of TMDL Script:
createOrReplace
table Calendar
column Date
dataType: dateTime

```
createOrReplace

    ref table Product

        column 'Unit Price'
            dataType: decimal
            is
            li
            su
            so
            an
            an
        column
            dataType: decimal
            is
            lineageTag: f89fa3e3-061d-4269-8cd3-aa6ce2a464d2
            summarizeBy: none
            sourceColumn: Unit Cost

            annotation SummarizationSetBy = Automatic

            annotation PBI_FormatHint = {"isGeneralNumber":true}

        column ProductKey
            dataType: int64
            is
            isHidden
            isKey
            formatString: 0
            lineageTag: 4184d53e-cd2d-4cbe-b8cb-04c72a750bc4
            summarizeBy: none
            sourceColumn: ProductKey
```

isAvailableInMdx
isDataTypeInferred
isDefaultImage
isDefaultLabel
isHidden
isKey
isNameInferred
isNullable
isUnique

Problems 3 Output

Report05 • Last saved: Today at 12:04 PM (Power BI Project)

Search

Rui Romano (MSIT)

File Home Help External tools

Share

Paste Cut Copy — Clipboard

Format Find Replace Command palette — Editing

Apply Preview

Share feedback

```
createOrReplace

    perspective Simple

        perspectiveTable Sales

            perspectiveMeasure 'Sales Amount'

        perspectiveTable Product

            perspectiveColumn Category

            perspectiveColumn Product

        perspectiveTable Calendar

            perspectiveColumn Date

            perspectiveColumn Year

            perspectiveColumn Month

            perspectiveColumn Day
```

Problems 0 Output

Script 1 Script 2 Script 3 +

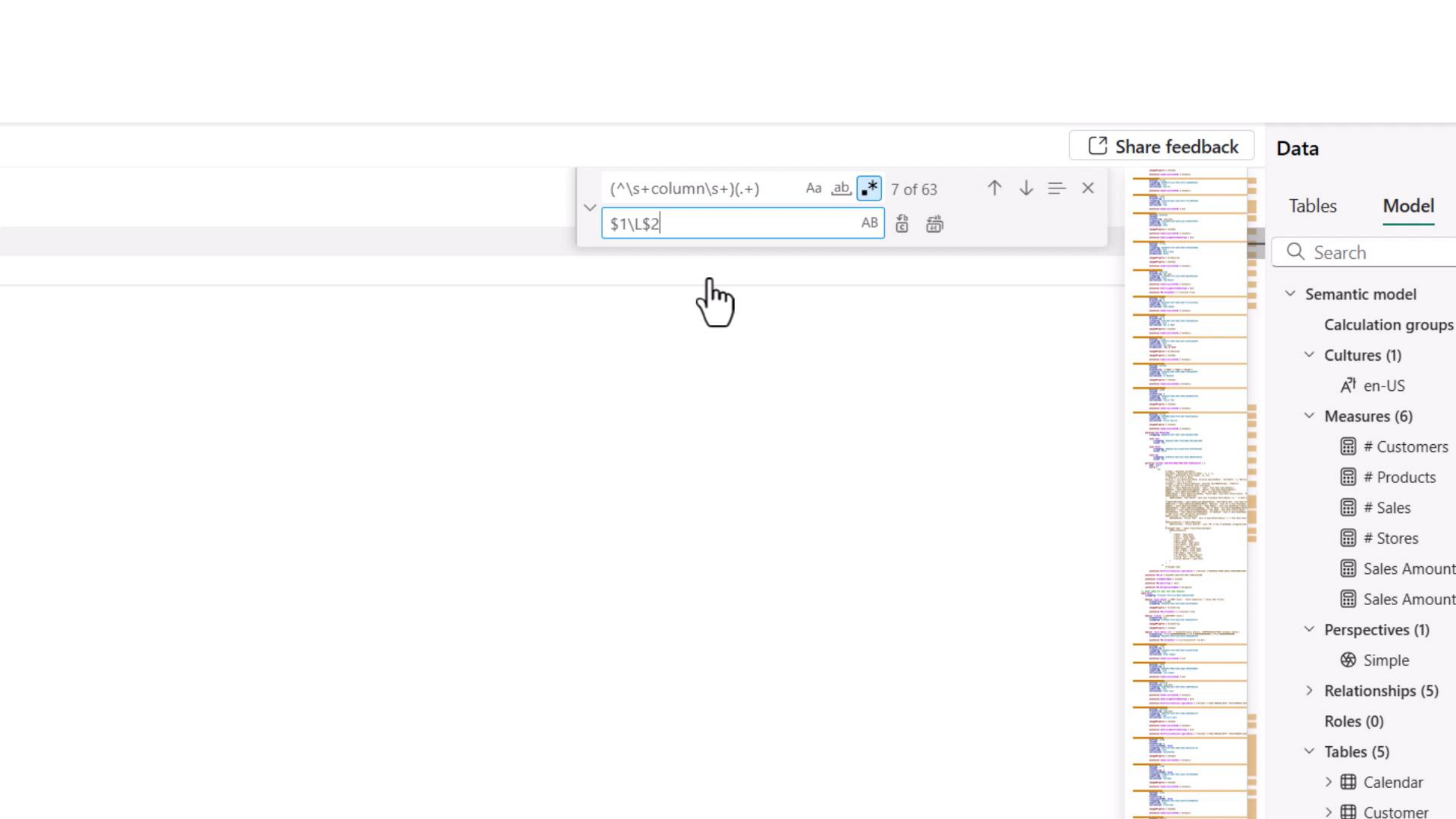
Data

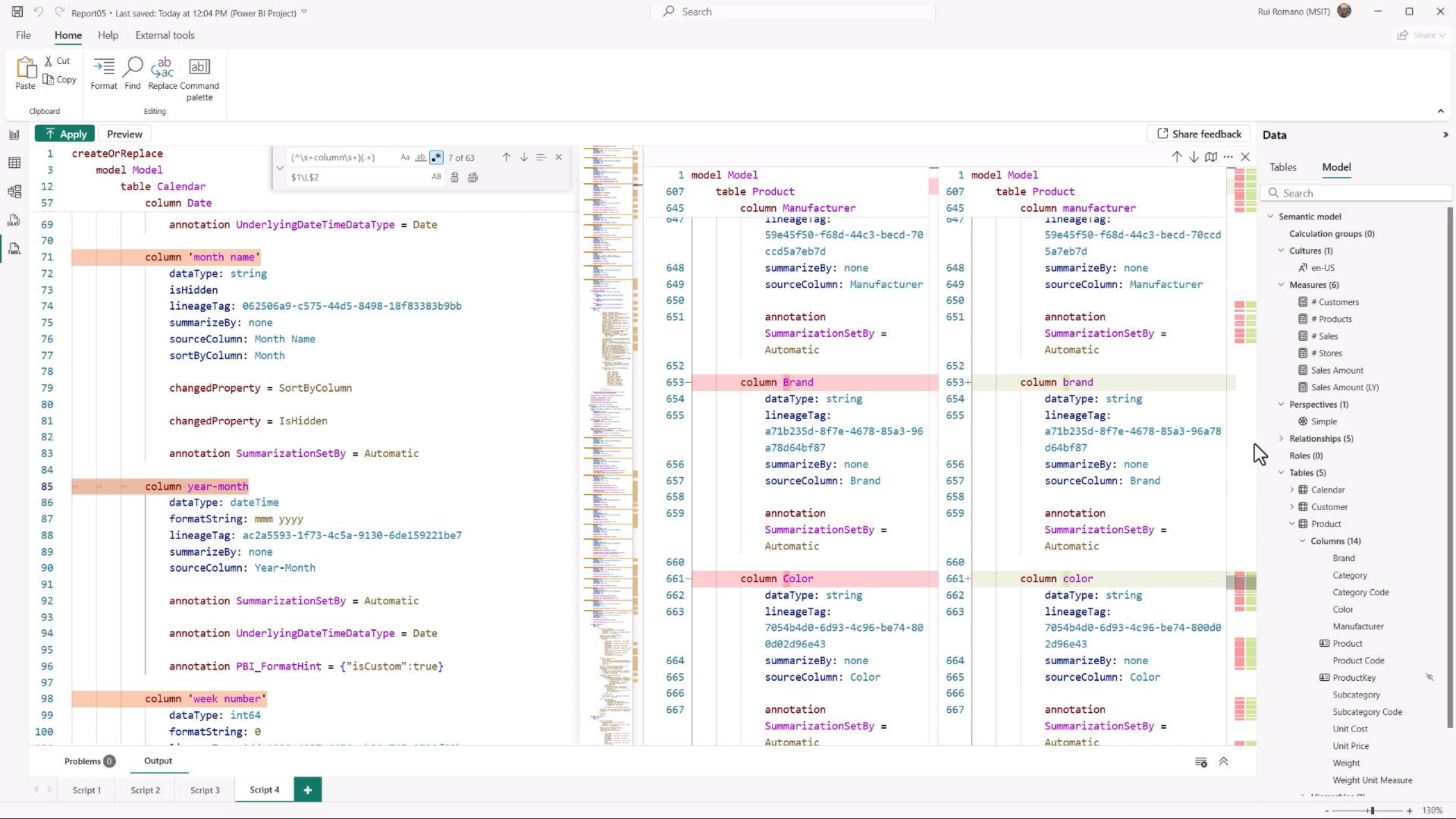
Tables Model

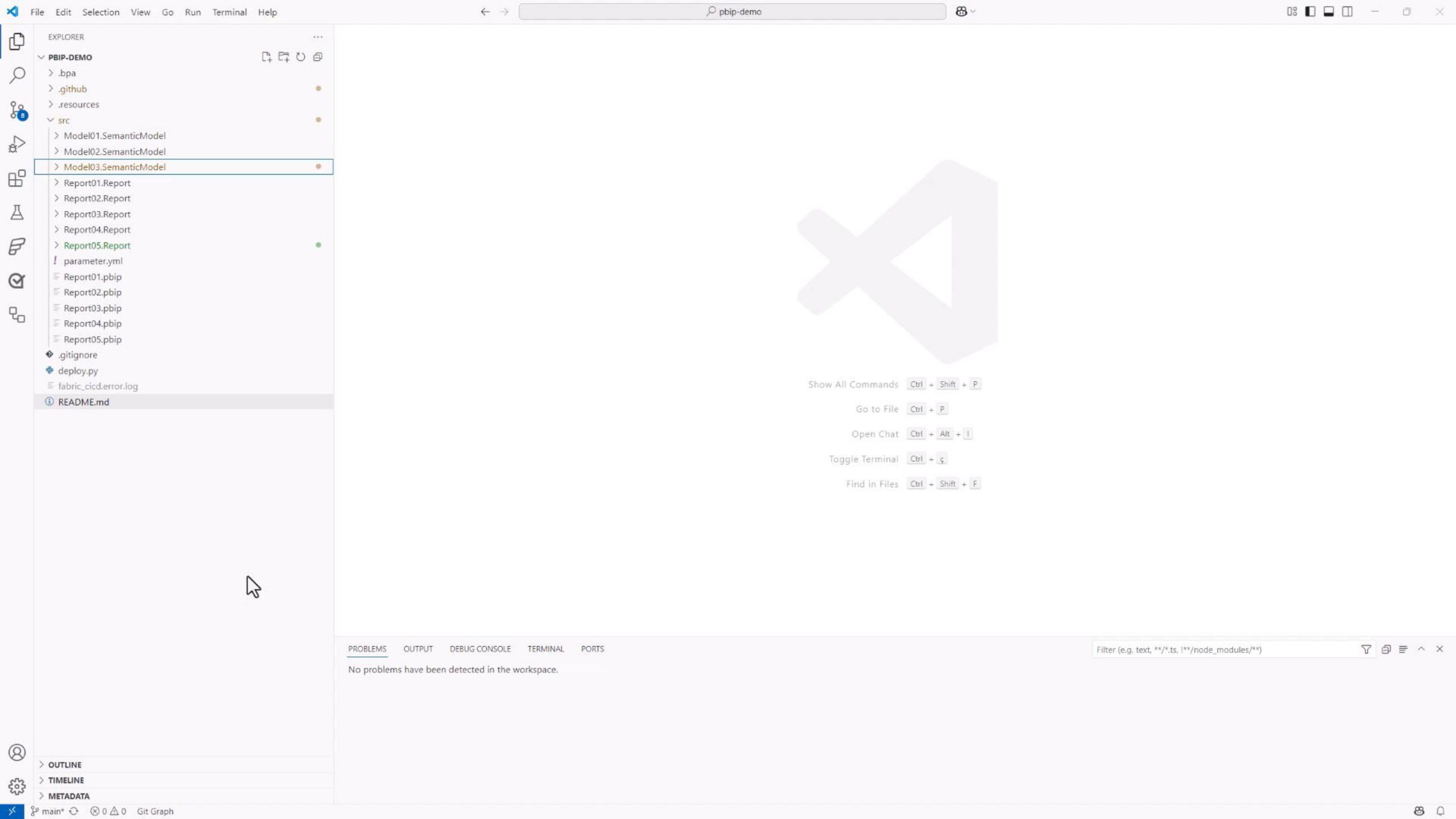
Search

- Measures (6)
  - # Customers
  - # Products
  - # Sales
  - # Stores
  - Sales Amount
  - Sales Amount (LY)
- Perspectives (1)
  - Simple
- Relationships (5)
- Roles (0)
- Tables (5)
  - Calendar
  - Customer
  - Product
    - Columns (14)
      - Brand
      - Category
      - Category Code
      - Color
      - Manufacturer
      - Product
      - Product Code
      - ProductKey
      - Subcategory
      - Subcategory Code
      - Unit Cost
      - Unit Price
      - Weight
      - Weight Unit Measure
    - Hierarchies (1)
    - Measures (1)
    - Partitions (1)
  - Sales
  - Store

130%


Share feedback
(^\s+column\s+)(.+)
Aa
ab
7 of 63
$1\L$2
AB
Data
Tables
Model
Search
Semantic model
Calculation groups
Cultures (1)
en-US
Measures (6)
Customers
Products
Sales
Stores
Sales Amount
Sales Amount
Perspectives (1)
Simple
Relationships (5)
Roles (0)
Tables (5)
Calendar
Customer


Report05 • Last saved: Today at 12:04 PM (Power BI Project)
Search
Rui Romano (MSIT)
File
Home
Help
External tools
Share
Cut
Copy
Paste
Format
Find
Replace
Command palette
Clipboard
Editing
Apply
Preview
Share feedback
Data
Tables
Model
Search
Semantic model
Calculation groups (0)
Cultures (1)
en-US
Measures (6)
Customers
Products
Sales
Stores
Sales Amount
Sales Amount (LY)
Perspectives (1)
Simple
Relationships (5)
Roles (0)
Tables (5)
Calendar
Customer
Product
Columns (14)
Brand
Category
Category Code
Color
Manufacturer
Product
Product Code
ProductKey
Subcategory
Subcategory Code
Unit Cost
Unit Price
Weight
Weight Unit Measure
Problems 0
Output
Script 1
Script 2
Script 3
Script 4
130%


File Edit Selection View Go Run Terminal Help
pbip-demo
EXPLORER
PBIP-DEMO
.bpa
.github
.resources
src
Model01.SemanticModel
Model02.SemanticModel
Model03.SemanticModel
Report01.Report
Report02.Report
Report03.Report
Report04.Report
Report05.Report
parameter.yml
Report01.pbip
Report02.pbip
Report03.pbip
Report04.pbip
Report05.pbip
.gitignore
deploy.py
fabric_cicd.error.log
README.md
Show All Commands Ctrl + Shift + P
Go to File Ctrl + P
Open Chat Ctrl + Alt + I
Toggle Terminal Ctrl + ç
Find in Files Ctrl + Shift + F
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Filter (e.g. text, **/*.ts, !**/node_modules/**)
No problems have been detected in the workspace.
OUTLINE
TIMELINE
METADATA
main*
0 0
Git Graph

Add Context... Script 5.tmdl Current file

suggest measures to this file.

Agent GPT-4.1

Ln 6, Col 1 Tab Size: 4 UTF-8 CRLF { } Tabular Model Definition Language

CHAT

RuiRomano

suggest measures to this file.

Script 5.tmdl:1-42

GitHub Copilot

Used 2 references

copilot-instructions.md .github

Script 5.tmdl src\Model03.SemanticModel\TMDLScripts

Based on the contents of `Script 5.tmdl` and following your repository and coding guidelines, here are suggested measures for the Sales table that would be useful for COMPANY X's business analysis, while avoiding duplication and keeping DAX simple:

1. /// Total quantity of products sold across all transactions, supporting analysis of sales volume trends for COMPANY X. measure 'Total Quantity' = SUM('Sales'[Quantity]) formatString: #,0

```
- When asked to create measures:
    - Suggest a measure description following the rules above.
    - Always include a formatString property appropriate for the measure type.
    - When summarizeBy = count use the DISTINCTCOUNT aggregation function.
    - Don't create measures for non aggregatable columns such as keys or descriptions. Unless they s
    - Don't create complex DAX. Keep it simple, most of the times I'm just trying to save some time
    - When creating measures look for existing measures and don't duplicate them.

- For all asks:
    - When creating new objects never include the lineageTag property.
    - Always learn from existing examples and patterns.
    - When creating new objects, look for existing objects of same type and follow the same naming c

- When asked to modify Power Query / M code only change the code inside the partition and source obj

    - You are an assistant to help Power Query developers comment their code.
    - Insert a comment above the code explaining what that piece of code is doing.
    - Do not start the comment with the word Step or a number
    - Do not copy code into the comment.
    - Keep the comments to a maximum of 225 characters.
    - Update the step name explaining what that piece of code is doing.
    - The step name should be enclosed in double quotes and preceded by the '#'
```

```
    changedProperty = IsHidden

measure 'Sales Amount (LY)' = CALCULATE([Sales Amount], SAMEPERIODLASTYEAR('Calendar'[Date]))
    formatString: "€"#,0.###############;("€"#,0.###############);"€"#,0.###############
    lineageTag: 94ea5e7c-6f4f-4c8e-b9c0-c85d4a65ef3d

    annotation PBI_FormatHint = {"currencyCulture":"en-US"}

/// Total quantity of products sold across all transactions, supporting analysis of sales volume
measure 'Total Quantity' = SUM('Sales'[Quantity])
    formatString: #,0

/// Average net price per unit sold, providing insight into pricing strategies and customer purc
measure 'Average Net Price' = AVERAGE('Sales'[Net Price])
    formatString: "€"#,0.00

/// Distinct count of orders, helping COMPANY X track unique sales transactions and customer eng
measure 'Distinct Orders' = DISTINCTCOUNT('Sales'[order number])
    formatString: #,0

/// Total sales cost, representing the sum of unit costs for all sold items, supporting margin a
measure 'Total Cost' = SUM('Sales'[unit cost])
    formatString: "€"#,0.00

/// Gross profit calculated as sales amount minus total cost, enabling COMPANY X to monitor prof
measure 'Gross Profit' = [Sales Amount] - [Total Cost]
    formatString: "€"#,0.00

column 'order number'
    dataType: int64
    formatString: 0
    lineageTag: e2e629f1-f1fb-4444-a627-b121d77fdc06
    summarizeBy: none
```

Keep Undo 1 of 1

GitHub Copilot

Used 1 reference

Script 5.tmdl +20 -0

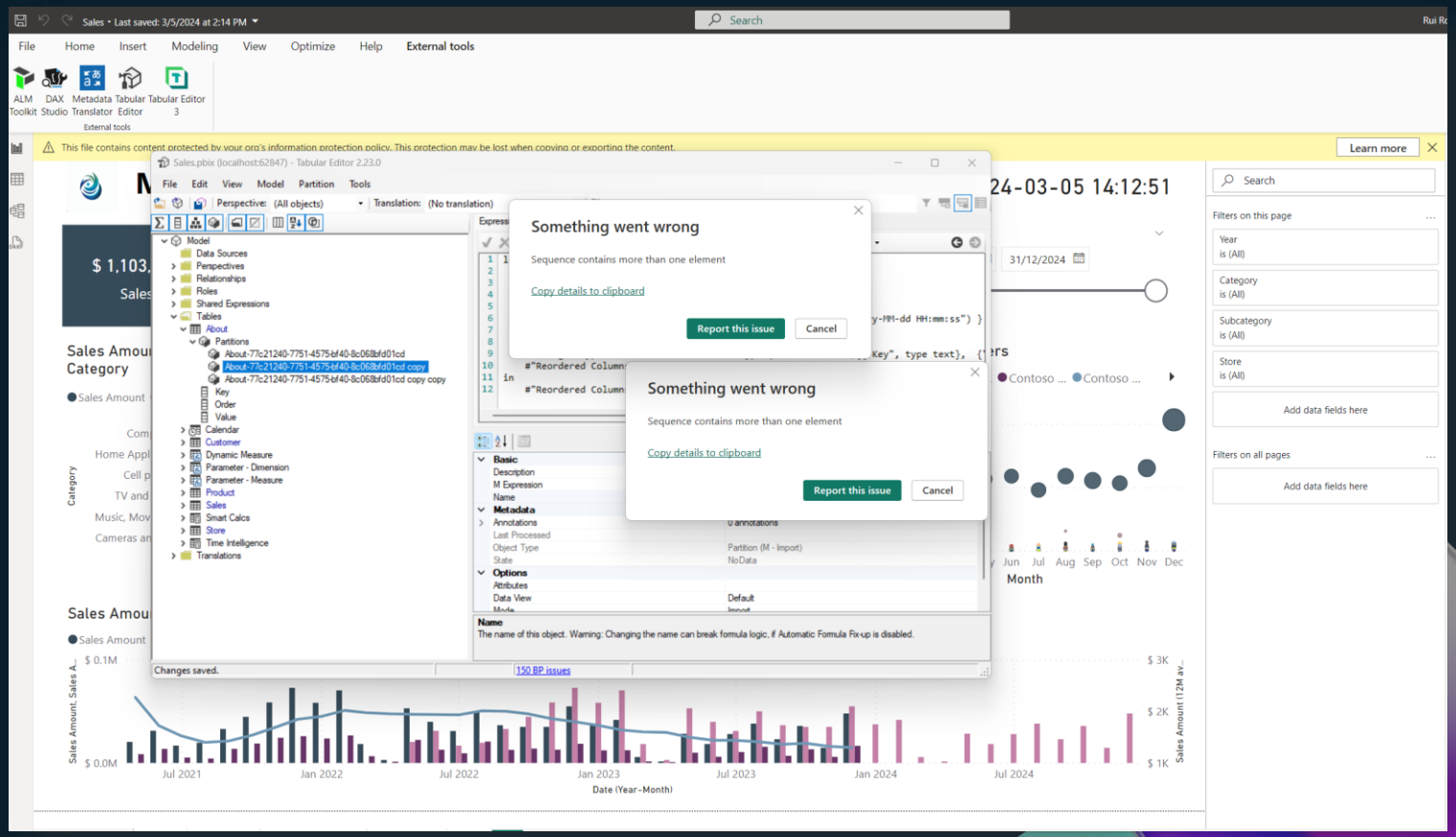
Five new measures have
description and appropr
know if you need furthe

Edit ANY semantic model using Power BI Desktop

Coming soon

Open and edit any semantic model running in Fabric.

- Edit partitions & data sources
- Model schema decoupled from queries
- Expand external tool writing operations



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Demo

TMDL view to edit any model

Report05 • Last saved: Today at 1:00 PM (Power BI Project)

Search

Rui Romano (MSIT)

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```
createOrReplace
    table Sales
        partition Sales-2024 = m
            source =
                    let
                                    List.Min({[Quantity], [Quantity] + ([Quantity] * Randomizer)})
                                )), Int64.Type),
                            "NewNetPrice",
                            each Number.Round(Number.RandomBetween(
                                List.Max({1, [Net Price] - ([Net Price] * Randomizer)}),
                                List.Min({[Net Price], [Net Price] + ([Net Price] * Randomizer)})
                            ), 3),
                            Currency.Type
                        ),
                        {{"Order Date", each _, type date}, {"Delivery Date", each _, type date}}
                    ),
                    Final = Table.AddColumn(
                        Table.RenameColumns(
                            Table.RemoveColumns(WithNewCols, {"Quantity", "Net Price", "Unit Price"}),
                            {{"NewQuantity", "Quantity"}, {"NewNetPrice", "Net Price"}}
                        ),
                        "Environment", each Environment, type text
                    ),
                DateFilter = Table.SelectRows(Final, each [Order Date] >= #date(2024, 1, 1) and [Order Date] <= #date(2024, 12, 31))
                in
                    DateFilter

        annotation PBI_Id = 975ddcc4-65e2-4eb0-9c9c-2c5ef0586f0e

        annotation LinkedQueryName = Sales

        annotation PBI_ResultType = Table

        annotation PBI_NavigationStepName = Navigation
```

Data

Tables Model

Search

- Semantic model
  - Calculation groups (0)
  - Cultures (1)
    - en-US
  - Measures (11)
    - # Customers
    - # Products
    - # Sales
    - # Stores
    - Average Net Price
    - Distinct Orders
    - Gross Profit
    - Sales Amount
    - Sales Amount (LY)
    - Total Cost
    - Total Quantity
  - Perspectives (1)
    - Simple
  - Relationships (5)
  - Roles (0)
  - Tables (5)
    - Calendar
    - Customer
    - Product
    - Sales
    - Store

Problems 0 Output

Script 1 Script 2 Script 3 Script 4 Script 5 +

130%

```
),

er Date] <= #date(2024, 12, 31))
```

- Product
- Sales
  - Columns (14)
  - Hierarch...0)
  - Measures (8)
  - Partitions (3)
    - Sales-2022
    - Sales-2023
    - Sales-2024
- Store

```
            each Number.RoundDown(Number.RandomBetween(
                List.Max({1, [Quantity] - ([Quantity] * Randomizer)}),
                List.Min({[Quantity], [Quantity] + ([Quantity] * Randomizer)})
            )), Int64.Type),
        "NewNetPrice",
        each Number.Round(Number.RandomBetween(
            List.Max({1, [Net Price] - ([Net Price] * Randomizer)}),
            List.Min({[Net Price], [Net Price] + ([Net Price] * Randomizer)})
        ), 3),
        Currency.Type
    ),
    {{"Order Date", each _, type date}, {"Delivery Date", each _, type date}}
),
Final = Table.AddColumn(
        Table.RenameColumns(
            Table.RemoveColumns(WithNewCols, {"Quantity", "Net Price", "Unit Price"}),
            {{"NewQuantity", "Quantity"}, {"NewNetPrice", "Net Price"}}
        ),
        "Environment", each Environment, type text
    ),
DateFilter = Table.SelectRows(Final, each [Order Date] >= #date(2024, 1, 1) and [Order Date] <= #date(2024, 12, 31))
in
    DateFilter
```

```
ion PBI Id = 975ddcc4-65e2-4eb0-9c9c-2c5ef0586f0e
```

ut

Script 3 | Script 4 | Script 5 | Script 6 | +

Script 5.tmdl U | Script 6.tmdl U

src > Model03.SemanticModel > TMDLScripts > Script 6.tmdl > createOrReplace > Sales > Sales-2024 > source

```
1   createOrReplace
4       table Sales
270         partition Sales-2023 = m
272             source =
326                     DateFilter
327
```

replace 2024 partition with quarter partitions

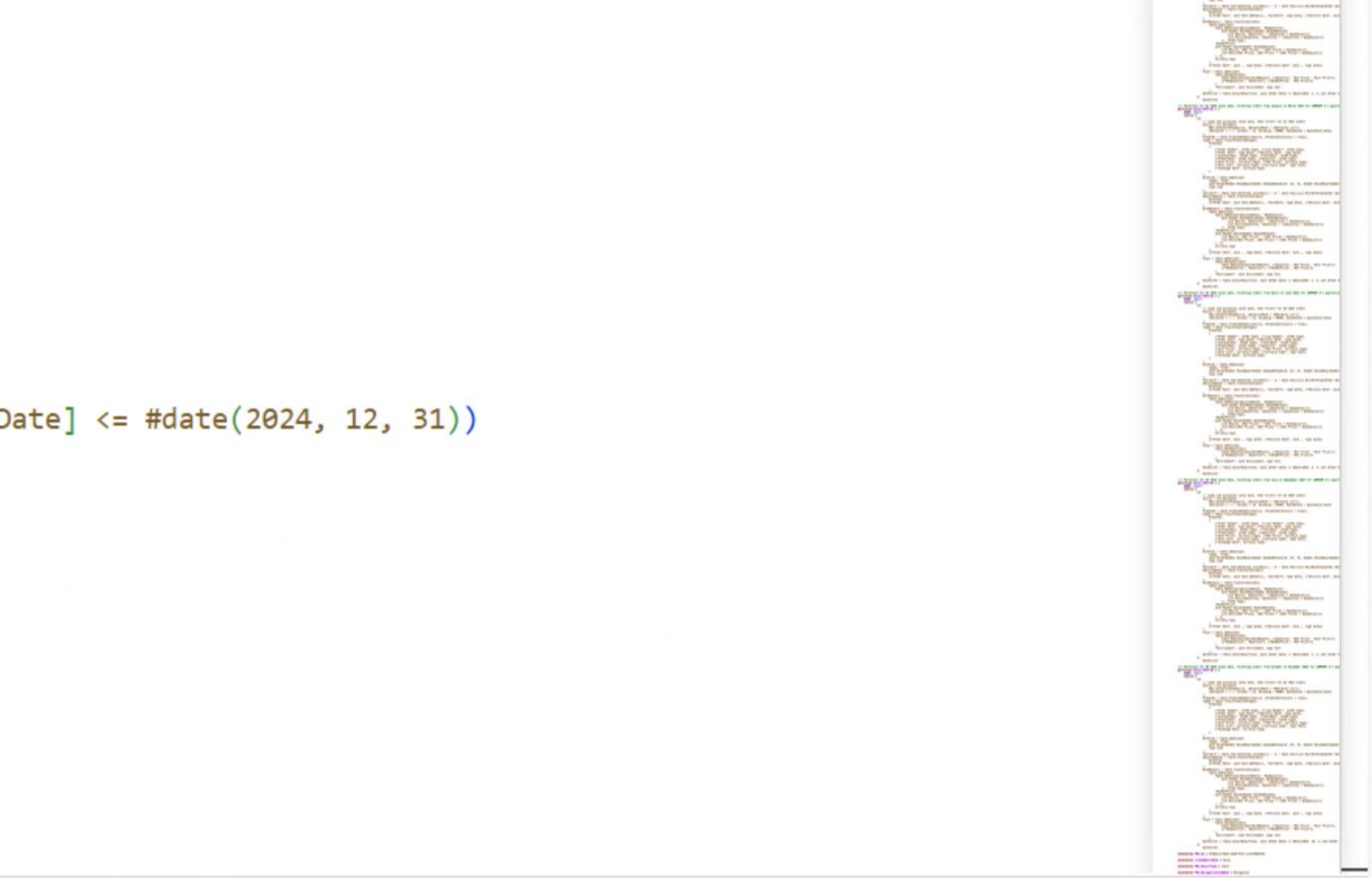
GPT-4.1

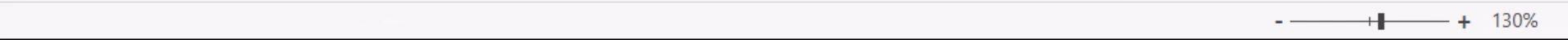
```
328         partition Sales-2024 = m
329             mode: import
330             source =
331                 let
332                     Source = Csv.Document(
333                         Web.Contents(HttpSource, [RelativePath = "RAW-Sales.csv"]),
334                         [Delimiter = ",", Columns = 13, Encoding = 65001, QuoteStyle = QuoteStyle.None]
335                     ),
336                     Promoted = Table.PromoteHeaders(Source, [PromoteAllScalars = true]),
337                     Typed = Table.TransformColumnTypes(
338                         Promoted,
339                         {
340                             {"Order Number", Int64.Type}, {"Line Number", Int64.Type},
341                             {"Order Date", type date}, {"Delivery Date", type date},
342                             {"CustomerKey", Int64.Type}, {"StoreKey", Int64.Type},
343                             {"ProductKey", Int64.Type}, {"Quantity", Int64.Type},
```

```
            each Number.RoundDown(Number.RandomBetween(
                List.Max({1, [Quantity] - ([Quantity] * Randomizer)}),
                List.Min({[Quantity], [Quantity] + ([Quantity] * Randomizer)})
            )), Int64.Type),
        "NewNetPrice",
        each Number.Round(Number.RandomBetween(
            List.Max({1, [Net Price] - ([Net Price] * Randomizer)}),
            List.Min({[Net Price], [Net Price] + ([Net Price] * Randomizer)})
        ), 3),
        Currency.Type
    ),
    {{"Order Date", each _, type date}, {"Delivery Date", each _, type date}}
),
Final = Table.AddColumn(
    Table.RenameColumns(
        Table.RemoveColumns(WithNewCols, {"Quantity", "Net Price", "Unit Price"}),
        {{"NewQuantity", "Quantity"}, {"NewNetPrice", "Net Price"}}
    ),
    "Environment", each Environment, type text
),
DateFilter = Table.SelectRows(Final, each [Order Date] >= #date(2024, 1, 1) and [Order Date] <= #date(2024, 3, 31))
in
    DateFilter

tion for Q2 2024 sales data, filtering orders from April to June 2024 for COMPANY X's quarterly analysis.
Sales-2024-Q2 = m
import
e =
    let
```

```
Date] <= #date(2024, 12, 31))
```

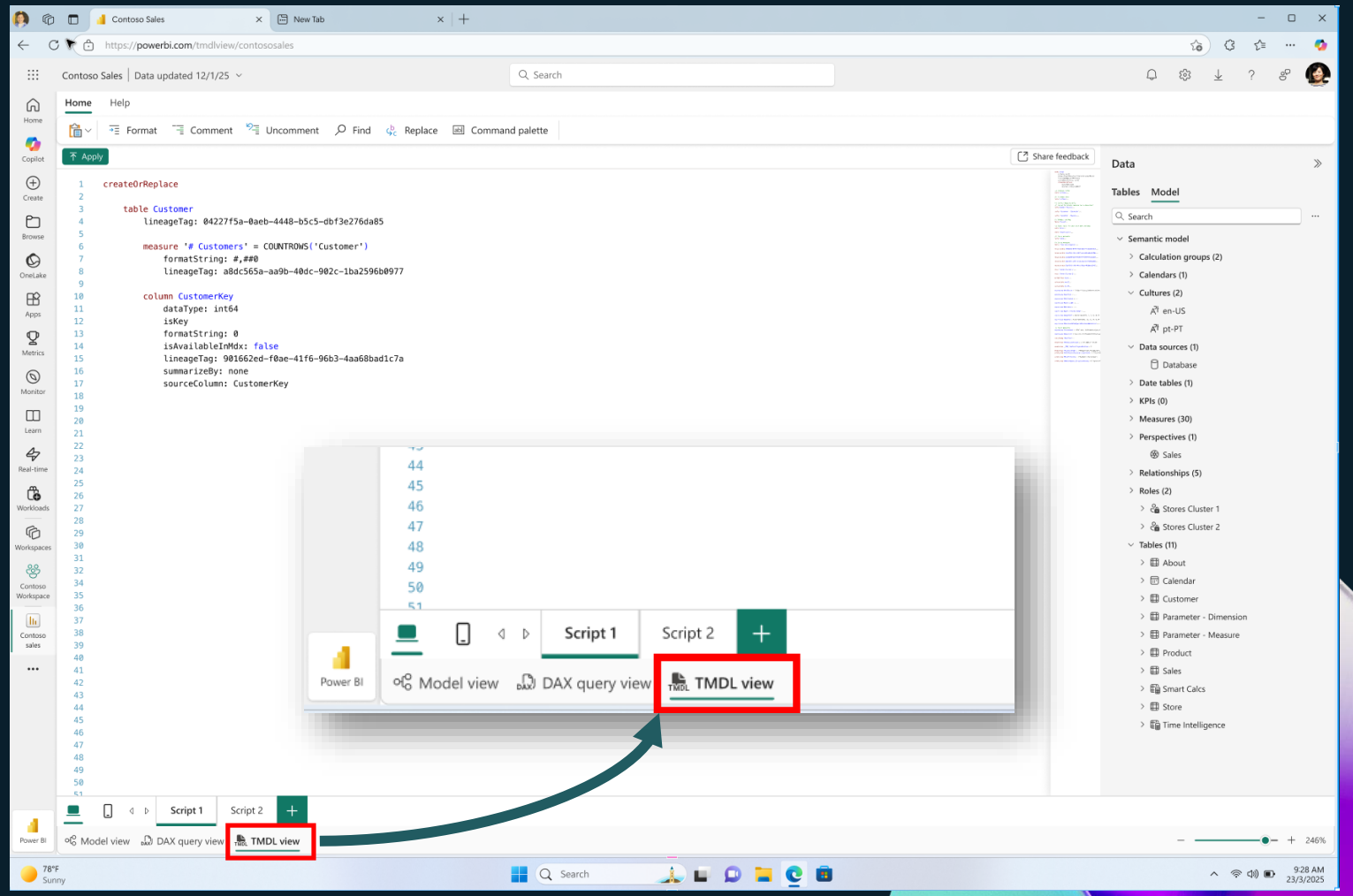




Planned

TMDL view in the Web

- Quickly inspect the metadata of any model running in service.
- Edit any object / property.
- Get and share semantic model snippets without opening Desktop.



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🧠 Recap

- TMDL is not only for source control
- Consider TMDL for development, specially for bulk changes
- Modify any object or property, beyond the UI.
- Gen AI works great with TMDL
- With Fabric Studio you can open and edit semantic models running in Service

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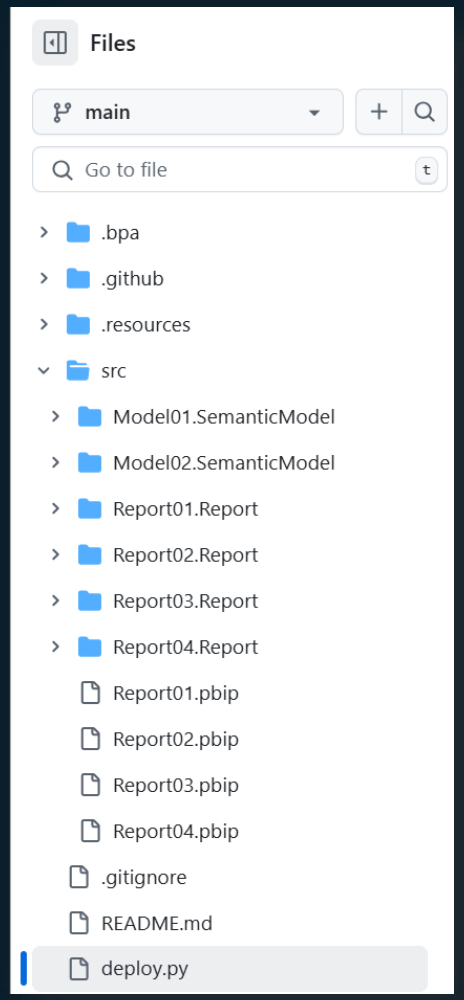
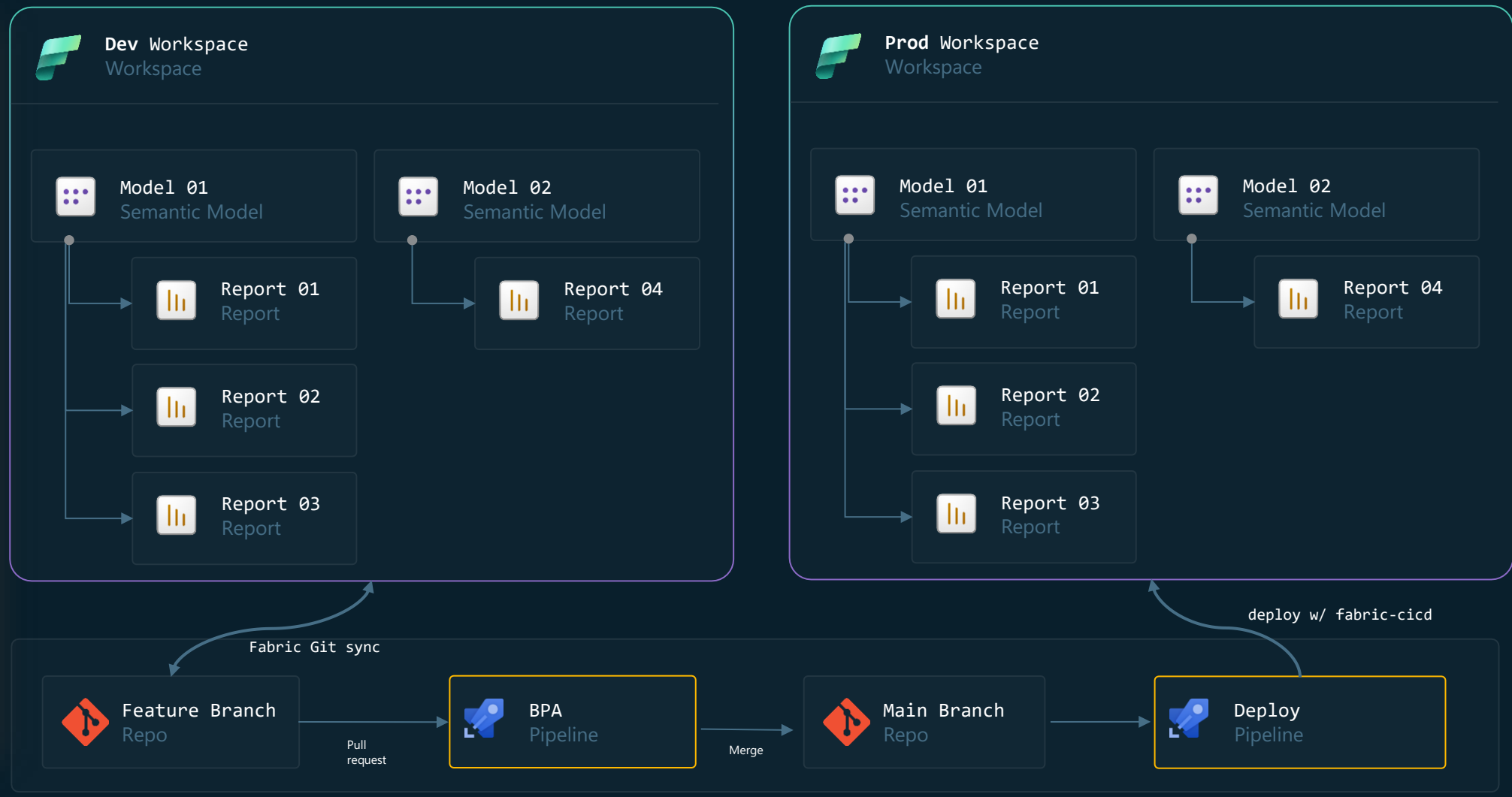
Power BI CI/CD – A possible pro-code approach





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Demo

Pro-code CI/CD approach – Setup Overview

pbip-demo/README.md at main | Getting Started - fabric-cicd

https://github.com/RuiRomano/pbip-demo/blob/main/README.md

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Files

main

Go to file t

- .bpa
- .github
- .resources
- src
- .gitignore
- README.md
- deploy.py

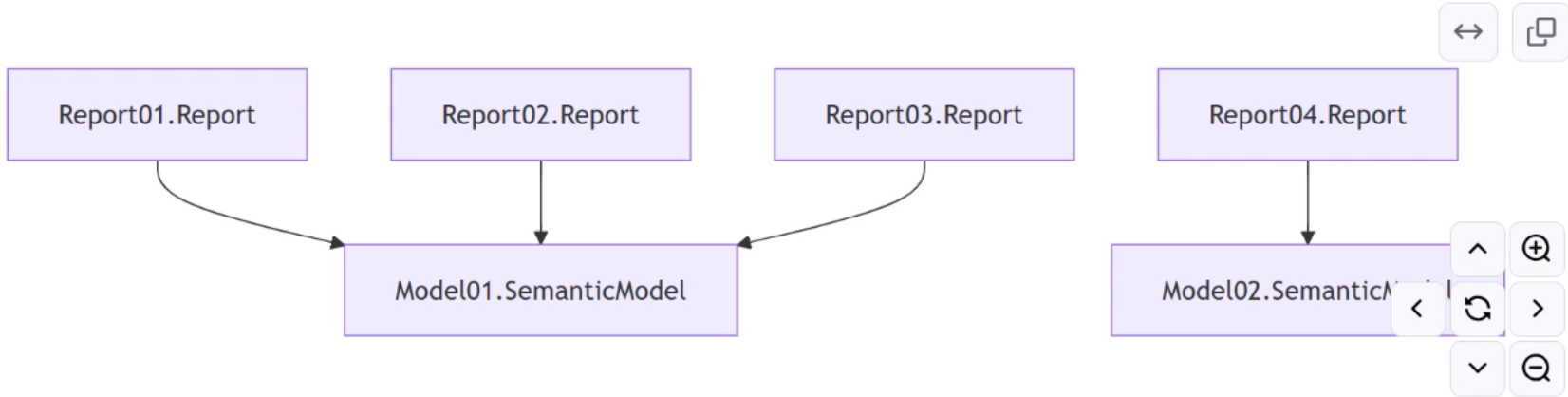
pbip-demo / README.md

RuiRomano readme — 8b42a5a · 5 days ago — History

Preview | Code | Blame — 45 lines (32 loc) · 2.6 KB — Raw

bpa passing | deploy passing

This repository showcases a Power BI Project (PBIP) folder structure in the src folder, featuring multiple reports and semantic models.



It also demonstrates how to PBIP to a Fabric workspace using Fabric CRUD REST APIs together with fabric-cicd and run how to automatically run best practice analysis.

Deployment with Github Actions and fabric-cicd

```
26      else:
27          client_id = os.getenv("FABRIC_CLIENT_ID")
28          client_secret = os.getenv("FABRIC_CLIENT_SECRET")
29          tenant_id = os.getenv("FABRIC_TENANT_ID")
30
31          credential = ClientSecretCredential(client_id=client_id, client_secret=client_sec
32
33      target_workspace = FabricWorkspace(
34          workspace_name = workspace_name,
35          environment = environment,
36          repository_directory = src_path,
37          item_type_in_scope = ["SemanticModel", "Report"],
38          token_credential = credential,
39      )
40
41      publish_all_items(fabric_workspace_obj = target_workspace)
```

```yaml
      uses: actions/checkout@v4

    - name: Set up Python
      uses: actions/setup-python@v2
      with:
        python-version: "3.12"

    - name: Install Fabric CICD
      run: |
        python -m pip install fabric-cicd

    - name: Run Deployment Script
      env:
        FABRIC_CLIENT_ID: ${{ secrets.FABRIC_CLIENT_ID }}
        FABRIC_CLIENT_SECRET: ${{ secrets.FABRIC_CLIENT_SECRET }}
        FABRIC_TENANT_ID: ${{ secrets.FABRIC_TENANT_ID }}
      run:
        python deploy.py --spn-auth true --workspace "$workspace_name" --environment $environment_name --
```

Files

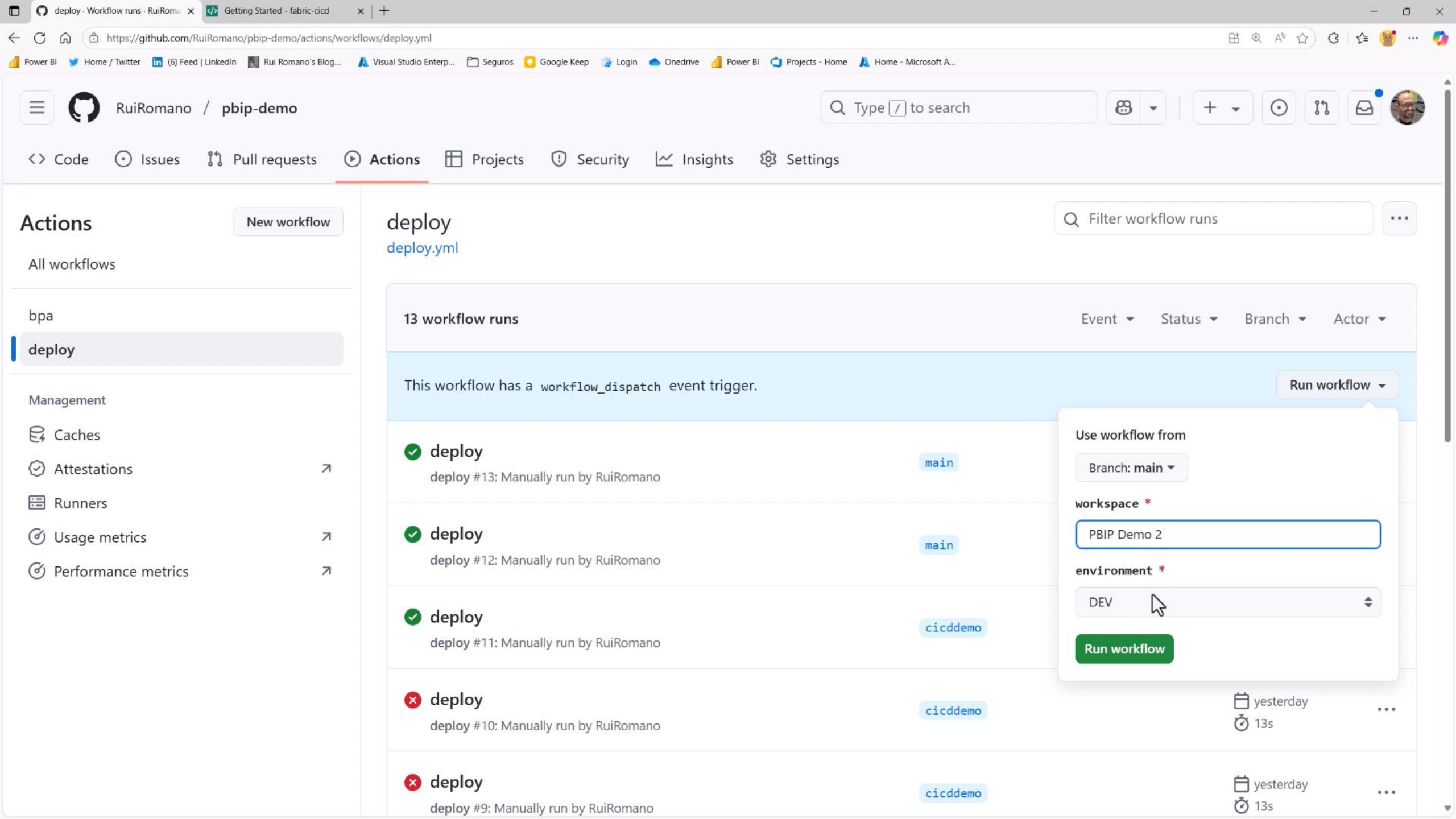
main

Go to file t

.bpa
bpa-rules-report.json
bpa-rules-semanticmodel.json
bpa.ps1
.github
workflows
bpa.yml
deploy.yml
copilot-instructions.md
.resources
src
.gitignore
README.md
deploy.py

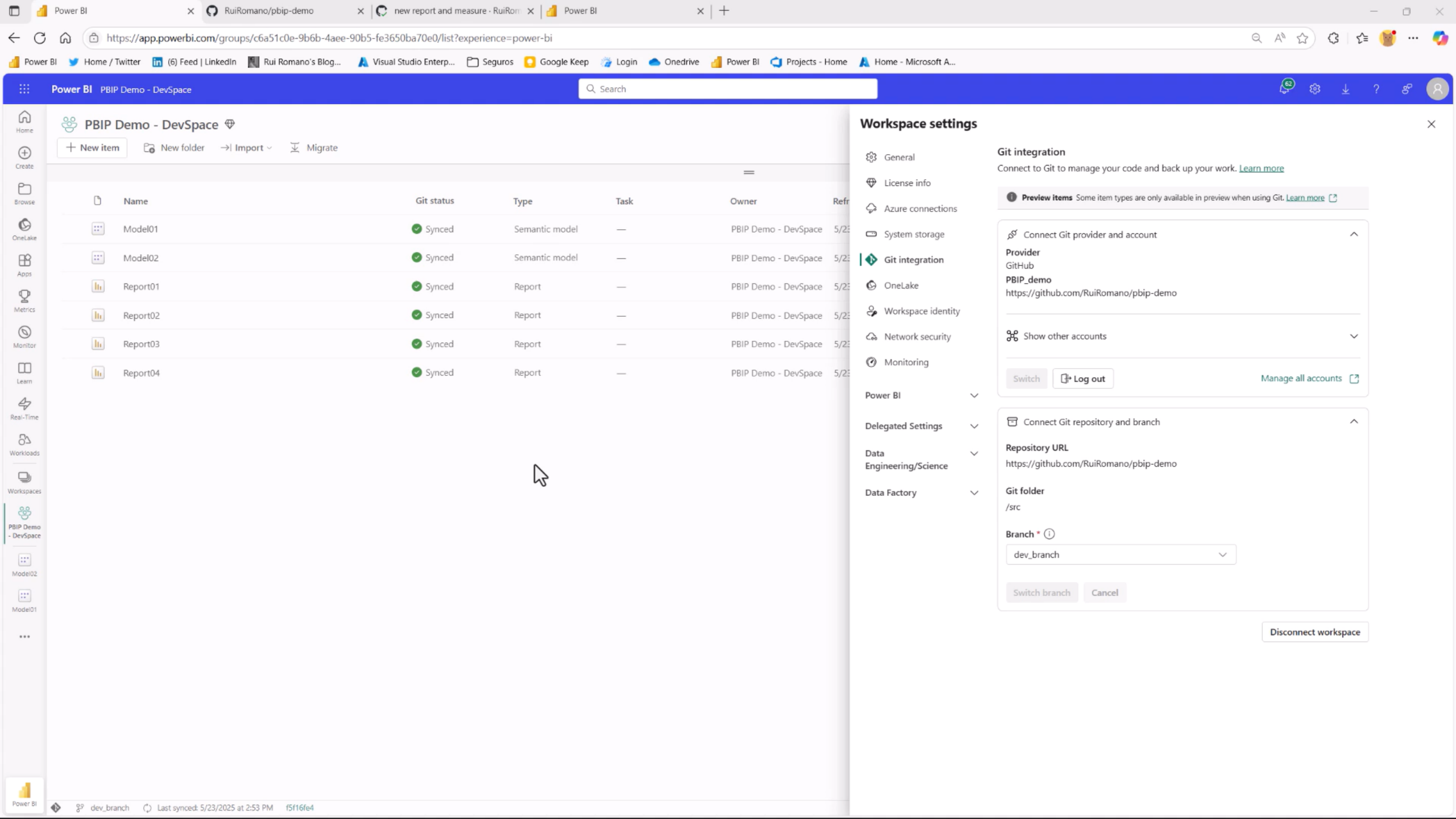

```yaml
    #   branches:
    #     - main
    #   paths:
    #     - '**/*.Report/**'
    #     - '**/*.SemanticModel/**'

    pull_request:
      branches:
        - main
      paths:
        - '**/*.Report/**'
        - '**/*.SemanticModel/**'

    workflow_dispatch:

jobs:

  bpa:
    runs-on: windows-latest

    steps:
        - name: Check out the code
          uses: actions/checkout@v4

        - name: BPA Semantic Models
          run: .\.bpa\bpa.ps1 -src @("./src/*.SemanticModel")

        - name: BPA Reports
          run: .\.bpa\bpa.ps1 -src @("./src/*.Report")
```

```
10 Loading model...
11 Running Best Practice Analyzer...
12 ================================
13 Warning:  Column 'Calendar'[Date] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
14 Warning:  Column 'Calendar'[Is Weekend] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
15 Warning:  Column 'Calendar'[Fiscal Year] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
16 Warning:  Column 'Sales'[Quantity] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
17 Warning:  Column 'About'[Key] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
18 Warning:  Column 'About'[Order] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
19 Warning:  Model Model violates rule "[Performance] Model should have a date table"
20 Warning:  Table (Import) 'Calendar' violates rule "[Performance] Date/calendar tables should be marked as a date table"
21 Warning:  Partition (M - Import) Sales-ddb4c40b-46fd-49ea-9a19-16e7e640a21a violates rule "[Performance] Minimize Power Query
   transformations"
22 Warning:  Relationship 'Sales'[Delivery Date] --> 'Calendar'[Date] violates rule "[DAX Expressions] Inactive relationships that are
   never activated"
23 Warning:  Column 'Calendar'[Month Name] violates rule "[Maintenance] Remove unnecessary columns"
24 Warning:  Column 'Calendar'[Is Weekend] violates rule "[Maintenance] Remove unnecessary columns"
25 Warning:  Column 'Calendar'[Fiscal Year] violates rule "[Maintenance] Remove unnecessary columns"
26 Warning:  Column 'About'[Key] violates rule "[Maintenance] Remove unnecessary columns"
27 Warning:  Column 'About'[Order] violates rule "[Maintenance] Remove unnecessary columns"
28 Warning:  Column 'Field Parameter'[Field Parameter Fields] violates rule "[Maintenance] Remove unnecessary columns"
29 Table (Import) 'About' violates rule "[Maintenance] Ensure tables have relationships"
30 Calculated Table 'Field Parameter' violates rule "[Maintenance] Ensure tables have relationships"
31 Column 'Calendar'[Date] violates rule "[Formatting] Provide format string for "Date" columns"
32 Column 'Sales'[Order Date] violates rule "[Formatting] Provide format string for "Date" columns"
33 Column 'Sales'[Delivery Date] violates rule "[Formatting] Provide format string for "Date" columns"
34 Column 'Store'[Open Date] violates rule "[Formatting] Provide format string for "Date" columns"
35 Column 'Store'[Close Date] violates rule "[Formatting] Provide format string for "Date" columns"
```


deploy · Workflow runs · RuiRoma
Getting Started - fabric-cicd
https://github.com/RuiRomano/pbip-demo/actions/workflows/deploy.yml
RuiRomano / pbip-demo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Security
Insights
Settings
Actions
New workflow
All workflows
bpa
deploy
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
deploy
deploy.yml
Filter workflow runs
13 workflow runs
Event
Status
Branch
Actor
This workflow has a workflow_dispatch event trigger.
Run workflow
deploy
deploy #13: Manually run by RuiRomano
main
deploy
deploy #12: Manually run by RuiRomano
main
deploy
deploy #11: Manually run by RuiRomano
cicddemo
deploy
deploy #10: Manually run by RuiRomano
cicddemo
yesterday
13s
deploy
deploy #9: Manually run by RuiRomano
cicddemo
yesterday
13s
Use workflow from
Branch: main
workspace *
PBIP Demo 2
environment *
DEV
Run workflow

Demo

Pro-code CI/CD approach – In Action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Power BI
https://app.powerbi.com/groups/c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0/list?experience=power-bi
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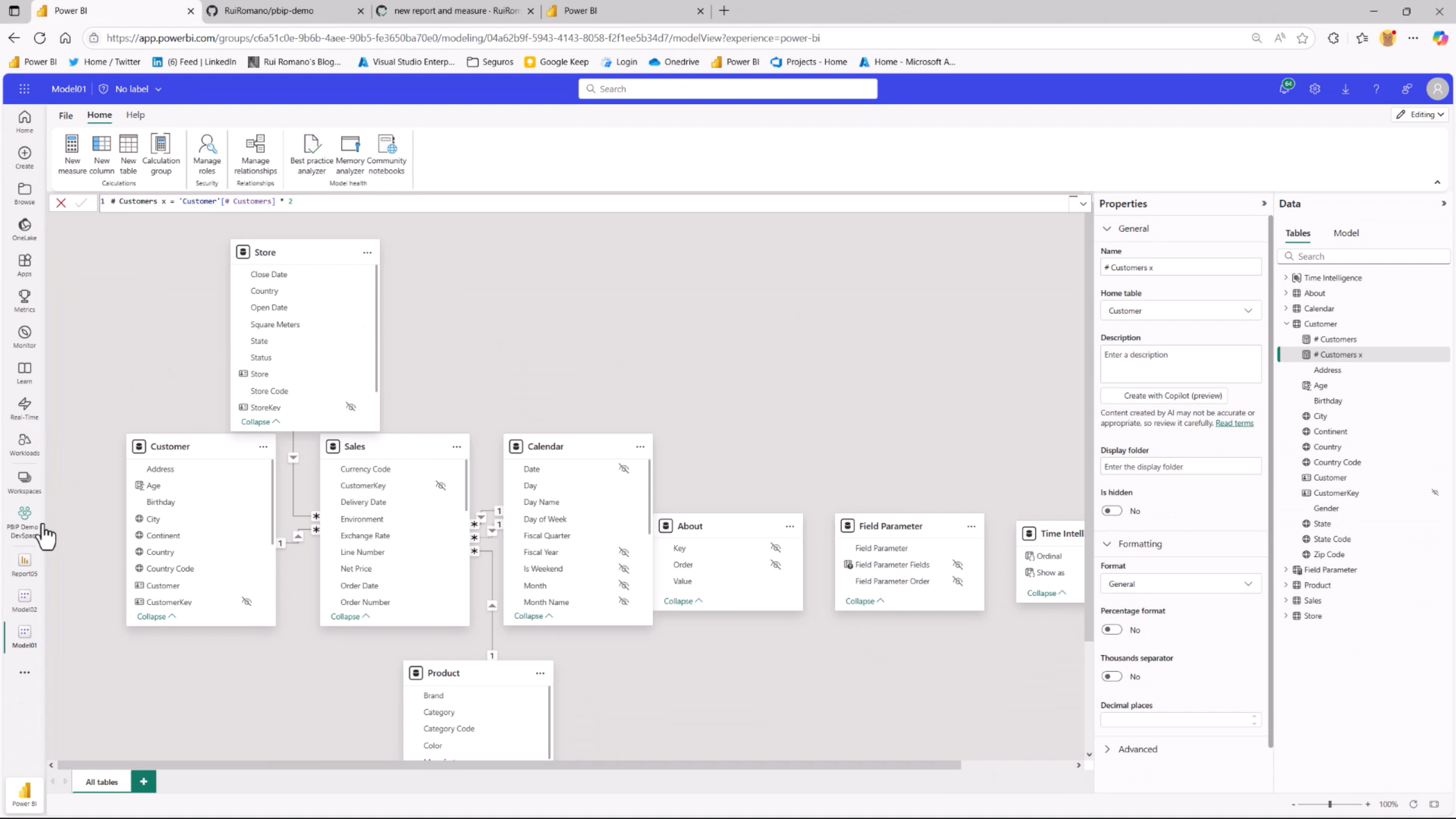
Home
Create
Browse
OneLake
Apps
Metrics
Monitor
Learn
Real-Time
Workloads
Workspaces
PBIP Demo - DevSpace
Model02
Model01
Power BI
PBIP Demo - DevSpace - DevSpace
New item
New folder
Import
Migrate
Name
Git status
Type
Task
Owner
Model01 Synced Semantic model — PBIP Demo - DevSpace
Model02 Synced Semantic model — PBIP Demo - DevSpace
Report01 Synced Report — PBIP Demo - DevSpace
Report02 Synced Report — PBIP Demo - DevSpace
Report03 Synced Report — PBIP Demo - DevSpace
Report04 Synced Report — PBIP Demo - DevSpace
Workspace settings
General
License info
Azure connections
System storage
Git integration
OneLake
Workspace identity
Network security
Monitoring
Power BI
Delegated Settings
Data Engineering/Science
Data Factory
Git integration
Connect to Git to manage your code and back up your work. Learn more
Preview items Some item types are only available in preview when using Git. Learn more
Connect Git provider and account
Provider
GitHub
PBIP_demo
https://github.com/RuiRomano/pbip-demo
Show other accounts
Switch
Log out
Manage all accounts
Connect Git repository and branch
Repository URL
https://github.com/RuiRomano/pbip-demo
Git folder
/src
Branch *
dev_branch
Switch branch
Cancel
Disconnect workspace
dev_branch
Last synced: 5/23/2025 at 2:53 PM
f5f16fe4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Power BI
https://app.powerbi.com/groups/c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0/modeling/04a62b9f-5943-4143-8058-f2f1ee5b34d7/modelView?experience=power-bi
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Model01
No label
Search
File
Home
Help
Editing
New measure
New column
New table
Calculation group
Calculations
Manage roles
Security
Manage relationships
Relationships
Best practice analyzer
Memory analyzer
Community notebooks
Model health
1 # Customers x = 'Customer'[# Customers] * 2
Store
Close Date
Country
Open Date
Square Meters
State
Status
Store
Store Code
StoreKey
Collapse
Customer
Address
Age
Birthday
City
Continent
Country
Country Code
Customer
CustomerKey
Collapse
Sales
Currency Code
CustomerKey
Delivery Date
Environment
Exchange Rate
Line Number
Net Price
Order Date
Order Number
Collapse
Calendar
Date
Day
Day Name
Day of Week
Fiscal Quarter
Fiscal Year
Is Weekend
Month
Month Name
Collapse
About
Key
Order
Value
Collapse
Field Parameter
Field Parameter
Field Parameter Fields
Field Parameter Order
Collapse
Time Intell
Ordinal
Show as
Collapse
Product
Brand
Category
Category Code
Color
Properties
General
Name
Customers x
Home table
Customer
Description
Enter a description
Create with Copilot (preview)
Content created by AI may not be accurate or appropriate, so review it carefully. Read terms
Display folder
Enter the display folder
Is hidden
No
Formatting
Format
General
Percentage format
No
Thousands separator
No
Decimal places
Advanced
Data
Tables
Model
Search
Time Intelligence
About
Calendar
Customer
Customers
Customers x
Address
Age
Birthday
City
Continent
Country
Country Code
Customer
CustomerKey
Gender
State
State Code
Zip Code
Field Parameter
Product
Sales
Store
All tables
100%

Summary

Jobs

bpa

Run details

Usage

Workflow file

bpa

failed now in 20s

Search logs | Explain error

BPA Semantic Models 11s

```
Tabular Editor 2.26.0 (build 2.26.9218.28510)
------------------------------
Loading model...
Running Best Practice Analyzer...
==============================
Warning:  Column 'Calendar'[Date] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
Warning:  Column 'Calendar'[Is Weekend] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
Warning:  Column 'Calendar'[Fiscal Year] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
Warning:  Column 'Sales'[Quantity] violates rule "[Performance] Set IsAvailableInMdx to false on non-attribute columns"
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Error:  Measure [# Customers x] violates rule "[DAX Expressions] Measure references should be unqualified"
Warning:  Relationship 'Sales'[Delivery Date] --> 'Calendar'[Date] violates rule "[DAX Expressions] Inactive relationships that are never activated"
Warning:  Column 'Calendar'[Month Name] violates rule "[Maintenance] Remove unnecessary columns"
Warning:  Column 'Calendar'[Is Weekend] violates rule "[Maintenance] Remove unnecessary columns"
Warning:  Column 'Calendar'[Fiscal Year] violates rule "[Maintenance] Remove unnecessary columns"
Warning:  Column 'About'[Key] violates rule "[Maintenance] Remove unnecessary columns"
Warning:  Column 'About'[Order] violates rule "[Maintenance] Remove unnecessary columns"
Warning:  Column 'Field Parameter'[Field Parameter Fields] violates rule "[Maintenance] Remove unnecessary columns"
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Calculated Table 'Field Parameter' violates rule "[Maintenance] Ensure tables have relationships"
Column 'Calendar'[Date] violates rule "[Formatting] Provide format string for "Date" columns"
Column 'Sales'[Order Date] violates rule "[Formatting] Provide format string for "Date" columns"
Column 'Sales'[Delivery Date] violates rule "[Formatting] Provide format string for "Date" columns"
Column 'Store'[Open Date] violates rule "[Formatting] Provide format string for "Date" columns"
Column 'Store'[Close Date] violates rule "[Formatting] Provide format string for "Date" columns"
Column 'Calendar'[Year-Month] violates rule "[Formatting] Provide format string for "Month" columns"
Warning:  Measure [# Customers x] violates rule "[Formatting] Provide format string for measures"
Warning:  Measure [Margin %] violates rule "[Formatting] Percentages should be formatted with thousands separators and 1 decimal"
Warning:  Measure [# Customers (w/ Sales)] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [# Products (w/ Sales)] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [Sales Qty] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [Sales Amount (LY)] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [# Sales] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [# Products] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [# Customers] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
Warning:  Measure [# Customers x] violates rule "[Formatting] Whole numbers should be formatted with thousands separators and no decimals"
```

Power BI | new report and measure by RuiR | RuiRomano/pbip-demo | Power BI

https://github.com/RuiRomano/pbip-demo/pull/4

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Open **new report and measure** #4

RuiRomano wants to merge 8 commits into `main` from `dev_branch`

RuiRomano and others added 8 commits 1 hour ago

new report		ebbed71
Committing 1 item from workspace c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0	✗	68e532c
Committing 1 item from workspace c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0	✓	9bc710a
Update Customer.tmdl	Verified ✓	6ef8aef
Committing 1 item from workspace c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0	✗	b2ba81f
Committing 2 items from workspace c6a51c0e-9b6b-4aee-90b5-fe3650ba70e0	✓	f5f16fe
new report and measure	✗	dd35b39
measure fix	✓	27d1341

All checks have passed

1 successful check

✓ bpa / bpa (pull_request) Successful in 45s

No conflicts with base branch

Merging can be performed automatically.

Merge pull request | You can also merge this with the command line. View command line instructions.

Add a comment

Write | Preview

Add your comment here...

Markdown is supported | Paste, drop, or click to add files

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

None yet

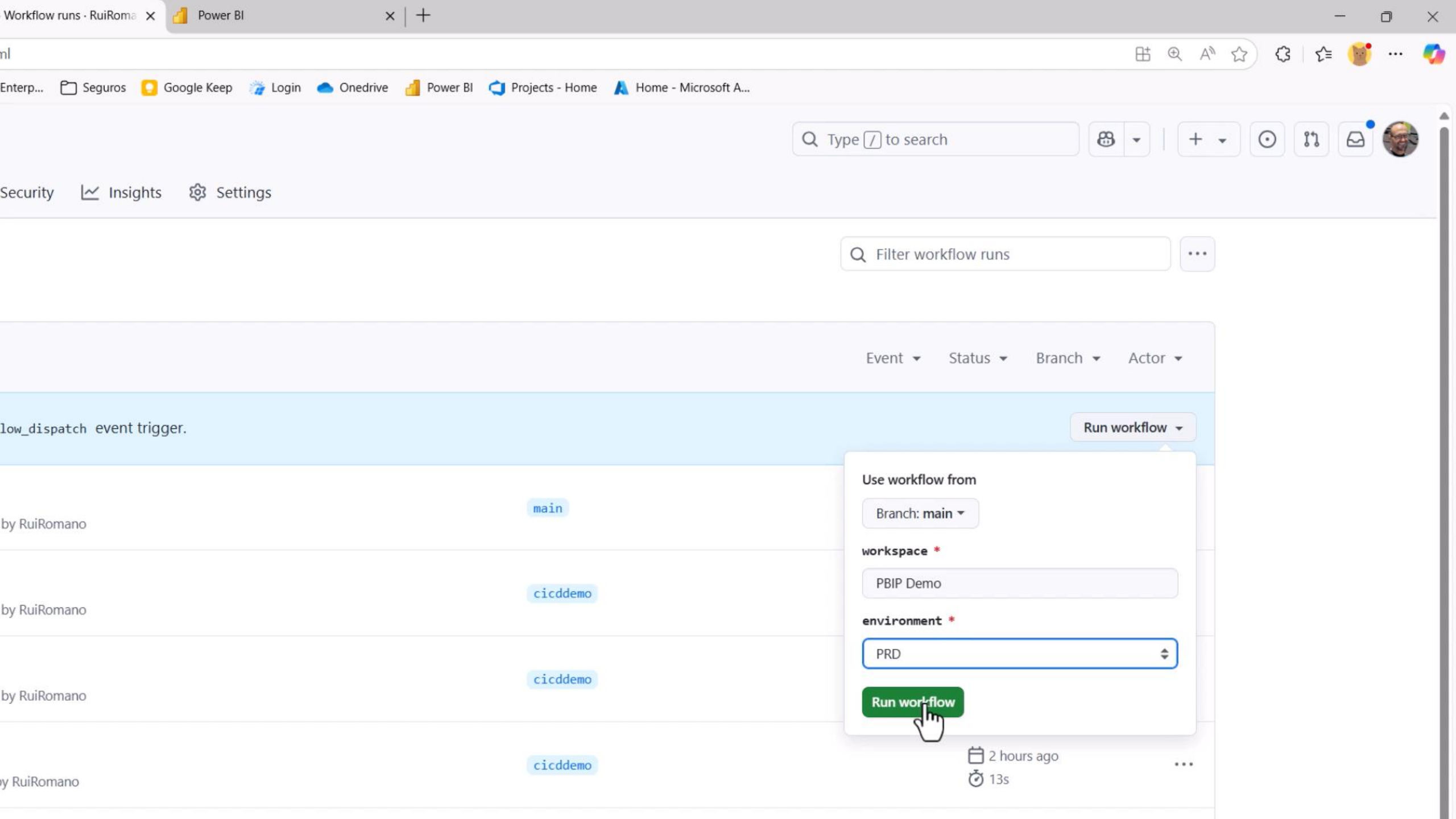
Notifications Customize

Unsubscribe

You're receiving notifications because you're watching this repository.

1 participant

Lock conversation


Workflow runs · RuiRoma
Power BI
Enterp...
Seguros
Google Keep
Login
Onedrive
Power BI
Projects - Home
Home - Microsoft A...
Type / to search
Security
Insights
Settings
Filter workflow runs
Event
Status
Branch
Actor
low_dispatch event trigger.
Run workflow
Use workflow from
Branch: main
workspace *
PBIP Demo
environment *
PRD
Run workflow
main
cicddemo
cicddemo
cicddemo
by RuiRomano
by RuiRomano
by RuiRomano
y RuiRomano
2 hours ago
13s

🧠 Recap

- Mix Fabric ALM with GitHub/ADO pipelines
- Invest in code quality gates, they are huge time savers!
- fabric-cicd greatly simplify code deployment options with GitHub/ADO
  - Abstracts Authentication, API calls, item relationship,…
- There are other options: Fabric CLI
- Learn from a sample: https://aka.ms/pbip-sample

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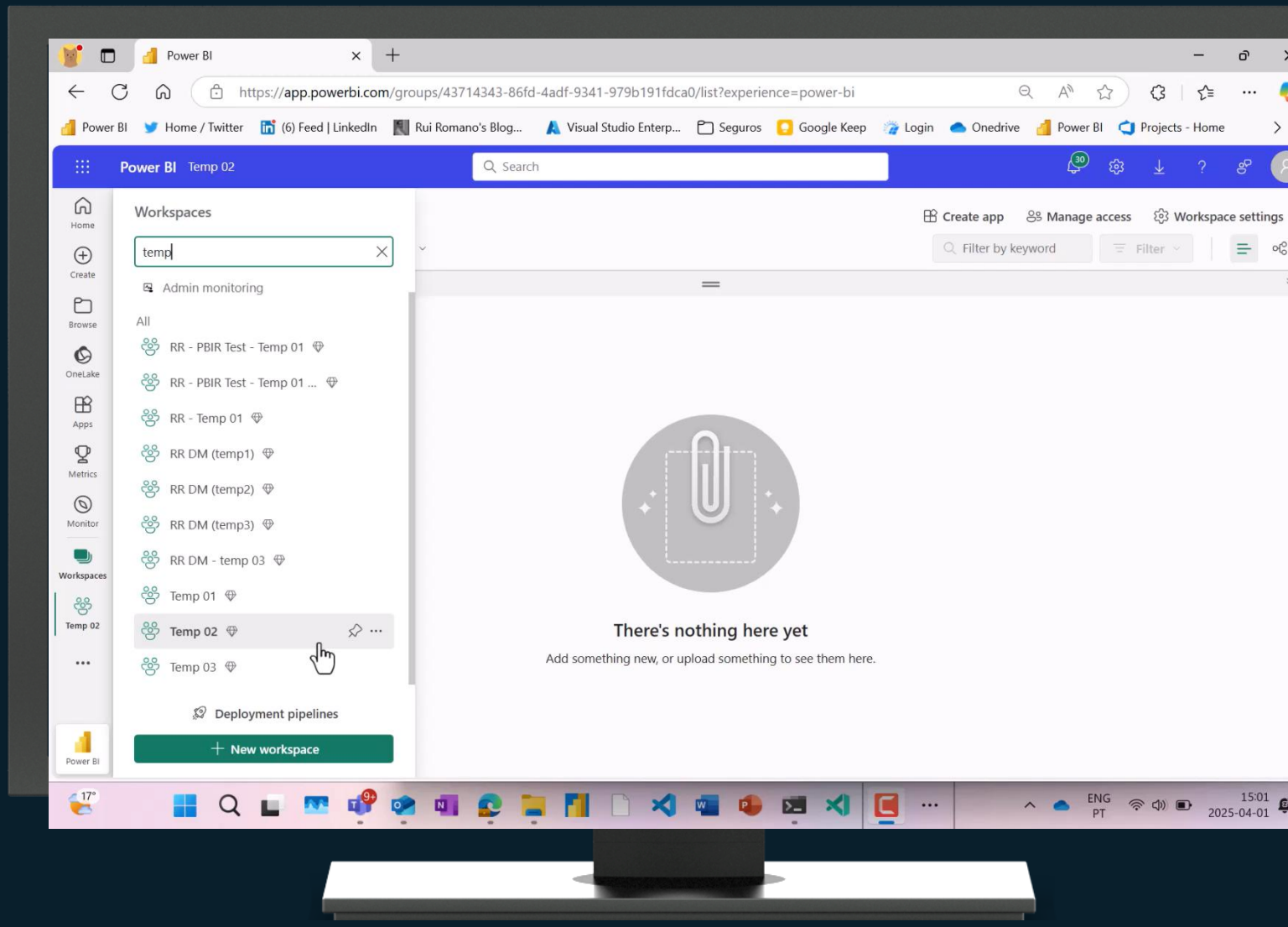
Fabric CLI

Available Now!

A command-line interface to Microsoft Fabric for use in your terminal or your scripts.

- Export/Import item definitions
- Bulk operations
- Call APIs
- Copy items between workspaces
- …



https://aka.ms/fabric-cli

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Translytical task flows

Available Now!

Go from insights to action directly in Power BI.

Call python code from your Power BI report leveraging Fabric Data Functions.

Scenarios:

- *Data Write-back*
- *Automation flows*
- ...

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Home
Copilot
Create
Workspaces
Translytical Task Flow ...
UpdateCompanyStatus
AutomateEmail
Translytical task flows
Translytical
...
Power BI

CONTOSO OUTDOORS | COLLABS

SALES | OPPORTUNITIES | FINANCE | COLLABS

EXPORT

Offers Proposed	Offers in Negotiation	Offers Approved	Offers Declined
7	2	2	1

FILTERS

Clear slicers

PRODUCT

ELECTRIC

MOUNTAIN

ROAD

REGION

- Germany
- India
- Ireland
- Italy
- Russia
- Spain
- Switzerland
- USA

PROFILE	NAME	COMPANY	STATUS	COMMENT
	John Smith	Acme Inc.	Initiated	03/21/25 2:28 PM - I emailed them an offer
	Sophia Lee	Food Delights	Accepted	03/21/25 2:27 PM - She accepted our offer and wants to buy 10 custom bikes
	Emma Davis	Tech Solutions Ltd.	In Review	03/21/25 2:09 PM - She's reviewing the design now
	Ava Müller	Swiss Precision	In Review	03/21/25 1:46 PM - She needs more time to review offer and design
	Sebastian Schmidt	AutoTech GmbH	Accepted	03/21/25 1:44 PM - They loved the design and want 4 custom bikes
	Maria Ivanova	GazProm Corp	Declined	03/21/25 1:41 PM - They don't want to collab at the moment
	Isabella Rossi	Fashionista Srl	Initiated	03/21/25 1:35 PM - She would be one of our first collabs in the fashion industry

ADD / EDIT AN OFFER STATUS

Step 1
Type the company name associated with the offer.
50 characters text limit

Step 2
Please select the status to assign to the offer.
Initiated | In Review | Accepted | Declined

Step 3
Type the comment you want to add for the offer.
50 characters text limit

Submit

Inbox

sunaraya@microsoft.com

Swiss Precision x Contoso collab time!

17:11

CONTOSO OUTDOORS

Thanks for partnering with us. We think you'll love your new bikes.

Contoso Outdoors is not just any bike company. We design and craft bikes to fit your company's unique branding and aesthetics.

Get ready to join the dozens of companies who have already made history with us!

Success Stories

This user data functions item has unsaved changes. Ensure all necessary libraries or connections are included and remember to publish to apply these changes.


```python
import fabric.functions as fn
import logging
import requests
from flask import Flask
from flask_mail import Mail, Message

udf = fn.UserDataFunctions()
app = Flask(__name__)
mail = Mail(app) # instantiate the mail class

# configuration of mail
app.config['MAIL_SERVER']='smtp.sendgrid.net'
app.config['MAIL_PORT'] = 587
app.config['MAIL_USERNAME'] = 'apikey'
app.config['MAIL_PASSWORD'] = '<API Key Here>'
app.config['MAIL_USE_TLS'] = True
app.config['MAIL_DEFAULT_SENDER'] = '<Your Email Here>'
mail = Mail(app)

html_email = """<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">
<html data-editor-version="2" class="sg-campaigns" xmlns="http://www.w3.org/1999/xhtml">
    <head>
      <meta http-equiv="Content-Type" content="text/html; charset=utf-8">
      <meta name="viewport" content="width=device-width, initial-scale=1, minimum-scale=1, maximum-scale=1">
      <!--[if !mso]><!-->
      <meta http-equiv="X-UA-Compatible" content="IE=Edge">
      <!--<![endif]-->
      <!--[if (gte mso 9)|(IE)]>
      <xml>
        <o:OfficeDocumentSettings>
          <o:AllowPNG/>
          <o:PixelsPerInch>96</o:PixelsPerInch>
```

With code, life it's just better…

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Thank you

Microsoft

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